



PFRMP- Implementation Guidelines

Billion Tree Afforestation Support Project

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ABBREVIATIONS

10-BTTP	10 Billion Tree Tsunami Program
APD	Assistant Project Director
BTAP	Billion Tree Afforestation Project
BTASP	Billion Tree Afforestation Support Project
CCFEWD	Climate Change, Forestry, Environment and Wildlife Department
CDE&GAD	Community Development, Extension and Gender and Development Department
CDO (m/f)	Community Development Officer (male/female)
CTA	Chief Technical Advisor
DCTA	Deputy Chief Technical Advisor
DFO	Divisional Forest Officer
DFS	Deutscher Forst Service GmbH
ESMF	Environmental and Social Management Framework
FFE	Female Forest Extensionist
FP&MC	Forest Planning and Monitoring Circle
KfW	Kreditanstalt für Wiederaufbau
NTFP	Non-Timber Forest Products
PD	Project Director
PFRMP	Participatory Forest Resources Management Plan
PIC	Project Implementation Consultant
PMU	Project Management Unit
PU	Planning Unit
SDFO	Sub-Divisional Forest Officer
VDC	Village Development Committee
WO	Women Organization
WWF	Worldwide Fund for Nature - Pakistan

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1. INTRODUCTION

Overview of BTAP, 10 BTTP, and BTASP Initiatives in Khyber Pakhtunkhwa

Under the Billion Tree Afforestation Project (BTAP) and the 10 Billion Tree Tsunami Programme (10 BTTP) in Khyber Pakhtunkhwa (KP), numerous activities—such as enclosures and plantations—have been carried out in rural areas on communal and private land. Community involvement was essential for ensuring effective and efficient implementation. This participation not only enhances project execution but also contributes to poverty reduction.

The BTASP (Billion Tree Afforestation Support Project) supports both BTAP and 10 BTTP. Its objectives include protection and sustainable management of BTAP and 10 BTTP plantations and enclosures, expanding forest resources, improving livelihoods, and mitigating the adverse impacts of climate change. The project aims to enhance the livelihoods of local populations.

Contributions of BTASP

BTASP focuses on the protection, sustainable use, and management of forests established under BTAP and 10 BTTP as part of the National Clean Green Pakistan Initiative and the global Bonn Challenge. It emphasizes improved governance, increased local participation in decision-making, and livelihood support. Key governance aspects include:

1. Prioritizing poor rural communities, small landowners, and informal users.
2. Delegating decision-making powers to local communities as forest owners and users.
3. Enhancing women's livelihoods and participation.
4. Strengthening the Forestry, Environment, and Wildlife Department (FEWD) as the Project Executing Department (PED).
5. Improving transparency, accountability, and efficiency in FEWD-managed programs.
6. Promoting the sustainability of FEWD's activities.

Goals and Measures of BTASP

The overarching aim of BTASP is to strengthen the participatory and sustainable management of forest landscapes restored through BTAP and 10 BTTP. Measures include:

- Improving the livelihoods of poor populations, particularly women and youth.
- Building the capacity of forestry departments.
- Enhancing the forestry management information system.
- Establishing a results-based payment system for farmers.
- Promoting forestry-based value chains to sustain afforestation activities.

Challenges in Past Projects

Previous forest projects in KP achieved significant milestones but often lacked sustainable management plans or exit strategies. Projects like the Watershed Management Project in Hazara and the Social Forestry Project in Malakand succeeded initially but failed to ensure long-term sustainability, leading to haphazard resource utilization. The BTAP Third Party Monitoring Report¹ provides not only the results achieved but also a SWAT analysis pointing out missing long term sustainable management strategies. Also, a case study report of 10BTTP analyses critical the impacts of short-term mega tree planting

¹ Third Party Monitoring of Ten Billion Tree Afforestation Project in KP, Phase II, July 2017, WWF

projects². BTASP aims to address these shortcomings by integrating best practice sustainable forest management practices for communities looking at all segments of the society with a high level of participation on decision taking in planning and execution also by small holders and women, activity contribution and establishment of NRM funds for long term financial operational viability.

Environmental and Socio-Economic Contributions: BTASP contributes to the creation of multifunctional forests that provide both ecological and economic benefits. These forests support biodiversity, regulate water runoff, and protect against erosion. They also offer resources such as firewood, fruits, nuts, and non-timber forest products (NTFPs), which are critical for local subsistence and income generation.

KP's forests are biodiversity-rich, hosting over 250 woody plant species. The region is renowned for wild fruit trees like wild apple, peach, and apricot, integral to its natural and cultural landscape. However, overuse and climate change have degraded forest sites, emphasizing the need for initiatives like BTASP to rehabilitate and expand forest areas.

Implementation and Impact: The BMZ-funded BTASP project, implemented by the Forest Department of KP in collaboration with DFS GmbH (Germany) and WWF Pakistan, supports activities across 13 forest divisions. The project emphasizes participatory forest management plans (PFRMP), which include:

- Silvicultural and agroforestry activities.
- Livelihood enhancement measures.
- Development of afforestation plans and guidelines for site selection, planting techniques, cost norms, and reward mechanisms.

The implementation framework ensures the involvement of local communities as forestland owners and users. While these guidelines are tailored to BTASP, they might also provide a model for joint forest management activities across KP in future.

BTASP builds on the achievements of BTAP and 10 BTTP, ensuring sustainable forest management and improving the livelihoods of local communities. By addressing governance issues, promoting participation, and enhancing forest resilience to climate change, the project is a cornerstone of KP's efforts to protect and expand its forest resources while combating poverty and environmental challenges.

Achievements of BTAP and 10BTTP From 2014, under the **BTAP** project about 593,292 ha of enclosures and plantation's have been established whereby 263,213 ha fall under new plantation's establishment, 307 ha under enclosure establishment and protection and ca 23,000ha under farm forestry and others. BTAP increased the KP forest cover by 6,3% from 20,3% to 26,6%. The **10BTTP** project achieved so far 713,55 million plants raised on all legal forest categories but most of these plantations are raised on communal and private lands.

How to use these guidelines: These guidelines provide essential information for the BTASP management on PMU and DFOs level how to implement selected PFRMP and ESMF measures on private and communal land in selected Planning Units. The document provides information of forest types found in the intervention area, provides guidelines on species to be used, beneficiaries to be selected, and seedling demand. The document also comprises the description of the PFRMP measures, responsibilities during implementation of the various stakeholders, essential agreements, cost norms and monitoring formats to be used.

² Participation and exclusion in mega-tree planting projects: A case study of the 10BTTP, Pakistan, 2022

2. FOREST SITUATION IN KHYBER PAKHTUNKHWA

2.1 Legal Classification of Forests in KP

Overview legal forest classification, ownership and area in hectare³

Legal Category	Ownership	Area in Mill. Ha	% Forest Dep.	% KP Province	BTASP
Reserved Forest	state	0,091	10	0,9	
Protected Forest	state	0,512	58	5,02	X
Un-classified Forest	state	0,006	0,6	0,06	
Resumed Land	state	0,036	4,4	0,35	
Guzara Forest	private	0,239	27	2,34	X
Private Forest	private	1,184	-	11,63	X
Project Plantations	n.n	0,675	-	6,45	X
Total		2,743	100	26,75	
Newly Merged Districts	n.n	0,5	Legal status is yet to be declared		

Table 1: Overview of legal forest classification, ownership and area

Description legal classification⁴

State Owned Forest Land:

1. Reserved Forests: These are Public Forest Lands, free of all land use rights except those admitted in the settlement process and recorded in Wajib-ul-Arz. Such rights are limited. At present limited management is done by the FD due to the felling ban. BTASP will not work in that forest category.

2. Protected Forests: This state forest land is open to all uses except those uses which may have been prohibited by special notification. The management is done by the FD; People having customary rights are entitled to 60% to 80% share of the sales from timber. In Malakand BTSAP activities on this category is limited to enclosures only.

3. Un-classed Forests: Public Forest land under the control of the Forest Department, which are neither Reserved Forests nor Protected Forests. Community or individually used land, mainly wasteland for grazing and firewood collection. Most of BTAP and 10BTTP measures have been carried out in this category therefore BTASP measures will also fall under this category.

4. Resumed Lands: Lands surrendered by larger owners following the fixing of a ceiling on the extent of land ownership under the Land Reforms Act of 1959. Concerned landowners retained cultivated lands but surrendered wooded lands in excess of the ceiling. BTASP will not work on this forest category.

³ Official source Forest Department, KP

⁴ The description does not contain the full official definition per Forest Ordinance of KP.

Privately Owned Forests:

1. **Guzara** (subsistence) forests and community forests: sizeable patches of wooded land or wasteland close to habitation, which were set aside to meet the “bona fide” domestic needs of local communities. They were designated from reserved government forests at the first land ownership settlement in Hazara, NWFP, 1872. Ownership is vested in local people either individually (private), or community owned as ‘village shamilat’. Partly with large landowners. Guzara forests and wasteland are eligible for BTASP, large landowners only with limited lands within a village. In Hazara area most of the BTASP enclosure related activities will be in Guzara Forests.

2. **Communal forests:** forest of village communities. Some forests of this category are eligible for BTASP project activities.

3. **Section 83:** Privately owned lands voluntarily and temporarily put under the control of FD for conservation and preservation of soil and vegetation. BTASP will work in this category.

4. **Farm Forest Areas:** Farm Forests are linear or compact plantings of trees on private farmlands owned individually or by a family. Farm forest areas are found throughout the barani (arid areas) and irrigated farming areas of Khyber Pakhtunkhwa. BTASP will work in that category.

2.2 Forest Types in KP

Based on a report from 2020 the area size of the KP forest types on designated forest have been assessed during a Green House Gas inventory of KP⁵. The total forest area in KP designated forest area according to (FAO definition of forest, with tree density more than 10 %) is in 2020 1,644,298 ha. The most dominant forest type in KP is the Dry-Temperate Forest followed by the Sub-Tropical Chir pine forest see Table 2.

Forest Type	2016	2020
	Area in ha	Area in ha
Sub-Alpine	15,636	19,562
Dry Temperate	812,749	838,036
Dry temperate Juniper and Chilghoza Forests	3,895	3,956
Moist Temperate	297,545	303,634
Sub-tropical Chir Pine	310,565	314,575
Subtropical broad leaved (Scrub)	157,568	162,692
Tropical Thorn	1,03	1,843
total	1,598,988	1,644,298

Table 2: Forest Types in KP and area coverage in hectare on designated forest only

⁵ GREEN HOUSE GAS INVENTORY OF FORESTRY SECTOR -KHYBER PAKHTUNKHWA, WWF Pakistan, 2022

2.3 Impact of Climate Change ⁶ in KP

Impacts of Climate Change in Khyber Pakhtunkhwa (KP), Pakistan

Khyber Pakhtunkhwa (KP), a region known for its mountainous landscapes, rich biodiversity, and reliance on natural resources, is increasingly vulnerable to the impacts of climate change. The province faces significant challenges that threaten its environment, economy, and communities. Below is an analysis of the impacts based on various reports and studies.

1. Rising Temperatures and Glacial Melt

The KP region is part of the Hindu Kush-Himalayan (HKH) belt, often referred to as the "Third Pole." Rising temperatures have accelerated the melting of glaciers, which poses serious threats to water security and increases the risk of glacial lake outburst floods (GLOFs). These floods endanger downstream communities and infrastructure. ([Nature](#), ADB)

2. Altered Precipitation Patterns

Changes in monsoon and winter rainfall patterns have led to unpredictable water availability, which affects agriculture, hydropower generation, and drinking water supplies. Reduced snowpack and erratic rainfall exacerbate water stress across the province. (ISAS)

3. Increased Frequency of Extreme Weather Events

KP has seen an increase in the frequency and intensity of floods, landslides, and droughts. Flash floods and heavy rains, intensified by climate change, have caused significant loss of life and property. For example, extreme monsoon events in recent years have devastated infrastructure and displaced communities. (Reuters)

4. Degradation of Forests and Biodiversity

KP's forests are crucial for regulating climate, conserving biodiversity, and providing livelihoods. Climate change has worsened forest degradation due to pests, diseases, and wildfires. Endemic species, such as wild fruit trees and medicinal plants, face growing threats. Reforestation efforts like the Billion Tree Afforestation Programme (BTAP) aim to counter these impacts but require sustained management. ([WWF](#))

5. Agriculture and Livelihood Vulnerability

Agriculture, the backbone of KP's economy, is highly sensitive to climate variability. Unpredictable rainfall, increased water scarcity, and soil degradation have reduced crop yields, threatening food security. Rural populations, especially smallholder farmers, are at heightened risk of poverty. (EarthArxiv)

6. Climate-Induced Migration

Natural disasters and resource scarcity have led to the displacement of communities in KP. Internal migration is becoming more common as people seek safer areas and better livelihoods, placing additional pressure on urban centers. (ADB)

7. Impacts on Human Health

The health systems in KP are under pressure from climate-related issues, including heatwaves, waterborne diseases, and malnutrition. Poor air quality and reduced access to clean water exacerbate these challenges, particularly for vulnerable populations. (OPML)

Ecosystem services to counter climate change impacts

As mentioned above ca. 26% of the KP areas is covered with diverse types of forests. Therefore, climate resilient forestry approaches are required to counter and limit the negative effects of climate change and its impacts. Forest land contributes in form of Eco-system services listed in Table 3.

⁶ **Citations:** Asian Development Bank (ADB): Climate Change Assessment, Nature: [Glacial Lake Outburst Risks](#), ISAS: Impacts of Climate Change in the Highlands, WWF Pakistan: [Forests and Biodiversity](#), EarthArxiv: Climate Change Impacts in KP, OPML: Health and Climate Change

Selected **Eco-System services of Forest** cover and its contribution (not exhaustive)

Macro Effects of forests	contributes to	Ecosystem services
Mico- and Meso Climate stabilization	reduces effects of extreme weather events	balancing temperature peaks, increase humidity and reducing impact on human health
Increase Air filter effect	filter of air pollution inc. dust	reduces risk of air borne diseases
Permanent ground cover by forests	Land stabilisation and Eco-system services available	reduces risk of landslides, increase water infiltration, soil fertility, etc.
Increased water infiltration rate	stabilise ground water tables, support to regular water discharge	available drinking and irrigation water
Increase Soil fertility	barren- and wasteland will be useable to support livelihoods	support availability of natural resources for livelihood support
Natural carbon sink	sequestration of CO ₂ in forest plant	support CO ₂ reduction to counter greenhouse effect and global warming
Forest biodiversity	stabilisation of forest systems against negative climate impacts and reduction of forest cover	ecosystem services delivery resilience increased against external factors

Table 3: Eco-System services of Forest in KP

2.3.1 Climate Resilient Forestry

Effects of climate change are in Pakistan and Central Asia will be more visible than in other regions of the world (ADB, 2015). The mean annual temperature has increased by 1.2 to 2.1°C in the period from 1950 to 2000 in Central Asia. For the end of the century, a warming in the range of 2.9 to 5.1°C (compared to the reference period from 1961 to 1990) is expected. Changes in precipitation are less predictable and so far, there was no clear trend observed (CSC, 2012). Overall, a strong increase in heat periods, catastrophic weather events (ADB, 2015) and shortening of cold periods are projected (CSC, 2012). The observed and projected changes require appropriate mitigation measures throughout the country.

Some general issues to consider in the context of **climate resilient afforestation** in Pakistan Climate change effects have to be considered during planning of the afforestation in order to create climate resilient and sustainable forests with multiple benefits (ecological, economic and social). Climate resilient forestry has to consider both, resilience of forest ecosystems (e.g. high biodiversity, connectivity of forests) and socio-economic systems (e.g. good governance, strong capacities and sustainable financing).

Guideline for climate resilient forestry:

1. **Native species** are to be used in afforestation of forests.
2. Site conditions (height above sea level, soil, exposition, small-scale site features and microclimate) are to be considered in order **to select the most suitable and site appropriate species**, which will increase potential resilience of the forest.
3. Wherever possible **no monocultures**, but establishment of **species mixed plantations** will be favoured. This is a key requirement for enhancing adaptive capacities of forest ecosystems. In case the changing leads to reduced survival rate or growth of one species and/or the species finally drops out, other species may still be able to cope with the changing conditions. **Basic**

mixing schemes shall be tested and if required adjusted to the field conditions and current understanding of local forestry.

4. Due to certain **mixtures synergy effects** can be utilized. One species may be able to protect another one against adverse effects (e.g. by shadowing) or will take the place (crown extension) when the lifespan of another already ends. This is one of the reasons for planting schemes consisting of a **frequent alteration between larger tree species and bushes**.
5. In future, the **management objectives of forests** should be made clear from the very beginning. Currently short-term economic objectives prevail the thinking in protection forestry. For this reason, the planting density should increase on sites. **Planting densities** of 1000–2000 or even more per hectare (ha) should be envisioned. The numbers of planted trees are generally higher in the beginning: thinning (e.g. for firewood) and competition can regulate the number during the stand dynamics and the growth process.

3. BTASP INTERVENTION AREA

3.1 BTASP Forest Regions and Forest Divisions

The BTASP intervention area comprises 13 Forest Divisions in the administrative Divisions of Peshawar, Hazara and Malakand. The Project area lies in the Forests Region-I, II and III in specified Forest Divisions (FD) as well as Watershed Divisions:

Region-I: FD Peshawar (1)

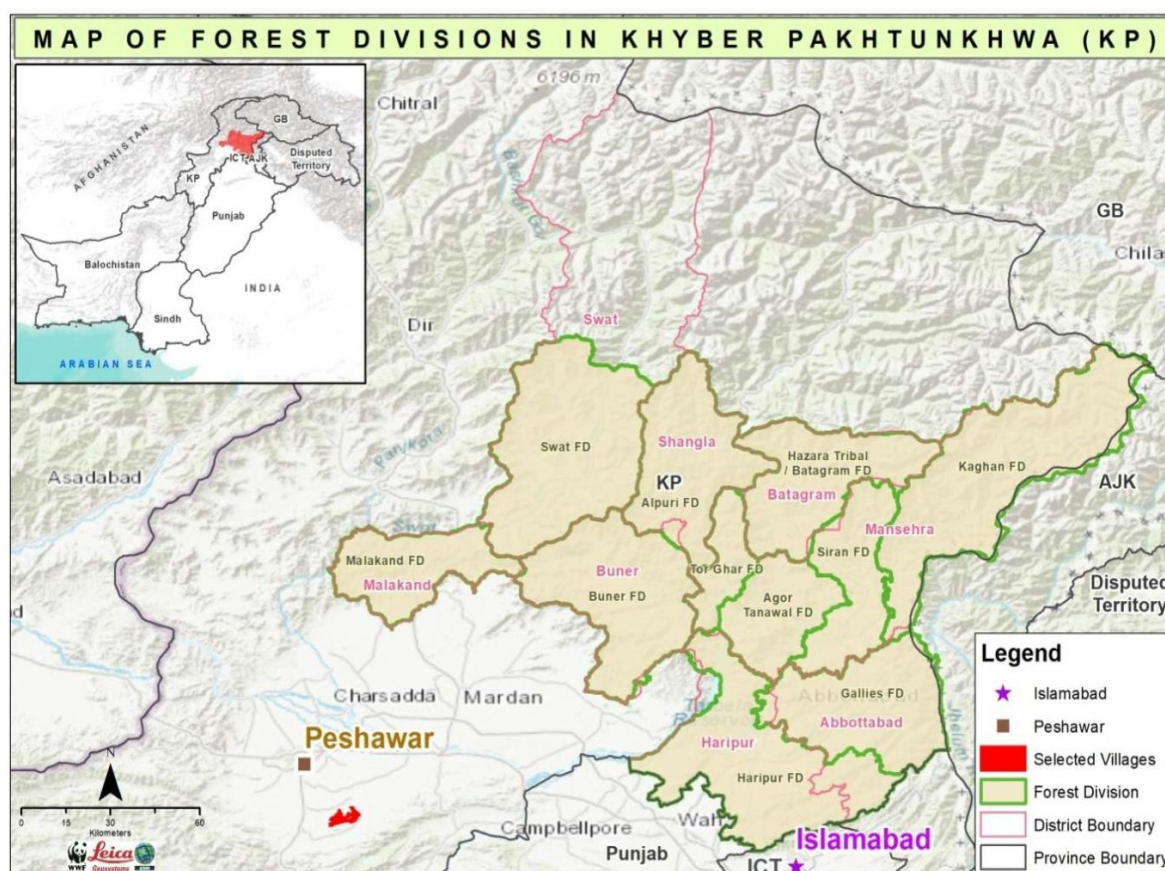
Region-II: FD of: Galies, Siran, Haripur, Kaghan, Agror Tanawal, Hazara Tribal, Torghar (7)

Region-III: FD of: Malakand, Alpuri, Upper Swat, Lower Swat, Buner (5)

Within those selected 13 FD at least 100 Planning Units have been selected based on criteria's⁷. The table of selected PUs is attached in Annex 3.

Forest Region	No. Forest Division	No of PUs	BTAP and 10 BTTAP Planation's & Enclosers ha
Region I	1	6	5,257
Region II	7	54	15,990
Region III	5	40	16,863
BTASP intervention total	13	100	38,109

Table 4: Overview Region, No Forest Divisions, No of PUs and corresponding forest data



Map 1: BTASP intervention area, 13 Forest Divisions incl. Peshawar

⁷ Report on Selection of Planning Unit (PUs), September 2023

The below Table 7 provides an overview of BTAP and 10BTAP area per ha established plantations and enclosures to be considered by BTASP to take in consideration for forest management. The full Table per Planning Unit is attached in Annex 3.

Region	Forest Division	No of PUs	Planations ha	Enclosers ha	Sum P+E ha	% total area
Region 1	Peshawar	6	5.257	0	5.257	14
					0	0
Region 2	Gallies	8	1.207	1.005	2.212	6
	Haripur	6	544	246	790	2
	Siran	8	481	2.248	2.729	7
	Hazara Tribal (Ba	8	1.632	240	1.872	5
	Kaghan	8	1.204	1.640	2.844	7
	Agror Tanawal	8	1.204	1.240	2.444	6
	Torgher	8	1.380	1.720	3.100	8
					0	0
Region 3	Malakand	8	4.873	1.658	6.531	17
	Lower Swat	8	929	200	1.129	3
	Upper Swat	8	547	4.245	4.792	13
	Alpuri	8	530	1.239	1.769	5
	Buner	8	2.003	640	2.643	7
R1+R2+R3		100	21.789	16.321		
Planations + Enclosuers					38.109	100

Table 5: BTASP Overview of BTAP and 10 BTAP Plantations and Enclosers⁸

3.2 Climatical Conditions in Hazara, Malakand and Peshawar Divisions

Hazara Forest Region: Climatically almost all of Hazara lies in the Monsoon range with warm to cool temperatures and rainy season in summer, while cold winter, with heavy snow falls in the upper ranges. Annual rainfall averages around 1200 mm (47 ft) but has been high as 1800 mm (71 ft). whilst in part of Mansehra District such as Balakot, the mean annual rainfall is as high as 1750 mm (69 ft). Due to its location on the boundary between the monsoon summer rainfall regime of East Asia and the winter-dominated mediterranean climate of West Asia, Hazara has an unusual bimodal rainfall regime, with one peak in February/March associated with the frontal southwest cloud bands and another monsoon peak in July /August. The driest month are October to December, even in the wettest parts theses month receive precipitation average around 40 mm (1.6 ft). Due to high altitude, temperature in Hazara area is cooler than on the plains, though Abbottabad at 1200 meter (3900 ft) still has a maxima of ca. 32°C (90 F). with high humidity in June/July. Further up, temperature is cooler, often cooler than the Northern Area valleys due to the cloudiness. In winter, temperature are cold with min. in January around 0°C (32F) and much lower in the high maintains. Snowfall is not uncommon even in the lower levels. (

⁸ Data provided by PMU taken from the BTAP and 10BTAP monitoring data base

Malakand Forest Region: The climate of Malakand region is harsh and severe with considerable differences among various Forest Divisions. The climate varies in each valley according to attitude. Each valley has its own micro-climate so that plants respond not to an average situation, but to the immediate, proximal prevailing conditions of the area. The climate at lower elevation has 2 extremes, cold in winter and hot in summer, becoming temperate at about 2000m above sea level. Most precipitation occurs during winter, mostly in the form of snow. Usually November-February are the coldest months while June till August are the hottest month with highly intensive sunlight. The annual total rainfall and temperature

Peshawar Forest Region: Peshawar has a hot semi-arid steppe climate, which is very dry with little rainfall. It can rain at any time of the year but the rain does not last long. As well as being arid, the climate is extremely hot in the summer but slightly cooler in the winter months. The warm season lasts from the mid of April to the mid of September with an average daily high temperature of above 34C. The hottest month of the year is June, with an average high of 39C and a low of 27C. The cold season lasts from December to the end of February with an average daily high temperature below 21C. The coldest month of the year is January with an average low of 5C and a high of 18C. The annual precipitation averages 384 mm. Due to western disturbances; the winter rainfall shows a higher record between the months of February and April.

3.3 Regulation of Forestry on Private Land in KP

The **Community Participation Rules (CPR)** of Khyber Pakhtunkhwa (KP), Pakistan, provide a legal framework for involving local communities in the sustainable management, protection, and development of forests and natural resources. These rules emphasize shared responsibilities and benefits between the government and local stakeholders on non-designated and designated forest land.

Woodlot Policy of Khyber Pakhtunkhwa (KP), Pakistan, regulates Forest Management on Forest lands which are non-designated communal and privately owned and not covered under regular forest management plans of KP FD. Key aspects of the policy include⁹:

1. **Sustainable Forest Management:** Encourages the establishment of small, community-managed woodlots to reduce pressure on natural forests while ensuring a steady supply of timber and fuelwood.
2. **Community Participation:** Empowers local communities by granting them management rights and involving them in decision-making processes related to woodlot management.
3. **Economic Benefits:** Aims to improve the socio-economic conditions of rural populations by creating income opportunities through the sale of timber, fuelwood, and non-timber forest products (NTFPs).
4. **Support Mechanisms:** Provides technical and financial support to communities for establishing and maintaining woodlots, including capacity building and training in sustainable forestry practices.

The policy aligns with broader goals of forest conservation, poverty alleviation, and climate change mitigation while fostering a balanced approach to resource use and ecological preservation.

⁹ The list is not exhaustive but summarizes with the author's own words the overall aspects of the Wood Lot Policy.

3.4 Forest Types in BTASP Intervention Area

There are many ways to describe or classify forests; they can range from very broad to very detailed classifications. These guidelines separate the forests of BTAP and 10BTTP into three categories: based on specie compositions and in five categories based on ecological zones.

Categories based of species compositions:

- Eucalyptus Forest,
- Mix Forest of Broadleaved species,
- Conifer Forests
 - Pure Chir
 - Mix Plantation with Bule Pine and Deodar

Categories based on Silvo-Ecological zones¹⁰, full map of KP map shown in Annex 4

Silvo-Ecological zones	Forest Division
Tropical forest,	Peshawar, Lower part Malakand
Sub-tropical broadleaved evergreen forests,	Malakand /Haripur
Sub-tropical Chir pine forest,	Haripur /Malakand /Lower Swat Upper Swat/Buner, Mansehra
Moist Temperate Forest (	Gullies, Kaghan, Alpuri, Batagram, TorGar)
Dry temperate forest.	Kaghan, Upper Swat, Batargram

Table 6: BTASP Forest Divisions and corresponding Silvo-Ecological zones

The first two categories are collectively known as **scrub forests**; however, the remaining three categories are called **conifers forests**.

3.4.1 Plantation categories based on species compositions

Eucalyptus forest plantations

In BTAP and 10BTTP mostly plantations were carried out on communal or private lands. Communities mostly give preference to those species which give immediate and quick returns. Eucalyptus is the first choice of most of the landowners because it's easy to grow, low maintenance cost and give quick returns. Eucalyptus is also planted as mix species with other broadleaved but 75% monoculture represents majority of Eucalyptus plantation on private and communal land.

BTASP approach: In new plantations or woodlots, the use of Eucalyptus is highly limited. The tree species composition in existing pure Eucalyptus stands will gradually shift towards mixed forest stands. While Eucalyptus will still be present in the midterm, its proportion will be reduced and replaced with more natural and fast-growing species that are beneficial for both the environment and livelihoods.

Mix Broadleaved Forest plantations

More than 27 different species have been planted in these plantations. Robinia, Phulai, Ailanthus, Kikar (A. nilotica), Eucalyptus, Shisham, Sanitha, Bakine, Siris, Poplar, Mulberry and A. farnesiana are the main species. Except Eucalyptus, Robinia, Ailanthus and Acacia farnesiana the rest are local and indigenous species. In mix forests, still there are few patches of pure species, mostly of eucalyptus.

¹⁰ Guidelines and Lists of Suitable Tree Species for Different Silvo-Ecological Zones in Pakistan; IUCN/FAO/WWF August, 2022

Conifer forest planation

In conifers forest the most abundant species is *Pinus roxburghii* (Chir Pine) followed by Kail and Deodar. Chir is mostly raised in pure stand while Kail and deodar are found in mix form along with few associated broadleaved such as walnut and horse chest nut.

3.4.2 Plantation categories based on Ecological Zones

Tropical forest planation: This category includes **Tropical thorn forest** and **Tropical dry deciduous forests**. These are low, open and pronouncedly xerophytic forests in which thorny leguminous species predominate. This type occupies the whole of the Indus plain except the driest parts. The major tree species are *Prosopis cineraria* (Jhand), *Zizyphus mauritiana* (Ber), *Tamarix aphylla* (Farash), *A nilotica* (Kikar), *Bombax ceiba* (Sumbal), *Dalbergia sissoo* (Shisham), *Cassia fistula* (Amaltas), *Albizia lebbek* (Siris), *Melia azederach*. Among them are a large number of shrubs of all sizes. The tree forest climax is very frequently degraded to a very open, low thorny scrub of *Euphorbia* (Thor), *Zizyphus* (Ber), etc. In BTASP this type of forests is available in Peshawar, Lower part of Haripur and lower part of Malakand Forest division.

Sub-tropical broadleaved evergreen forest planation: These are xerophytic forests of thorny and small-leaved evergreen species. This type occurs on the foothills and lower slopes of the Himalayas, Hindukush Range. The typical species are; *Olea ferruginea* (Kau) and *Acacia modesta* (Phulai), the two species occurring mixed or pure, and the shrub *Dodonaea viscosa* (sanitha) which is particularly abundant in the most degraded areas. In BTASP such type of forests is available in all twelve forest divisions except Peshawar.

Sub-tropical Chir pine forest planation: These are open inflammable pine forests sometimes with, but often without, a dry evergreen shrub layer and little or no underwood. This type consists of Chir pine (*Pinus roxburghii*) forests found between 900 m and 1700 m elevation in the Western Himalayas within the range of the south-west summer monsoon. It is the only pine of these forests though there is a small overlap with *Pinus wallichiana* (Kail, Biar) at the upper limit. In BTASP such type of forests is available in Haripur, Gallies, Siran, AT, Lower Swat and Bunner Forest Divisions. In BTAP and TBTP most of enclosures are also established in Chir pine forests.

Moist Temperate Forest planation: The evergreen forests of conifers, locally with some admixture of oak and deciduous broad-leaved trees fall in this category. Their undergrowth is rarely dense, and consists of both evergreen and deciduous species. These forests occur between 1500 m and 3000 m elevation in the Western Himalayas except where the rainfall falls below about 1000 mm in the inner ranges, especially in the extreme north-west. These forests are divided into a lower and an upper zone, in each of which definite species of conifers and/or oaks dominate. In the lower zone, *Cedrus deodara* (Deodar, diar), *Pinus wallichiana*, *Picea smithiana* and *Abies pindrow* (Partal) are the main conifer species in order of increasing altitude, with *Quercus incana* (rin, rinj) at lower altitudes and *Q. dilatata* above 2130 m. In the upper zone *Abies pindrow* and *Q. semecarpifolia* are the dominant tree species. In BTASP such type of forests is available in, Gallies, Kaghan, Hazara Tribal, Siran, AT, upper Swat, Tor Ghar and Alpuri Forest divisions.

Dry temperate forest plantation: These are open evergreen forest with open scrub undergrowth. Both coniferous and broad-leaved species are present. This type occurs on the inner ranges throughout their length and are mainly represented in the north-west. Dry zone deodar, *Pinus gerardiana* (Chalghoza) and/or *Quercus ilex* are the main species. Higher up, blue pine communities occur and in the driest inner tracts, forests of blue pine, *Juniperus macropoda* (Abhal, Shupa, Shur) and some *Picea smithiana* are

found locally. In BTASP such type of forests is available in Kaghan, Hazara Tribal, upper Swat, Tor Ghar and Alpuri Forest divisions. Though very limited plantation has been carried out in this zone under BTAP and TBTP, however enclosures were established to encourage natural regeneration of Deodar and other conifer species.

3.5 BTASP Participatory Forest Resource Management Planning

The Community Participation Rules 2004 specify the establishment of Village Development Committees and Women Organizations. The VDC and WO are responsible to develop a Village Plan through a Village Land Use Planning Process. In BTASP a so called PFRMP will be elaborated.

Khyber Pakhtunkhwa has a Forest Policy 1999 and a Rangeland Policy 2014. Both policies promote the sustainable management of forest and rangeland resources through engagement of local communities and the development of the participatory development of Village Land Use Plans.

The 10 BTTP Project supported by BTASP will prepare Participatory Forest Resource Management Plans¹¹ in the context of the Forest Ordinance, Community Participation Rules, and the Forest and Rangeland Policies.

The PFRMP objectives:

1. To introduce sustainable community-based forest- and rangeland management systems for the maintenance and improvement of existing forest and rangeland resources of the planning unit, maintaining and possibly strengthening its role in local economy;
2. To promote farm forestry to reduce pressure from (uncontrolled) use of forest and rangeland resources to meet local timber, fuelwood and fodder requirements;
3. To prevent degradation of, and if possible, to enhance the watershed values of the forests and rangelands, by sustainable use and conservation practices;
4. To sustain income from forest and rangeland resources by controlled use by the community.

The **PFRMP methodology** comprises 6 steps, detailed step-by-step approach and corresponding timeline in weeks are shown in Table 3. Step 1-5 is planned for a total duration of 12 weeks with up to 30 field days and 20 office days.

- Step 1: Preparation for Forest Resource Management Planning
- Step 2: Participatory mapping of forest/range lands (bio-physical/social)
- Step 3: Social organization
- Step 4: Drafting of the overall framework Forest Resource Management Plan (zoning plan)
- Step 5: Drafting of a detailed operational plan for each Management Unit (technical plan)
- Step 6: Implementation and monitoring of FRMP for each Management Unit

The process can be summarised in community mobilisation (Step 1), data collection and analysis (Steps 2-3) planning (Steps 4-5) and implantation (Step 6).

The guiding principles and tools used during the PFRMP are participation and inclusion, social organization, integrated approach, controlled use, agreed upon distribution of rights, obligations and benefits, gender based, poverty orientation and free, prior and informed consent (FPIC)¹².

¹¹ Participatory Forest Resource Management Plan Manual, BTASP 2025

¹² PFRMP Manual 1st Draft, Section 1.2 Guiding Principles for Participatory Forest Resource Management Planning

Community Forestry related issues to consider for Operational Planning

During the PFRMP process steps 4 and 5 consist of the planning of activities and interventions for sustainable **community forestry**. Basis for the planning is the current situation on the ground (steps 2-3) and the overall PFRMP management objective. Based on that a forest site specific objective needs to be formulated between the land owner(s), land users and technical staff of the Forest Division staff. This process is one of the most crucial participatory approaches where the need and the technical expertise shall be combined for a long-term management of up to 10 years. Technical staff needs to be trained to serve the purpose as advisors to the land owners and de-facto decision makers.

Inclusion of all forest sites within the Planning Unit

Initially, Result 4 of the Result Matrix¹³, was limited to (BTAP and) 10BTTP afforestation activities only but has been extended to cover all forest sites within the PUs regardless of when and which program / project supported the developed of these forest resources (**MoM KfW Mission 2023**). With that, the diversity and quantity of forest sites per Planning Unit for planning and measures has been increased. At this point in time, it is not yet possible to quantify (ha) all the total forest-based activity sites. From the PMU data base only the information for BTAP and 10BTTP sites of plantations and enclosures are available per Region, Forest Divisions and envisaged 100 PUs in Annex 3. A summarized area calculation is shown in Table 6 and is calculated with ca. 38,000ha. With the extension to all forest sites per PU a diverse mix of plantations, enclosures and natural forest patches in different age classes, species compositions und usage stages have to be assessed and measures proposed.

3.6 Guideline on Tree Species used by BTASP

Most of the tree species have a wide distribution and are found naturally or can be cultivated across many ecological zones. However, every species has an ecological zone where its growth is vigorous and have close association with other biodiversity especially wildlife. The below list was developed by a consortium of IUCN, WWF-Pakistan and FAO during the third-party monitoring of TBTP¹⁴. The list was developed after a series of consultation with ecologist, foresters and NRM specialist. It is recommended to follow the same list during BTASP plantation attached in Annex 4.1.

The lists of suitable trees for every ecological zone are organized in two parts – main species and subsidiary species. Main species include the species that are best suited for the zone and are commonly planted or found in that zone. The subsidiary species are those that are also found or planted in the zone but are not very common. Similarly, the list for every zone includes trees that are part of the natural vegetation of that zone, many of which are also cultivated outside of the natural forests as well.

3.6.1 Forest Tree Seedlings Needs

BTASP will require for the implementation of its envisaged PFRMP measures, as described in chapter 4, ca. 17,635,413 seedlings see Table 7 and 8. The forest tree seedling will be provided in 2 different ways:

1. **Open market private nurseries:** The number and species to be procured are based on the PFRMP planting models. The price for seedling type is specified in the cost norms Annex 1 SW 11 and the procurement procedure specified by amount in the approved Financial Management

¹³ Separate Agreement, Annex 1, page 5: "Result 4 10BTAP afforestation measures are supported in all BTASP area"

¹⁴ Guidelines and Lists of Suitable Tree Species for Different Silvo-Ecological Zones in Pakistan

Guidelines (FMG). This type of nursery will provide bare rooted seedlings and potted seedlings. The transport cost to the nearest planting site shall be included in the procurement contract. During delivery a delivery note and inspection on quantity and quality shall be issued. The cost for seedlings for P-bags and bare rooted plants are based on a market survey done over the last year.

2. **Women nurseries:** BTASP will promote women run nurseries for potted seedlings in P-bags. The cost for seedlings is specified in the performance-based contract. The transport of the seedlings to the plantation site within the PU is the responsibility of the VDC/WO under supervision of the concerned Forester/Forest Guard and the Management committee. The established women nurseries shall be used also after BTASP for seedling propagation and livelihood support.

The estimated need for seedlings

Nursery type	Cost / fit plant	No. of plants to be raised	Cost in PKR and Euro ¹⁵
Women nurseries (370)	10 PKR	9,250,000 (370*25.000 plants)	PKR 92,500,000 EUR 308,000
(73)	12 PKR (pine)	1,825,000 (73*25.000 plants)	PKR 21,900,000 EUR 73,000
Private nurseries	18 PKR /P-bag	5,196,000	PKR93,528,000 EUR311,760
	25PKR / bare root	1,000,000	PKR 25,000,000 EUR 83,333
Seedling's demand	100 PU x 17, 635,413 seedlings	17,271,000 plants	PKR 232,928,000 EUR 775,426

Table 7: Calculation of seedling demand and supply

Estimate of required seed material BTASP 100 PU standard PFRMP activities¹⁶

Seedling required BTASP					
S. No	Intervention	Spacing	Per ha	Qty.	Total
1	Block Plantation (incl. 2x 25% beating up)	10x10 feet	1613	4305	6941813
2	Block Plantation (incl. 2x 25% beating up)	8x8 feet	2520	2555	6438600
3	Enrichment planting		300	2450	735000
4	Woodlot	8x8 feet	2520	1000	2520000
5	Farm Forestry -Tree seedlings		10000	100	1000000
Total					17,635,413

Table 8: Estimation of seedling demand per standard Planning Unit

3.6.2 SOP Seed Material

Seed material for private, women nurseries and forest department nursery establishment shall be acquired according to the following criteria and procedure:

1. The DFO is responsible for quality seed collection and/or acquisition.
2. Seed species selection is based on the tree species selected during the PFRMP process.

¹⁵ Exchange rate of 1: 300 is used (1 EUR=300 PKR)

¹⁶ The estimate can only be verified and confirmed after elaborated PFRMP plan.

3. Seed material for forest species shall, if possible, be obtained from the forest within the same silvo-ecological zone and climate where the saplings later shall be planted.
4. Seed stands shall be marked in a map and the GIS data taken and send to PMU GIS Unit
5. Mother trees shall be selected according to the following criteria: age, form vitality and vigor to ensure viable seed material.
6. Required seeds shall be collected according to the collection season in Pakistan. A list and calendar for species wise collection and ripening shall be applied and is attached in Annex 8.
7. If possible, samples of collected seeds shall be certified by a specified institute R&D or PFI to ensure that the seed is free of diseases and insects and have good germination rate. The DFO will send a testing sample to the available testing lab. The cost for testing is reimbursed by the BTASP. The cost for testing is stipulated in Annex 12.
8. The seed material shall be stored in a dry cool place secure to rodents, insects and other seed consumer at the DFOs office until handing over to women nursery contractor.
9. Once tested, the seeds shall be handed over to the contracted nurseries. A delivery note is signed by the nursery owner and stored in the DFOs office.

Costs: The cost norm for seed material and testing is listed in Annex 1, No.: 26

4. IMPLEMENTATION OF PFRMP MEASURES

The implementation of PFRMP plans include the following:

- | | |
|---|-------------------------|
| 1. Agreements for implementation of measures, | Annex 5.1-5.4 |
| 2. Guidelines for site selection, | Chapter 4.1.7.1 |
| 3. Guidelines for beneficiary selection, | PFRMP manual Annex 15 |
| 4. Guidelines for species selection, | Chapter 3.6 and Annex 5 |
| 5. Measures to be implemented, | Chapter 4.1 |
| 6. Implementation calendar, | Annex 9 |
| 7. Cost Norms for implementation, | Annex 1 |
| 8. Contribution schemes, | Chapter 8 |
| 9. Community NRM Fund, | Chapter 9 |
| 10. ESMF conformity / compliance: | Chapter 10 |
| 11. Monitoring of implementation, | Annex 6.1-6.5 |

Tasks/ role of stakeholders during implementation are described in each PFRMP measure under chapter 4.1. below summarised in Table 9

Stakeholder	Task during implementation
PMU (supported by PIC)	- Overall steering, monitoring and reporting of implementation according to ESMF, FMG and PFRMP Implementation Guidelines; - Procurement of inputs, - Coordination of BTASP actions between ISUs and DFOs, - Provision of funds for ISUs and Forest Divisions, - Quality control of planning and implementation, - Provide technical advice to ISUs and DFOs,
PMU GIS Unit	- Provide maps for DFO teams - Collect and/or receive geo-reference data on all PFRMP measures sites, - Develop and maintain monitoring data base for BTASP - Provide quarterly summary result reports
CDE&GED	- Backstopping regional planning teams in all aspects on social VDC/WO issues, quality check on beneficiary selection, agreements, NRM Fund development and community contribution, - Quality control ESMF conform implementation, - Monitoring visits and feedback reporting on monthly basis to PMU,
IHR&D	- Provide capacity building and trainings according to GCBP and PFRMP, - Contracting trainers, develop training curricula, - Execution of trainings and payments, - Monitoring and reporting to DFO, PMU
NTFP Directorate	- PFRMP -level: provide technical input on action plans for NTFP and value chain support to DFO, - Provide action plans and implementation proposals for regional value chain support to PMU, - Implement action plan on regional level,
R&D Directorate	- Testing of seed material sent by DFOs
Regional Level Conservator of Forest	- Supervise the work of DFO and SDFO through regular meetings and monitoring reports,

	- Monitoring of PFRMP measures in the field, - Reporting of BTASP actions to CCF,
Regional Planning Teams (CDE&GAD)	- Backstopping CDO staff on FD level, - Advise on ESMF conform implementation, - Assess gaps and propose solutions for improvement on VDC/WO performance - Report on monthly basis to CDE&GED Peshawar
DFO (FD) teams	- Planning and supervision of PFRMP implementation - Ensure quantitative implementation of measures according to PFRMP Implementation Guidelines, - Procurement of inputs according to FMG, Contracting of selected beneficiaries according to beneficiary selection criteria, - Provide advice to PU and Mgt. Committee on silvicultural measures, - Coordinate with IHR&D for trainings, - Timely payment according to contracts, - Monitoring of progress, results and finances to PMU and Regional level (CF), participate on quarterly PFRMP progress meetings, - Ensure that the field staff of CDE&GAD and Forest Division have followed the FPIC and Environmental and Social Safeguard framework in the field. - Ensure that all the steps are properly followed during establishment of community-based organizations. - Responsible for reporting and monitoring through MIS system developed under the BTASP project.
SDFO	- Provide technical Inputs and advise for selection of plantation sites, selection of species and afforestation model, - Development of planting design in groups and troops, - Supervise the Forest Guard and Forester and check quality on plantation and enclosure Journal development and documentation, - Supervision of Forest Guard and Forester on all PFRMP measures, - Supervision of plantation work together with VDC/WO - Monitoring of progress and results,
Male/Female CDOs	- Liaise and coordinate with VDC/WO on all implementation issues, - Prepare agreements for signature between DFO and VDC/WO, Mgt. Committees and/or beneficiaries, - Support VDC/WO to be functional and operational, check VDC/WO documentation and reporting at least once per month, - Focal persons for livelihood projects for VDC/WO, - Participate and/or check criteria for beneficiary selection, - Co-facilitate trainings of VDC/WO and evaluate functionality, - Monitor community contribution, develop SOPs for community Fund,
Forester/Forest Guard	- Facilitate, plan and document the plantation and enclosure journal for old and new plantation, enclosures and forest sites incl. advise on harvest quotas and sustainable use-plans for all types of forest resources, cost benefit calculation and maintenance plans, - Facilitate with VDC/WO on plantation site and species selection for new plantation sites, - Facilitate with VDC/WO and Mgt. Committee planting actions,

	- Facilitate with VDC/VO all measures for rangeland mgt. and livelihood support, - Monitoring field implementation of all measures and report to SDFO
VDC/VO	- Ensure communication with all sections of community, - Resolve conflicts arising during PFRMP implementation - Signing of agreements, - Selection of beneficiaries, - Selection of new plantation sites with support of the Mgt. committee and forest guard/forester, - Fund management in case VDC account is available; timely payments according to contracts and agreements, - Mobilise community contribution, - Set up NRM fund, develop NRM bylaws, participate in trainings, - Maintain VDC/VO documentation file for PFRMP implementation, - Participate in survival rate assessments, - Monitoring of progress, results and finances
Management Committee (representative of land owners)	- Develop forest site specific micro management plans (Journals) - Participate in forest site inventory and formulate development objectives, - Maintain and update plantation and enclosures Journals annually, - Signing the Hand-Over Agreements and ensure sustainable management - Control of the forest warden, finance forest measures, - Participate in trainings, participate in the plantation works and survival rate assessments, - Report to VDC/VO,

Table 9: Tasks of stakeholders during PFRMP implementation

4.1 Description of Selected PFRMP Measures

Under the BTASP different interventions will be implemented broadly categorized in Forest Resource Management interventions, Range Management interventions, Livelihood Interventions and Mitigation measures adopted under ESMF.

The implementation calendar shown in Annex 9 provides the time during the year for implementation taking in consideration annual rain pattern for plantation work, and peak agricultural work of the community beneficiaries.

Below in Table 10 showing different interventions under BTASP. Qualitative and qualitative details of each intervention will be available in PFRMP of each PU. Information on participatory approach and management of environmental and social issues are detailed in the PFRMP manual and ESMF documents, respectively.

	Planning unit community project
1.	Community support project in the NRM sector
	Forest resources management
2.	Update plantation and enclosure journals
3.	Protection of existing enclosures and plantations
4.	Enrichment of plantations
5.	Thinning of aged plantations
6.	Establishment of fire-line's
7.	BTASP new plantations
8.	Wood Lots
9.	Spring protection
	Rangeland management
10.	Controlled grazing
11.	Provide fodder seeds on agricultural and marginal lands
	Livelihood activities
12.	Farm Forestry - tree seedlings provision
13.	Women livelihood improvement
14.	Management of NTFP
15.	Women tree nursery
	Capacity building
16.	Training VDC, WO and Management Committees
17.	Technical trainings on Basic Forest Management

Table 10: List of PFRMP framework measures

4.1.1 Community Project

In the context of BTASP with its PFRMP approach a **community measure** refers to an initiative that shall support a wide fraction of the community of the planning unit. The approach puts the new created or reactivated VDC/VO in the centre of the process. The identification is already built into the PFRMP process in step 3 and 4. These actions are designed to build momentum, and foster community engagement early in the PFRMP process. The selected project does not necessarily need to be directly be connected to forest management but shall support unified decision-making lead by the VDC/VO.

Criteria community project:

1. The PU is selected and an the PFRMP cooperation agreement has been signed
2. The VDC and the WO together with advice of the CDO and DFO will select and agree thru a selection process some urged needs from the NRM sector focusing on livelihood improvement, water or communal small-scale investments.
3. A larger section of the PU will benefit from the project or will support local governance,
4. A budget sealing of 600,000 PKR is been allocated to each PUs,
5. CDOs screen the projects against social and environmental risks,
1. VDC/VO ensures the community contribution of at least 10%,
2. FD will provide the funds for execution and will coordinate the implementation
3. VDC/VO will mobilize the agreed community contribution
4. FD will monitor the progress of the implementation,
5. VDC/VO will monitor the progress from PU side,
6. Project completion protocol shall be signed by both parties and send to PMU.

Costs: The cost norm for that measure is listed in Annex 1, No.: 14

4.1.2 Forest Resource Management Measures

Forest Resource Management and silviculture is a subject that describes the procedures how to raise trees, establish and maintain forests, manipulating forest vegetation to enhance forest growth and establish new forest stands. Silvicultural operation means the growing of trees for private, commercial, recreational or ecological use, including preparing and layout the land, planting (spacing and pit size), growing, selection of species, and harvesting of trees.

These chapters comprise all relevant aspects of silvicultural works to be done to establish new plantations, to maintain existing plantations established during BTAP and 10 BTAP and to finally harvest main forest products. The chapter also contain the following silvicultural categories:

Planting site preparation, planting of seedlings/seeds, schedule for planting, planting pattern, planting designs, tree species composition in various areas, maintenance, fencing, watering, silvicultural operations and monitoring. General aspects of BTASP concerning seedling demand and species to be used under BTASP are documented in chapter 3.6. For the silvicultural works done by selected community workers and supervised by the Forest Guard and Forester will follow basic guidelines descript in the below chapters and in the Operational Guidelines for Field Foresters¹⁷.

¹⁷ Operational Guide for Field Foresters, Booklet, IHR&D Department, year of publication unkwon

4.1.3 Update Plantation and Enclosure Management Plans

Description: All existing plantations and enclosures established under BTAP and 10BTTP had initially a 3-year plan. These plans need to be updated and relevant activities documented and followed up. The update of the site-specific plan will include: thinning, pruning, fire protection, regeneration or replanting and sustainable harvesting cycles and cost-benefit calculation for at least 10 years. The activities required and controlled usage of resources will be discussed and agreed with the management committee of the site and put under controlled and sustainable management monitored by the VDC/WO. Up to 10% of the future cash income should be deposited in the VDC- NRM (saving) account for the management of natural resources in the PU. Based on the participatory rapid assessment and inventory of forest and rangeland resources the interventions are detailed in the operational management plan per unit as part of the planation and enclosure journal.

Objectives of the measure:

Forest Management activities at selected forest sites are planned and documented in a participatory manner.

Expected results:

- For each forest site a specific planation or enclosure journal is available containing information on sustainable forest management for the improvement, maintenance, regeneration and controlled use/harvest and cost-benefit data.
- To train local forest owners in basic forest planning and documentation
- To introduce a participatory process and dialog of planning and documentation between private- and communal forest owners and Forest Department

Criteria for beneficiary selection of plantations or enclosures under this intervention are not applicable, sites already have owners and users.

Responsibilities:

- FD together with the management committee will update the mgt. plans according to the agreed Management objectives and silviculture activities required and agreed.
- FD will document the results in the Plantation- and Enclosures Journals, attached in Annex 2. And will hand over a copy of the plan incl. maps to the Mgt. Committee for signature.
- The update includes the main silvicultural activities in time over a period of at least 10 years, the maintenance cost involved and expected revenue or volumes of extraction.
- Management committee agrees to follow and implement the silvicultural activities with advisory support of the FD. The agreement is part of the Journal.
- FD will advise the Management Committee in implementation of silviculture activities
- VDC/WO to monitor the update process and implementation of the silviculture activities
- VDC ensures that 10% of all revenue generated will be saved into the NRM (saving) account.

Benefit: Site specific management plans for the next 10 years
 Cost – benefit calculation provides overview for the next 20 years is available,
 Basis for site specific (compartment) silvicultural activity planning up to 10 years,
 Revenue saving into NRM fund for community activities

Costs: Cost involved see cost norms in Annex 1 No.: 4

Hand Over Contracts – A draft of the hand over contract is been attached in Annex 11

4.1.4 Protection of Existing Enclosers and Plantations

Description: During previous projects (BTAP and 10BTTP), enclosures and plantations were established in degraded forest patches to encourage natural regeneration. To protect those areas community forest wardens were employed for protection and support aided regeneration. If observed during the PFRMP forest and rangeland assessment that the existing enclosures need further support for protection and restoration, the **BTASP** will extend the **support for up to 4 forest wardens with 24-month salaries**. The concept, TORs and payment procedure of the community forest warden needs to be agreed also beyond the support of BTASP and financially agreed. It is envisaged that BTASP will finance for the 1st year 100% payment and in the 2nd and 3rd year 50% payment. The balance of 50% payment for 2nd and 3rd years shall be taken over by the VDC/WO or Mgt. Committee financed by harvest revenues. The work description and performance-based payment of the forest warden is attached. Standard SOP for forest warden has been drafted and is attached to the contract. BTASP will not identify new enclosures. In PUs where more than 4 enclosures and or plantations exist locally adapted protection strategies have to be developed and agreed with the VDC/WO and Management committees how to ensure protection. In new plantations up to 2 new forest warden shall be employed with provision of up to 36 month.

Objective of the measure:

Selected BTAP and 10BTTP enclosers and plantations are sustainably managed and protected

Expected benefits/results

To secure initiated investments and protection efforts of BTAP and 10BTTP enclosures,
To encourage efforts on enclosures protection as a viable future forest resources and controlled use,
To introduce new tasks of the forest warden and combine it with a performance-based payment beyond project duration

Criteria for site selection are not applicable due to its already existence of up to 4 enclosures.

Criteria for forest warden selection: see PFRMP manual Annex 14 for beneficiary selection;

Contract, TORs and performance-based payments: see Annex 5.2

Responsibilities:

1. VDC/WO and Mgt. committee will confirm existing forest warden,
2. BTASP will provide a new 3 party agreement between DFO, VDC/WO and/or Mgt. Committee and forest warden incl. job description, a performance-based payment and the salary scheme (FD Rate 30.000PKR) of the forest warden for 24 month.
2. Forest warden will (if not yet done) demarcate the area, patrol and protect site against uncontrolled use and report natural hazards (according to TORs).
3. Management Committee will check regularly and randomly on Negheban performance and will report (format Annex 6.2) any progress and development to VDC. The VDC will report the monitored activities to the DFO on a monthly basis (format Annex 6.5). The DFO will then release the payment of the forest warden to the VDC for payment. The payment shall be reported to the DFO.
4. If possible, VDC will ensure 50 % of forest warden payment in year 2 and 3.
5. The VDC and /or Mgt. Committee will develop with support of BTASP a strategy to ensure an forest warden payment system for year 2 and 3 and future protection mechanism independent from FD.
6. VDC will monitor and report to DFO that all activities set out in the enclosure management plan will be executed on a monthly basis.

Costs: The cost norm for that measure is listed in Annex 1, No.: 5 and 17

4.1.5 Enrichment Planting of Forest Sites

Description: Enrichment planting is a silvicultural technique used to enhance the growth and diversity of a forest stand by adding new tree species or individuals to an existing forest. BTASP will contribute to the enrichment of existing plantation with regards to density per ha and diversity. During enrichment, the blank patches in the existing plantation will be identified and filled with native tree species with an estimate of ca. 200 plants per ha across the plantation. BTASP puts a limit of its support of up to 25 ha.

Objective of the measure:

Upgrade of degraded or depleted plantations or forest sites

Expected benefits/results

1. To improve the overall health, structure, and composition of the forest site,
2. To close wide forest gaps and forest site openings,
3. To increase diversity in monotonous stands.

Criteria for site and

- Old or mid aged BTAP and 10 BTPP plantation and other forest sites on PU area
- Forest cover of sub-blocks of the forest site is **less than 50%** with an overall plantation **density of 60%**.
- Willingness of land owner of planting native species based on the species list attached
- Gaps and openings in the forest area of **≥0,5 ha**.

Beneficiary selection: owners and users of the identified forest side,

Responsibilities:

- A general agreement with the VDC/VO and the Mgt. committee will be signed to define the activity and the stakeholder responsibilities of the FD and VDC/VO and Mgt. committee.
- FD and Mgt. Committee /land owner will together assess the forest site including reasons for plantation failures, gaps and openings are identified, measured and documented,
- FD will use GPS to track the size and location of the enrichment sites,
- FD technical staff (SDFO/forester) together with the Mgt. Committee will define the objectives of the enrichment planting, such as improving biodiversity, increasing timber production, or enhancing wildlife habitat.
- FD will advise the landowner on the native species to plant for enrichment according to landowner usage requirements based on the species list in Annex 5
- FD will develop based on the area size a planting design inc. number of trees and species compositions,
- The Mgt. committee will organise labour force,
- Forest Department jointly with the management committee will carry out the planting
- Planting will be checked by FD and VDC/VO sign pro-forma (see Annex 6.1)
- The FD will provide plants and technical assistance for the enrichment planting,
- Management committee will ensure 10% contribution for enrichment planting,
- FD cost norms will be used and applied,
- Implementation will be monitored by VDC/VO.

Costs: The cost norm for that measure is listed in Annex 1, No.: 9

4.1.6 Thinning and Pruning of Forest Sites

Description: In most of selected Planning Units selected by BTASP plantations with different age classes do exist either from BTAP or other previous project (Social Forestry, Watershed Management or FDF). Those plantations aging from 10-35 years have a potential and need for thinning, pruning and sustainable harvest. BTASP will assess the resource through the PFRMP inventory and will prescribe thinning and pruning of the selected plantation on sustainable bases. In forestry, thinning refers to the selective removal of some trees in a forest stand to improve the growth, health, and quality of the remaining trees. It is a common silvicultural practice used to manage forest density and promote the desired development of the stand. The thinning will apply the 3D principle as well as the selection and free thinning where the stand development objective is used to define the desired tree species, tree forms or biodiversity. Together with thinning tree pruning will be applied in younger stands for improved height growth and control branch density.

Objective of the measure:

To develop productive and healthy plantations and forest sites

Expected benefits/results

1. To enhancing tree growth by reducing competition for resources like sunlight, water, and nutrients, the remaining trees can grow faster and healthier.
2. To improving tree quality by encourages the development of trees with straighter trunks, fewer branches, and better wood quality.
3. To reducing risk of disease or pests: Decreases overcrowding, which can lower the risk of disease outbreaks and pest infestations.
4. To Improving Forest Health: Helps to maintain a balance in forest ecosystems and reduce the likelihood of stress-related tree mortality.
5. To increase Economic Benefits: Harvesting thinned trees can provide an intermediate source of income before the final harvest.

Criteria for site selection: based on the result of the PFRMP rapid assessment

Beneficiary selection: owner and users of the plantation or forest site

Responsibilities:

- A general agreement with the VDC/VO and the Mgt. committee will be signed to define the activity and the stakeholder responsibilities of the FD and VDC/VO and Mgt. committee.
- FD will carry out training of selected members of the different Mgt. Committees of the PU on thinning and pruning based on management objectives,
- FD will together with the forest warden mark the trees to be thinned,
- FD will together with the forest warden demarcate the stands for pruning,
- The FD will provide a set of tools (hand saw) for pruning in young stands to the VDC/VO
- The Mgt. Committee will provide the tools (hiring chainsaw operator) for tree thinning,
- Mgt. Committee will be responsible for evacuation of pruned and thinned products,
- VDC/VO will monitor the planning and implementation of the measure and will report to DFO.

Costs: The cost norm for that measure is listed in Annex 1, No.:

4.1.7 New Plantation Establishment on Barren Land

During BTASP, up to 10,000 ha area will be brought under tree covers in 100 PUs with an average of 100 ha in each PU including all types of planting activities (enrichment planting, block plantation, woodlots). Block plantation will be carried out on private and communal wasteland or barren land on units of 20-40 ha, individual land ownership shall not exceed 5 ha. The BTASP will provide planting stock, labour cost (90%) and other relevant cost. BTASP will provide up to 2 forest wardens responsible for the watch & ward of the plantation for up to 3 years. The plantation will be carried out with 10 x 10 feet spacing in sub-tropical and tropical zone while in Moist temperate the spacing will be 8 x 8 feet. The planting stock will be 1- to 2-year-old with size ranging from 18 inches in conifers and 18-30 inches in broadleaved. For each new plantation a BTASP plantation journal will be developed inc. at least 10-year site specific management plan (see 4.1.2). After 3 years the new plantations will be handed over to the Mgt. Committee with a Hand-Over Agreement Annex 11.

Forest Plantation Establishment

BTASP target to establish up to 10,000 hectare of forest plantations including enrichment planting /gab closure of existing plantation's of BTAP and 10BTTP. The forest plantations can be characterised by medium size planting areas varying from 5-70 hectare. For BTASP the average area per Planning Unit for new plantations is estimated with up to 50-70ha but can vary based on land viability and need. The plantations will be established on communal and private barren land according to site selection criteria.

Objective of the measure:

To establish climate resilient new forest plantations based on the objectives of the beneficiaries needs, using silvo-ecological relevant forest species.

Expected benefit/ results

To build up a natural resource base for fire wood, timber, poles to support livelihood
 To increase tree cover, protect soil from erosion and improve biodiversity
 To develop climate resilient forest sites

Criteria for site selection

- Location on waste and barren
- Site selection according to site selection criteria: chapter 4.1.7.1
- Species selection according to species selection criteria Annex 5
- No major disputes of area
- Willingness for contribution by landowners/users in watch and ward
- Signed resolution by VDC/VO

Criteria for Beneficiaries selection:

- Beneficiaries' selection according to selection criteria PFRMP Manual Annex 14.
- Mgt. Committee available
- Max. 5ha for individual land ownership

Responsibilities:

- A 3-party agreement will be signed between FD, VDC/VO, Mgt. committee Annex 5.1
- FD technical staff (SDFO/forester) together with the Mgt. Committee will define the objectives and silviculture activities and sustainable controlled usage of the new plantation,

- The FD will provide technical assistance in the species selection process and plantation model and planting schemes (groups and troops),
- FD will provide suitable plants for the block plantation according to agreed species,
- The VDC/VO and the Mgt. committee will ensure labour availability,
- VDC/VO and the Mgt. committee will ensure 10% of community contribution
- VDC will monitor the daily planting process and will report to DFO on weekly basis
- SDFO will monitor the planting activities on weekly basis and report to DFO
- 3rd party monitoring will check annually the survival rate
- VDC/VO will monitor the defined responsibilities of the Mgt. committee
- The Mgt. Committee will monitor the performance of the forest warden

Costs: The cost norm for that measure is listed in Annex 1, No.:7 and 8

4.1.7.1 Guidelines for forest site selection

Following are the guidelines for further selection of species from above lists. The following site-specific guidelines are already stipulated in the PC-1¹⁸: In general, only afforestation and sowing of mixed species stands under block plantation are supported by the project with the updated rates of the PC-I 10 BTTP. The criteria and standards for “mixed plantations” are as following:

Technical plot selection / exclusion criteria:

1. Afforestation sites will not be on land that was covered by forests during the past 5 years
2. Max. 5 ha per private owner will be included in the BTASP afforestation measures, any additional hectare has to be financed by the private owner.
3. Areas are to be excluded where surface- and groundwater is scarce and the project would require a water scarcity assessment in order to avoid further water scarcity.
4. Afforestation sites will not be on land used for agricultural production or on land which serves vulnerable community groups as an important source for their livelihoods and incomes.
5. Afforestation sites will not be on land that is dedicated for infrastructure projects that could adversely affect the sustainability of afforestation measures (e.g., industrial production, mining etc.);
6. Afforestation sites should only be selected in areas where they could form a resilient ecosystem to allow for long-term afforestation benefits.
7. The afforestation must be under a PFRMP safeguarding the protection and sustainable management, especially considering microsites of specific importance and value for the village ecology, culture and economy.

¹⁸ PC-1: Planning commission document for BTASP project

4.1.7.2 Guidelines for Species Composition

The below guidelines for species selection ensure a balanced tree species composition serving the need of the economic and livelihood relevance on one side and the ecological diversity and stability of forests for its ecological function on the other side.

The species selection for a specific planting site in BTASP will be done according the silvo-ecological zone and corresponding relevant tree species list (see chapter 2.4 and Annex 5). The forester is free and flexible to discuss and agree with the VDC / private owner on the tree species selection and composition as long as the following criteria are met:

1. The species list is divided in main/**key species** and **subsidiary species**. The proportion of main species to subsidiary species shall be as follows:
 for Tropical Forest and Sub-tropical broadleaved evergreen forests: 50:50
 for Sub-tropical Chir pine forest, Moist Temperate Forest and Dry temperate forest: 60:40
2. Mixed forests have a maximum of short-rotation species of 50%. The purpose of the short rotation species is to generate income for the VDCs/WO to start forest management and to pay for forest wardens and other investments in forests or for the community.
3. A minimum of 3 species/ha will be established in groups or troops.
4. At least 80 % of the total planted seedling should be native to the area
5. In plantation's ≥ 10 ha area at least 5 different species should be planted with at least 2 main species
6. In plantation's with less than < 10 ha area least 4 species shall be planted with at least 1 main species
7. Monoculture block is not permissible in any plantation, incase site is not suitable to plant in groups or troops with at least 3 species/ha, prior approval has to be taken from PMU.
8. Selected species shall be seasonal drought tolerant

4.1.7.3 Plantation Layout and Planting Models

The Plantation layout contain the spacing, pit design and the species composition.

Planting in groups and troops

Spatial distribution and planting scheme: According to the PC-1 the mixing of the species shall be carried out in troops (12 to 20 trees), groups (20 to 30 trees) or blocks. This approach shall help to avoid negative competition effects between the species, as slower growing species are protected from faster growing species. Single tree species mixture shall only be implemented in fruit tree dominated stands. If the groups are planted with respect to microsite and relief (e.g. focusing on depressions), the planting scheme may have higher influence on the success of the afforestation in terms of capacity of buffering climate effects.

In forestry and plantation design, the concepts of "groups" and "troops" are approaches to planting trees in specific patterns within a plantation. These planting designs are used for various purposes, including timber production, biodiversity conservation, and ecological restoration. Here's a description of plantation design in groups and troops:

Group planting involves planting trees in clusters or groups, with multiple trees planted close together in the same area. The number of trees in each group and their spacing can vary based on the plantation's objectives. Group planting allows for efficient use of space and resources. It can enhance resource utilization, such as light, water, and nutrients, leading to potentially faster tree growth and establishment. Group planting can be advantageous for timber production, as it can result in straighter, taller trees with fewer branches. When trees are grown closely together, they compete for light, which can encourage them to grow tall and straight as they reach for the sunlight.

Biodiversity and Wildlife Habitat: group planting can create microhabitats and enhance biodiversity. Clusters of trees provide shelter, nesting sites, and foraging areas for wildlife. By varying the size and arrangement of groups, you can create diverse habitats within the plantation.

Ecological Restoration: In ecological restoration projects, group planting can mimic natural forest patterns. It helps recreate the complex structure and biodiversity of native ecosystems, particularly in areas where natural regeneration is challenging.

Responsibility:

DFO/SDFO: to advise community and design a planting scheme based on best knowledge and to design a planting layout of groups and troops.

DFO/SDFO: to develop based on the below layout planting designs for each new planation including tree species according tree species list.

DFO/SDFO: to develop planation layout to comply with planting in groups and troops

Planting models: the below planation models serve an example. BTASP will not provide predefined planting models across the 3 regions and the 5 silvo-ecological zones. The DFO/SDFO shall, based on the guidelines provided in this document, develop in a participatory format the site-specific planting model.

Planting models: Block Planation and Wood Lots in different spacing and composition in Region I, II and III

	meters a.s.l.	exposition (example area)	Soil characteristics	climate /precipitation	main tree species for the plantation	Auxilliary species (min 15 %)	planting season	planting scheme (number per ha)	main management objectives
REGION -1 (Peshawar Forest Division)									
Block Plantation	350 m. to 400 m	Plain	Silty clay loam	The temperature averages 22.3 °C or 72.1 °F. Precipitation ranges from 817 mm to 32.2 inches per year	Acacia nilotica Prosopis cineraria Tamarix aphyla Zizyphus mauritiana Dalbergia sissoo Cassia fistula Eucalyptus	Acacia nilotica Prosopis cineraria Tamarix aphyla Zizyphus mauritiana	Mid Jan to end Feb) and Monsoon (July-Aug)	10 X 10 feet (1075)	To increase the ecological services of forest for environment and ensure sustainability of BTAP and TBTTP - To generate income through utilization of plantation (trees; shrubs; grasses) - To fulfill the fuelwood and fodder requirement of locals
Wood Lots	350 m. to 400 m	Plain	Silty clay loam	The temperature averages 22.3 °C or 72.1 °F. Precipitation ranges from 817 mm to 32.2 inches per year	Poplar Morus Alba Melia azedarach Eucalyptus	Poplar Morus Alba Melia azedarach Eucalyptus	Mid Jan to end Feb) and Monsoon (July-Aug)	8 x 8 feet. (1680)	

Region II (Hazara Region) and Region III (Malakand Region)									
Block Plantation	800 m. (around 450 m.) Sub-Tropical Broad Leaved	preferably northern exposition	General barren land has thin soil, and rocky	Summer with monsoon rain and hot, max. temp.: +25 to +30; during winter some precipitation, min . temp. to -5 °C.	<i>Acacia modesta</i> (Palosa) <i>Olea ferruginea</i> (Zatoon) <i>Dodonaea viscosa</i> (Ghuriskai) <i>Eucalyptus</i> <i>Robinia</i> <i>Ailanthus</i>	<i>Acacia modesta</i> <i>Olea ferruginea</i> <i>Dodonaea viscosa</i>	Spring (mid Feb to end March) and Monsoon (July-Aug)	10 X 10 feet (1075)	
Block Plantation	1500 m. (around 800 m.) Sub-Tropical Chir Pine	preferably northern exposition	General barren land has thin soil, and rocky	Summer with monsoon rain and hot, max. temp.: +25 to +30; during winter some precipitation, min . temp. to -5 °C.	<i>Chir</i> <i>Acacia modesta</i> (Palosa) <i>Eucalyptus</i> <i>Robinia</i> <i>Ailanthus</i>	<i>Mulberry</i> , <i>Bakine</i> and <i>Fig</i>	Spring (mid Feb to end March) and Monsoon (July-Aug)	10 X 10 feet (1075)	
Block Plantation (Moist Temperate)	2400 m. (around 1500 m.) Moist temperate Forest	preferably northern exposition	Mull soils, which have a high level of organic matte	Moderate mean annual temperatures, with average monthly temperatures above 10°C in their warmest months and above -3°C in their colder months From Mid Nov to Mid-January receive Snow	<i>Cedrus deodara</i> <i>Pinus wallichiana</i> <i>Abies Pindrow</i> <i>Quercus dilatata</i> (Oak) <i>Aesculus indica</i> (Bankhor) <i>Walnut</i>	<i>Pinus wallichiana</i> <i>Abies</i> <i>Pindrow</i>	Spring (mid Feb to end March) and Monsoon (July-Aug) & November for Sowing and Dibbling before Snow	8 x 8 feet. (1680)	

Wood Lots	Sub-Tropical	preferably northern exposition	General barren land has thin soil, and rocky	Summer with monsoon rain and hot, max. temp.: +25 to +30; during winter some precipitation, min . temp. to -5 °C.	Poplar Morus Alba Melia azedarach Bauhinia variegata Robinia pseudoacacia Eucalyptus Ailanthus	Poplar Morus Alba Bauhinia variegata Robinia pseudoacacia	Spring (mid Feb to end March) and Monsoon (July-Aug)	8 x 8 feet. (1680)	
Wood Lots	Temperate	preferably northern exposition	Mull soils, which have a high level of organic matte	Moderate mean annual temperatures, with average monthly temperatures above 10°C in their warmest months and above -3°C in their colder months From Mid Nov to Mid-January receive Snow	Poplar Robinia pseudoacacia Eucalyptus Ailanthus Walnut Aesculus indica	Poplar Robinia pseudoacacia Eucalyptus Ailanthus Walnut Aesculus indica	Spring (mid Feb to end March) and Monsoon (July-Aug) & November for Sowing and Dibbling before Snow	8 x 6 feet. (2240)	
Bioengineering (Close Plantation)	1200 m. (around 800 m.)	Eroded Land			Cutting of Salix, Mulberry, Robinia and poplar	Supplement plantation with sowing of Phulai, Olive, Oak and Sanitha	Spring (mid Feb to end March) and Monsoon (July-Aug)	5 x 5 feet. (4300)	

Table 11 Planting models examples for BTASP Region I, II, III

4.1.7.4 Spacing

BTASP will apply a standard spacing for all standard plantation types in BTASP:

Ideally it is suggested to provide 100 sq. ft area for each plant i.e., spacing between plant to plant and row to row should be **10 X 10 ft** provided soil / space is available for pit excavation while in case of big rock or hard surface the spacing may differ with ± 3 ft. i.e., ranging from 7 ft to 13 ft.

A closer spacing of **8x8 ft** or even **5x5 ft** shall be agreed on need base. A closer spacing can be used by very sloppy area to increase the soil stabilisation efforts.

Spacing categories used in BTASP:

Spacing in ft	Slope/Ecological Zone	Trees/ha	used for
10x10	Sub-tropical	1075	standard afforestation
8x8 ft.	Temperate	1680	wood lots
5x5	Steep slopes	4300	spring protection

Table 12: Planting spacing for different zones and PFRMP measures

4.1.7.5 Pit Design

Standard pit

Trees will be planted in circular pits with 10 ft spacing in staggered form with upslope V-shaped embankment of surrounding stones and gravels etc. This will help to harvest maximum water during rainfall. The Pits size for Bare Rooted Plants shall be 2'x1.5'x1.5' and 1.5'x1'x1' for potted plants

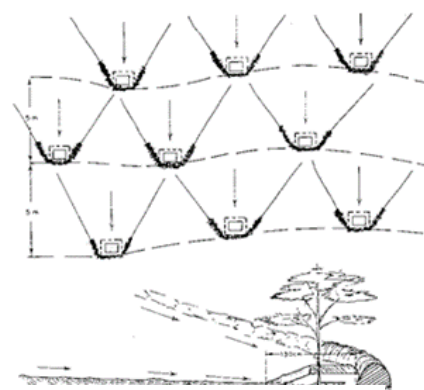


Figure 1. Pit Design

Eyebrow shape pit

Eyebrow shaped pits or terraces are a form of micro catchment water harvesting system for plantation of steep slope. They are handmade micro basins, in the shape of an eyebrow, and are often made from soil and stones. Their main goal is to facilitate vegetation growth on steep slopes by collecting runoff water from a small catchment area and having it infiltrate the soil. This way, it is stored in the root zone and will be available for crops. They are usually situated on hillsides and can be used on up to 50% of slopes. The steeper the slope, the more the bunds have to be reinforced with stones. Length of Eyebrow pit is 3 ft with a width and depth of 1.5 feet.

4.1.7.6 Planting Stock

The plants at delivery on site ready for planting have to fulfil clear criteria. The sizes recommended will ensure maximum survival rate in xerophytic conditions with high wind speed and scarcity of water. All the selected seedlings should be "fit" plants. Detailed specifications are given in Table 13 below: The delivery note signed by delivery shall ensure the above. The delivery note shall be signed by FD and Mgt. Committee.

Species	Type of seedlings	Age (years)	Height (inch)	Length of root (Inch)
Conifers	Container (P-Bags)	2	≥ 18	≥ 9
Broadleaved	Container (P-Bags)	1	≥ 18	≥ 9
Broadleaved	Bear Rooted	1	≥ 30	≥ 12

Table 13: Seedlings planting stock conditions and specifications

Plants to be purchased in any of the 3 different nursery types have to fulfil the criteria for **FIT plants**: The criteria are also part of the contracts and hand over protocols.

- 18 inches for broad leaved and 12 inches for Conifers
- The root shoot ratio should be 1:2
- About 60 % foliage (leaves) are intact (evergreen)
- Pest free, undamaged

4.1.7.7 Transportation of Seedlings from Nursery

The responsibility of transport of seedlings depends on the type and location of the nursery.

type of nursery	responsible	Agreement
Women nursery within PU	VDC/VO	3 Party Women nursery agreement
Private nurseries	Contractor	Transport to planting site part of procurement contract

Table 14: Transport of seedlings

For container (bio-degradable bag or P-bags) seedlings, seedlings should be watered 2-3 days before shifting to site to enable sufficient water to roots so that the soil is intact with the roots system. For bare root seedlings, before transport, seedlings should be dipped into mud, and packed with the Jute bags by bundles. Some mud shall be put in the bags to keep the roots moist during transport. Container seedling have to be loaded and transported in an upright position. Both, container and bare rooted seedlings, have to be protected during transport from wind and plain sunshine. In particular, the roots of the bare rooted seedlings must be completely protected against wind and sunshine in order to prevent them from getting dry. Bare rooted tree seedlings need to be grouped in bundles of 50 to 100 tree seedlings. Before loading the tree seedlings on the lorry, the roots should be dipped in mud. Then the bundles shall be placed in bags or boxes and then loaded.

Disposal of P-bags:

1. Used P-bags at nursery sites are collected
2. Burning is NOT an option: Avoid burning plastic bags, as it releases toxic fumes harmful to health and the environment.
3. Burring in the ground is NOT an option: the P-bags will not rot,
4. The collected P-bags shall be delivered to plastic collection centres for plastic recycling

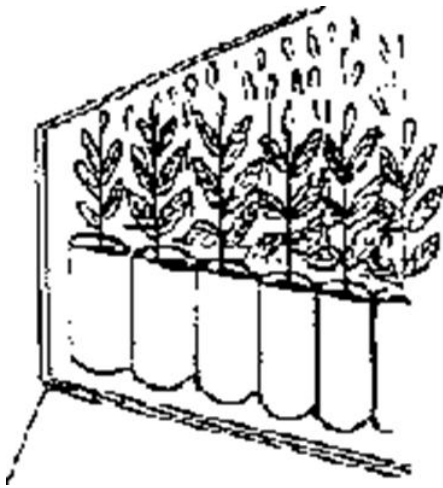


Figure 2. Proper method of seedling transportation



Figure 4. Water the seedlings before transportation



Figure 3. Transportation of Bare Rooted in Jute Bag

4.1.7.8 Storing of Seedlings

Sometime seedlings have to be stored for a few days at plantation site because all the seedling can't be planted on one day. Keep them in a dark and cool place, a cool cellar or elsewhere in the shade. If bare-rooted seedlings are packed in bags, the bags should not be opened. Bare-rooted seedlings which cannot be planted in a few days must be healed-in close to the planting site to minimize later transport. "Healing-in" means temporarily putting bare-rooted seedlings in moist soil, under shade, until they can be used for planting. Healing-in should be avoided but it may sometimes be necessary.

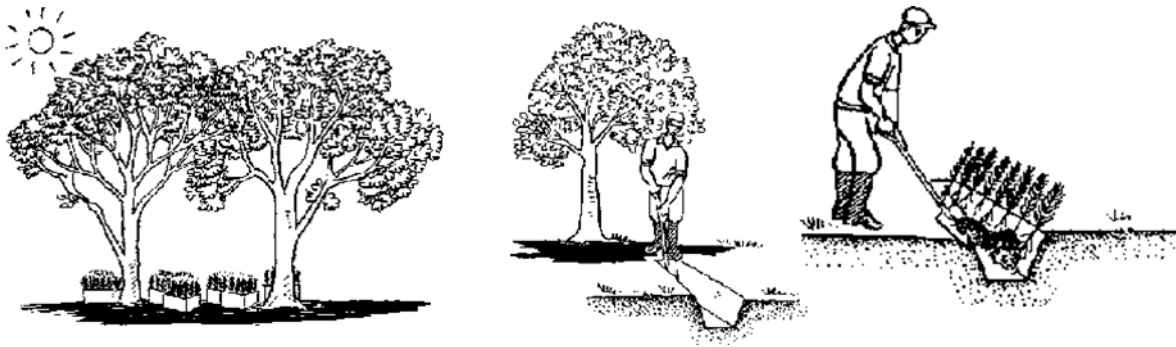


Figure 5. Storing of Seedling

4.1.7.9 Planting of Saplings

Plants can be planted in many ways, in a pit or in tranches. It is recommended to plant trees on cloudy days. In one pit only single plant should be planted. The B-bags should be opened on the bottom before plants are transplanted into the ground. The P bag will be removed by cutting the bag with a sharp razor with inverted “T” shape cut. All the P bags must be collected and properly disposed of from the wild after plantation. it is recommended to press the bag to intact the soil with a root system. Plants should neither be planted too deep nor too shallow.



Figure 7. Removal of P Bag

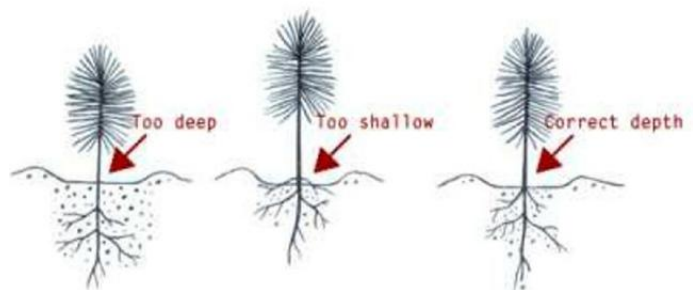


Figure 6. Proper depth of Plantation

4.1.7.10 Sapling Support

There is no sapling support foreseen in BTASP.

4.1.7.11 Watering

There is no watering foreseen in BTASP.

4.1.7.12 Protection

For the new forest plantations, up to 2 community wardens can be employed based on a 3-party agreement including a performance-based payment scheme. The Mgt. committee has the responsibility to monitor and document the forest warden's performance and report to VDC/VO. A specimen of the performance assessment sheet attached in Annex 6.2.

4.1.7.13 Mulching

During mass plantation, mulching might not be possible, however if feasible, 1.5-inch mulching with Jantar or saw dust can be applied. Plants are also protected by stones to prevent evaporation and rodents to damage the plant.

4.1.7.14 Beating up of Failure

Usually in forest plantations up to 25% failure is acceptable. It is recommended to replace the failed plants with healthy stock in proper season preferably in beginning of the monsoon. The species composition in beating up will remain the same as per initial plan of plantation.

1st year up to 25%
2nd years up to 25%
Target survival rate 75%

BTASP specifies the overall survival rate target with 75% in the 3rd year, as the target to achieve based on the FD standard procedure and target see also chapter 11.2 Table 25. ¹⁹

4.1.7.15 Documentation of Plantation Works

For each new plantation a separate final planation protocol shall be signed and forms part of the **planation journal**. The journal, attached in a separate document as Annex 2 will contain the following information:

Location, size and overview map in 1: 5000 and for each compartment.
Date of the establishment, owners and users,
Tree species composition and planting scheme
Plantation maintenance plan for at least 10 years
Plantation harvesting plan for at least 10 years
The plantation regeneration scheme foreseen
The cost – benefit calculation for at least 10 years,
Monitoring plan for Mgt committee and VDC/VO
Survival rate assessment report

Responsibilities:

- Final planation hand -over protocol shall be signed after completion of planation works by Mgt. Committee, VDC/VO and DFO,
- The VDC/VO has the responsibility to monitor the full documentation and annual review of the Mgt. committee's planation journal.

¹⁹ Forest Ordinance Code, year 2022

4.1.7.16 Time schedule for plantation work

Site preparation	Before start of monsoon
Fencing	Before start of monsoon
Site clearing	Before start of monsoon
Planting	After first showers of pre-monsoon/monsoon
Monitoring	1 st year monthly
Beating up	2-3 rd year short before monsoon
Survival rate assessment	Short before monsoon / before beating up

Table 15: Time schedule for plantation work

4.1.8 Woodlots (0.1 ha to 2ha)

Description: Woodlots are the small plantations for firewood, animal fodder or poles on private land. To enable smallholders with access to very little land the area size ranges from 0.1-2ha. Wood lots will be allotted to small landholders with a max. size of 10 ha per PU. In plantation, the multipurpose native species will be preferably planted with about 20 % (max) eucalyptus after signed resolution by VDC/VO.

Objective of the measure:

Smallholders are able to grow trees for livelihood support

Expected benefit/result

To enable smallholders to establish tree plots for livelihood support

To increase the tree cover in the community area

To reduce the fire wood pressure from native forest

Criteria for site selection:

- Availability of area up to 0.1-2 ha area that is not used for current agriculture
- Area that is close to farm or homestead to be secured

Criteria for beneficiary selection:

- Preference for women headed households recommended by VDC/VO
- Families with low socio-economic status recommended by VDC/VO
- Willingness to plant multipurpose fast growing tree species

Responsibilities:

1. A simple agreement will be established between FD, VDC and Woodlot/Owner
2. Beneficiary demarcates the plot area, VDC/VO confirms the site
3. Beneficiary prepares the planting site and planting pits in a spacing 8x8 feet,
4. FD provides the number of saplings required – after digging of planting pits 50%
5. The beneficiary is responsible for planting work, protection and maintenance,
6. Beneficiary, VDC and DFO sign final plantation protocol,
7. Payment is based on survival rate assessment see Annex 5.3
9. After plantation the owner will solely be responsible for watch & ward,
10. Owner agree to maintain the woodlot for up to 10 years.

Costs: The cost norm for that measure is listed in Annex 1, No.: 10

4.1.9 Spring Protection and Management

Description: In most of PUs natural springs are found and form the major source of domestic water source. In some areas it has been observed that the discharge of spring water is reducing. This can have multiple courses but often the main reason generally the recharge area of the springs are not effective anymore due to increased water runoff and a subsequent reduced water infiltration rate often caused by forest degradation and unsustainable land management.

The BTASP spring protection measure will assess the spring shed area in a simple form and will propose active counter measures e.g. close plantation and/or suitable water retention measures. This will improve the vegetation cover near the spring and surrounding area from erosion and increase the infiltration rate. Species like Willow, Poplar, Mulberry and Robinia having a potential to propagate vegetatively will be planted with 5 x 5 feet spacing through cutting.

The spring shed mgt. approach is actively propagated by ICIMOD an organisation from Nepal already active in Pakistan in the SAFER project. The BTASP spring protection approach is based on the main principals of spring management; however, the approach will be more simplified and is part of the PFRMP.

Objective of the measure:

Spring shed management measures for the improved water retention are implemented

Expected benefits/results:

- Increased Forest cover and soil cover for:
- Increased erosion control
- Increased rain water penetration
- Higher water infiltration rate
- Less soil and water pollution of spring site
- Increase stable spring water spill
- Increase water availability also for forest plants

Criteria for site selection:

- Spring areas and recharge area identified within the PU
- Springs used for community or Khel water supply

Beneficiaries:

- Khel or community of the PU using the spring water

Responsibilities:

- FD provides **training and technical inputs** on spring shed management
- VDC/VO assign community members to participate in spring shed assessment
- FD provides **local expert to assess spring shed system** and define recharge area,
- FD and VDC/VO agree on activities to increase water infiltration rate in recharge area
- FD provide funds to carry out activities of **close plantations, trenches and rain water harvesting** structures, see below descriptions,
- VDC/VO carry out activity monitoring and assign spring mgt. committee
- A committee for spring mgt shall be set up and trained on spring mgt. tasks

Cost: The cost norm for that measure is listed in Annex 1, No.: 11

4.1.9.1 Contour trenches with seed sowing / or vegetative propagation

Contour trenches are earth ditches placed on zero or near zero grades for the purpose of exerting some control on runoff and erosion. This control is obtained by catching and storing run off from the area above the trench. Associated with this control of surface runoff is the improvement of soil moisture conditions with its subsequent benefit to vegetative cover. The trenches are ca. 10 feet long, 2 feet width and 1 feet depth. The trenches are staggered from one line to the other. Up to 5 ha with trenches and rain water harvesting schemes have been planned with up to 50,000 PKR/ ha.

- FD and VDC – site identification for layout area for contour trenches,
- FD - layout of the contour line shall be done with the help of A-Frame,
- VDC/WO – provide daily labor against payment,
- FD will provide seed or cuttings for planting material e.g., willow, poplar, mulberry and robinia
- FD will provide payment for daily labor work,
- VDC will monitor on a weekly base the progress of the implementation,
- DFO will monitor the progress on a monthly basis,
- VDC/WO, DFO will sign a completion protocol
-

4.1.9.2 Rain water harvesting and retention schemes

As adaptation to climate change and fallback against droughts, rain water harvesting schemes in the spring shed areas shall be developed and implemented. Expertise of technicians of relevant disciplines shall also be availed on need basis. These schemes not only cater for the watering needs of the plantations but also help increasing biodiversity of the area. Further these would equally help recharging the ground water with increased infiltration rate. Identifying forest-based rainwater harvesting schemes involves the assessment of forest and their capacity to capture, store, and regulate rainfall for various purposes. Up to 5 ha with trenches and rain water harvesting schemes have been planned with up to 50,000 PKR/ ha.

4.1.10 Fire Prevention and Fire fighting

Plantations are most vulnerable to fire hazard. For protection and sustainable use of forest resources, establishment of fire lines before dry or fire season combined with awareness raising campaigns are the most effective measure against forest fires. A firebreak is defined as a natural or already existing barrier artificially constructed before a fire occurs from which all or most of inflammable material have removed. The BTASP will develop the fire management plan and organize training for community watchers (Chowkidars) to establish fire lines. During the BTASP, about 6-meter-wide fire line will also be established in the fire prone areas of Plantations. Mechanised fire line cleaning with tractors shall be used.

Objective of the measure:

The selected forest sites are better protected against forest fire

Expected benefit/results:

- To control and reduce fire incidents in forest planation and other forest sites
- Introduce forest fire management on communal level

Criteria for site selection:

BTAP, 10BTAP plantations, enclosures and other forest sites within the PU area

Responsibilities:

- FD and Mgt. committee: during Update process of planation journal – fire lines or other fire prevention or protections measures are planned
- FD will provide fire mgt. training on fire prevention and firefighting for community's
- FD will provide simple firefighting tools (e.g. fire patches) to be available for the forest warden,
- VDC/VO and Mgt Committee will select community members to be trained in fire prevention and fire fighting
- Mgt. committee will provide men power for cutting the fire lines, the Mgt. Committee shall compensate the labourers,
- VDC/VO will monitor the execution of the actions and will report to DFO.

Costs The cost norm for that measure is listed in Annex 1, No.: 14,15

4.1.11 Harvesting of Forest Products

Description: BTASP will support the sustainable planning of harvest and harvest measures of forest and non-forest products with harvesting plans and quotas agreed for controlled use over time. This is part of the of the plantation and enclosure management plan and description. Based on the rapid assessment inventory an annual increment can be estimated. Ca.80% of that estimated increment will be used to calculate the harvest quota. Additionally, land owners' *needs* shall be taken in consideration for harvesting plans.

Objective of the measures: Sustainable harvest of forests and non-timber forest products (NTFPs) through various strategies and initiatives to balance ecological conservation, economic viability, and community development are planned, executed and monitored.

Benefits of the measure:Harvest quotas for the plantations and other forest sites are available and agreed, Harvesting has been planned and is been executed sustainably, VDC and Mgt. committees monitor the harvesting in the PU area

Beneficiaries: Forest land owners and users, Mgt. Committee
VDC/VO and Community

Responsibility:

- FD and Mgt Committee members carry out participatory forest inventory (step 2 PFRMP)
- FD and Mgt committee agree, based on inventory result on harvesting quota
- FD support the Mgt committee with training on the job with in harvesting techniques
- VDC/VO monitor the harvesting plan / sustainable use plan in the PU on 12-month basis
- Chain saw operators are trained in operations safety

This is not a standalone activity but include in other measures. There is no cost involved

Eucalyptus Production Cycle

Eucalyptus camaldulensis

The eucalyptus plantation site is initially cleared prior to the planting. The most important management operation if required is weeding and protection in non-intercropped areas and at the age when intercropping ceases. The competition of grass and other weeds can be so intense as to cause growth cessation in eucalypts. Fire alone is a sufficient reason for total failure if not managed properly.

The rotation is generally fixed at 10 years with 2-3 (or more) rotations of coppice regeneration and then replanted with eucalypts.

1. **Harvesting Practices:** The timing of Eucalyptus harvesting depends on the intended use of the wood. Generally, Eucalyptus trees can be harvested for timber between 5 to 15 years of age, depending on the desired size and quality of the logs. Pulpwood may be harvested at a younger age.
2. **Timber Production tree age of >10 years:** Eucalyptus is harvested for its timber, which is used in various construction and woodworking applications. The fast growth of Eucalyptus trees makes them an attractive option for timber production. With an age of 10 years Eucalyptus can reach an “big pole size” of up to 25-30 cm diameter.
3. **Pulpwood or MDF (Medium Density Fibre) <5 years old:** Eucalyptus trees are also used for pulpwood production, primarily for making paper and related products as well as for the MDF production. Here the wood harvesting is demand driven and the wood is sawn according to demand.

Eucalyptus can be coppiced up to 3-4 times without losing biomass growth. The trunk is cut at little above surface. Shortly after 1-2 weeks multiple shoots will come up. After 4-5 months the shoots are up to 2-meter high and can be used already for support sticks for vegetables etc. Selection and pruning of 3-4 main coppicing stems

4. **Agroforestry:** Eucalyptus is sometimes integrated into agroforestry systems in Pakistan, where it is grown alongside crops (linear plantations) or other tree species. This can provide additional income for farmers and contribute to soil conservation and improved land use.
5. **Selective Harvesting:** Sustainable harvesting practices in Eucalyptus plantations often involve selective harvesting, where only a portion of the mature trees is cut during each harvest cycle. This approach allows the remaining trees to continue growing and ensures a continuous supply of wood over time.
6. **Logging Methods:** The logging methods used for Eucalyptus harvesting in Pakistan may include chainsaw felling, skidding, and loading onto trucks or other transport vehicles. Mechanized equipment is often used to expedite the harvesting process.
7. **Replanting:** Sustainable Eucalyptus management practices often include replanting or reforestation. After a harvest cycle of 20-25 years, new Eucalyptus saplings or seedlings are planted in the cleared areas to ensure the continuation of the plantation and maintain forest cover.

Selected harvesting figures:

Based on data of the PAK- WWF Report 2020 *Role of MDF & Particleboard Industry in protection of Natural Forests and Promotion of Agroforestry in Pakistan*. It was calculated that with a 5 years rotation and a 5x10 ft. spacing the per hectare cost of production of Eucalyptus is about PKRs 289,628 while the **net annual return PKRs 272,506** assuming that only 80% of the trees will survive with an average **weight per tree of 120kg**. The study shows that the net annual return from Eucalyptus is ca 20-25% higher than any other cash crop combination in KP. Detailed cost of production and net annual return are shown in tables in Annex 7.

4.2 Rangeland Management

4.2.1 Controlled Grazing

Description: Under BTASP the VDC/WO will develop with the livestock owners a controlled grazing plan (rotational grazing) and grass cutting, based on a pasture analysis. This includes training for livestock owners, herders and nomads, if they are visiting the PU in controlled grazing and pasture management techniques. Mostly the local farmers are now practicing stall feeding for cows and only small ruminates kept on rangeland. However, the BTASP in partnership with the KP Agriculture, Livestock and Forest Departments will conduct training for the local herders with sessions on grazing practices, pasture assessment and monitoring techniques and assessment of carrying capacity. The recommendation for that approach comes from report of the Social Forestry Project²⁰ and Environmental Rehabilitation²¹ Project.

Objective of the measure:

The available rangeland is used in a sustainable and controlled manner.

Criteria for selection:

- Households having livestock and grazing their animals in the hillsides
- Willingness of VDC/WO to support rotational grazing plan

Responsibilities

- VDC/WO will nominate the participants for rangeland mgt. training,
- VDC/WO is responsible for the participation of the participants,
- FD will organize training sessions on rotational grazing in an ToT approach,
- FD together with herders will develop a rotational grazing plan and herder scheme,
- FD with external support will draft a simple rotational plan and monitoring,
- VDC/WO will monitor the progress of rotational grazing plan development and its implantation over time,

Expected Benefit: Reduction of overuse and underuse of rangeland area
 Reduce area degradation and erosion
 Improve regeneration capacity of rangeland sites
 Increase feed gras and herbs diversity due recovery time
 Indirectly improve animal health due to feed diversity

Costs The cost norm for that measure is listed in Annex 1, No.:18

²⁰ Basic Concepts of Rangeland Management and the Implications for Malakand Division, SFP, 1993

²¹ Rangeland Utilisation in Malakand Division – Under USE or Over USE, ERP, 1997

4.2.2 Provision of Fodder Seeds

The Provision of fodder seeds for cultivation on agricultural and marginal lands is a compensation measure or incentive for the rational grazing activities under 4.2.1.

BTASP will provide on test basis fodder seeds for use on agricultural and marginal lands, preferably with the agricultural extension department. On average 500 kg seed per year for 2 years for 100 ha shall be provided in each PU.

Objective of the measure:

Short term losses on animal fodder from the start of the rotational grazing is been compensated

Expected benefit/result

To increase fodder plant volume and diversity on degraded sites,

Criteria beneficiaries selection:

- Households with livestock on rangeland sites
- Households with land available for cultivation of fodder

Responsibility

- SDFO & CDO (M/F) will communicate to the VDC and WO about the incentives along with criteria
- VDC and WO will provide list of beneficiaries along with formal resolution
- The SDFO and CDO (M/F) will randomly reverify few of beneficiaries from the list
- VDC & WO will provide information and detail of the beneficiaries,
- VDC will monitor of seeded sites and will report to DFO
- SDFO Forest Guard / Forester will do random check of improvement and report to DFO

Costs The cost norm for that measure is listed in Annex 1, No.: 19

4.3 Livelihood Support Measures

4.3.1 Farm forestry -tree seedlings

Description: In most of PUs the land holding size is low. The smallholder farmers or landless member of the communities may not benefit from block plantation or woodlots. Therefore, the distribution of tree seedlings to private persons for planting along the edges of their farmland and near houses for fuelwood and fodder has been perceived successful action also under 10BTAP. BTASP plans to provide for 2 years 10,000 plants be distributed in the PU during spring and monsoon season over the project period. A max of 100 seedlings will be provided to each selected household.

Objective of the measure:

Improve the firewood and animal fodder situation for landless or smallholders

Criteria for beneficiary selection

- Preference households who do not benefit from block plantations and fuel woodlots
- 50% of the seedlings distributed through the WO
- Up to 100 plants per beneficiary HH

Responsibilities

- VDC and WO will provide list of beneficiaries along with formal resolution,
- FD will advise the beneficiaries on tree species suitable and available,
- SDFO and CDO (M/F) will randomly verify beneficiaries from the list,
- VDC/WO will produce a map where the planting lines are indicated for monitoring purpose, the Forest Guard or Forester will verify the locations,
- VDC/ WO and FD will agree on date and venue for seedling delivery & distribution,
- FD will provide viable and healthy seedlings according to standard norms,
- Forest Guard/Forester will monitor the planting activities
- VDC/WO will monitor and report the planting progress to the DFO,
- Beneficiaries are responsible for planting and maintenance of seedlings free of charge
- FD will inspect planting site and prepared pits before distribution of plants,
- Beneficiaries shall contribute 2 PKR / plant to the VDC/WO to the saving account.

Species composition:

Seedling for Farm forestry: fire wood, animal fodder, wild fruits and nuts e.g. Poplar, Mulberry, Alianthus, Bakine, Kachnar, etc.

Guidelines

1. Demand from the potential Farmer shall be collected prior to the planting season.
2. 30% Pits dug for the desired plants shall be verified by concerned forester before handing over the desired plants.
3. Proper storage and planting shall be ensured by concerned VDC/WO forester/forest guard.
4. Record of beneficiaries/farmers shall properly be maintained as per format (to be developed).
5. Photography should be carried out at the time of handing over Plants to Farmers.
6. The distributed plants shall be frequently monitored by VDC concerned supported by local forest staff and shall submit the same to PMU on monthly basis with intimation to DFO concerned,

Costs The cost norm for that measure is listed in Annex 1, No.:12

4.3.2 Women Livelihood Project

Description: Due to cultural norms in KP women are more or less confined to the home stead or close by area. Women in KP are often only indirectly via their husband's decision involved in forest and other NRM activities. Therefore, BTASP will provide a flexible measure to involve and target women in activities with the aim to support the household livelihood. The identification process will be led by the female CDO and the WO. Based on the criteria's below and a budget ceiling a project will be designed and shall include either training or inputs or a combination of inputs and training. Concept: introduce trainers and train local women about marketable handicrafts or improving the existing procedures and quality.

Example: Kitchen gardening: An initial training plus some packets of improved variety of vegetable seeds will be provided to women for kitchen gardening. About 500 packets will be provided to families nominated by VDC/WO. Each packet will contain 6 varieties of vegetable seeds. Criteria for selection is availability of at least 200 feet² area, women headed family (Preference for widows and disabled) and availability of water source near the house.

Objective of the measure:

To increase women involvement in livelihood (NRM) measures

Criteria for beneficiary selection

- Availability of at least 200 Sq. Ft area,
- Women headed family (preference for widows and disabled)
- Families with low socio-economic status
- Availability of water source near the house
- Recommendation by WO.

Responsibilities:

- VDC and WO will provide list of selected beneficiaries along with formal resolution,
- The SDFO and CDO (M/F) will randomly reverify beneficiaries and criteria,
- FD will provide the training and seed packages,
- VDC/ WO and FD will agree on date and venue for training,
- VDC/ WO and FD will agree on date and venue for seedling delivery & distribution,
- The WO members will monitor the planting activities and progress,
- VDC/WO will monitor and report the progress to the DFO,
- Beneficiaries are responsible for planting and maintenance,

Benefit e.g. Kitchen gardening

- Approximately a family can save about 3000 to 3500 PKRs through kitchen gardening.
- Fresh vegetables consumption will improve the food diversity, the nutrient and vitamin consumption and may lead to better health condition within the family and esp. children

Costs The cost norm for that measure is listed in Annex 1, No.:20

4.3.3 Sustainable Management of NTFPs:

Description: Non-Timber Forest Products (NTFPs) play a crucial role in the daily livelihood of rural communities. Some of the NTFPs are used purely for household consumption and others are harvested and traded for cash income. During the PFRMP process the types, amounts and locations of the NTFPs are identified and documented. Based on the assessment the PFRMP will suggest measures for sustainable NTFP harvest and management.

At PU level pre- and post-harvest training of existing NTFPs for local collectors will be conducted. There are some NTFPs, which may not be economically important for national and international markets, however ecologically and at local level they are equally important. The BTASP will take care of both the components. The economic important NTFPs will be taken under the value chain component while at PU level for conservation a pre- and post-harvest and sustainable collection training will be imparted.

Objective of the measure:

Identified NTFP's in the PU area are sustainable managed

Criteria for beneficiary selection

- Households involved in collecting and/or trading in NTFPs
- Recommendation from the VDC/VO

Responsibilities

- VDC and WO will provide list of selected beneficiaries along with formal resolution
- VDC and WO will get wiliness and agreement from the HH of NTFP.
- FD will facilitate the training through VDC and WO.
- FD and VDC/WO will agree on a harvesting quota per year and harvesting site,
- FD will if possible support marketing through NTFP directorate.
- VDC/WO will monitor the progress of the measuer and the harvesting quota,
- VDC/WO will report on a monthly basis to the DFO,

Benefit

- Enhance livelihood of low socio-economic HH at community level
- Sustainable use of NTFPs.

Costs The cost norm for that measure is listed in Annex 1, No.: 22

4.3.4 Women Nurseries

Description: The main PFRMP forest activities focusing on planation and enclosure management and new planation establishment. Therefore, large numbers of seedlings are required for distribution among local population and for plantation see also chapter 3.6.1. BTASP will establish **women run private nurseries** on average of 3 units in each PU, in collaboration with VDC/VO. Each unit of nursery will contain 25,000 plants. In total 370 broad-leafed and 73 chir-pine women nurseries shall be established with 25,000 seedlings per contract. The model contract, mode of payment, monitoring and detailed input specifications are attached in Annex 5.5. The species composition in those women nurseries varies and are based on agreed PU specific PFRMP planting models.

Objective of the measure:

Women nurseries provide seedlings for BTASP PFRMP measures

To support livelihood actions for women

To support the seedling supply and raise qualified seedlings for planting activities of BTASP

To increase women involvement in the BTASP NRM and forest measures

Criteria for beneficiary selection:

- Woman willing to work in nursery
- Recommendation VDC/VO
- Availability of at least up to 250 square feet area
- Accessibility of site through road
- Availability of water source near the location for watering
- Willingness to participate in nursery raising training
- Bank account holder or willing to open the bank account

Responsibilities:

1. VDC/VO recommends a woman willing to do the nursery
2. SDFO site inspection to check above criteria for site and nursery person,
3. Establishment of three-party performance-based contract between DFO, VDC/VO and women nursery contractor Annex 5.5.
4. SDFO provides nursery training (based on existing procedure) and inputs as required,
5. Contractor starts nursery work and receives 1st payment installment based on payment plan,
6. VDC/VO inspects nursery on monthly basis based on monitoring sheet Annex 6.4
7. SDFO inspects nursery very 3 month and before each installment Annex 6.5
8. 3rd Party Monitoring and SDFO and VDC/VO inspects nursery short before delivery to the planting site and issues a report on quality and quantity raised. Based on that final payment can be released
9. SDFO inspects readiness of planting site and agrees on date of seedlings transport
10. FD will be responsible for seedling transport.

Costs The cost norm for that measure is listed in Annex 1, No.: 21

4.3.5 Capacity Building, Training and Workshops

Description: In BTASP capacity building is a cross cutting issue and is required on different levels, thematises and topics. Therefore, the IHR&D Department carried out a Training Needs Assessment²². The TNA assessed proposes a list of trainings for FD staff and community members. This list forms the base for the BTASP Global Capacity Building plan.

The PFRMP operational plan defines thematic measures including Capacity Building on PU level for VDC and WO, Mgt. committees and selected beneficiaries. These trainings are part of the BTASP Global Capacity Building plan.

Important for BTASP is the involvement of the VDC and WO as main official community link and access point for planning, implementation coordination, monitoring and financial management. Therefore, the VDC and WO functioning is important and require trainings on organisational management, office record keeping, resolution development, accounts management, project monitoring and reporting. Technical trainings will be included in the different PFRMP measures to improve forest- rangeland- and livelihood management in selected measures.

The training approaches used will be based on adult learning: short sessions fitting into the working circle of the trainees and respecting agricultural production circles. The training approaches include exchange visits, direct group trainings and ToT trainings. The trainings will be organized by IHR&D Directorate with internal and external resource persons.

Responsibilities

- VDC/WO will nominate participants according to technical intervention
- DFO will request training with IHR&D directorate or PMU
- FD will facilitate the trainings through I&HRD directorate
- FD will pay for training transport, venue and food and resource person,
- VDC/WO will ensure participation of nominees.
- VDC/WO will monitor the training sessions and report to DFO

Costs The cost norm for those trainings are listed in Annex 1, No.: 23, 24, 24.1-24.3

²² BTASP Draft TNA Report, IHR&D Department Dec. 2024

Overview of PU relevant trainings foreseen in the PFRMP (no exhaustive)

Topic	Training target audience	Training type	Content
CMST and LMST ²³	VDC and WO	direct training in various sessions, exchange visits	see foot note
Financial Management BTASP	VDC and WO	direct sessions	Account management, financial documentation and accounting
Basic Forestry	Forest warden and selected members of Mgt. committee and forest land owners	ToT topics around 1 year, exchange visits, training on the job, guided format with FD	Forest planning, Planation and Enclosure Mgt. Sustainable harvest, Fire prevention and firefighting, Regeneration of forest sites, Cost-Benefit calculation Updating micro forest site plan
Spring (shed) management	Selected community members	ToT, exchange visit, training on the job,	Various sessions on assessment and measure identification
Rangeland management	Selected HH, herders	TOT and direct training exchange visits	Rotational Grazing plan
Nursery management	Selected Women	Direct training, exchange visits	how to set up, run and manage a nursery
Women livelihood support	Selected Women	ToT approach and direct trainings	how to set up a kitchen garden, seeding, plant mgt. harvest
NTFP management	Selected Women and men	direct training	Sustainable pre- and post-harvest NTFP management

Table 16: Overview of PU relevant trainings foreseen in the PFRMP

²³ • CMST: Community Management Skills Training

- This training is aimed at building the capacity of community organizations and their leaders.
- Focuses on improving management, decision-making, and leadership skills at the community level.
- Typically includes topics like group dynamics, financial management, conflict resolution, and effective communication.
- Often conducted in the context of rural development or social mobilization programs.

□ LMST: Leadership Management Skills Training

- Designed to develop leadership and advanced management skills for community leaders and office bearers.
- Emphasizes strategic planning, organizational development, problem-solving, and motivational skills.
- Aims to strengthen the role of leaders in managing and guiding their community-based organizations effectively.

5. BONUS PAYMENT SYSTEM

To encourage VDC/WO, management committee, the community warden and concern Forester/Forster Guard for their good work and for future success of the newly raised plantation under BTASP, a bonus payment or reward mechanism shall be introduced in BTASP. Newly raised block plantation under BTASP will only be eligible for bonus payment for next three years. The total bonus amount for next 3 years will be about up to 10% of the overall afforestation costs. Based on survival rate, the plantation will be graded into Grade-A & B. Grade-A plantation will get 10% bonus payment, which is 10,000 PKR per ha in three years, however Grade-B will get 6% bonus payment, which is 6000 per ha in three years. Following is the years wise detail of both Grade A & B according to survival rates.

Year	Year-1	Year-2	Year-3
Grade A	100-80% Survival Rate	100-75% Survival Rate	100-70% Survival Rate
Grade B	79-70% Survival Rate	74-65% Survival Rate	69-65% Survival Rate

Table 17: Bonus payment calculation

Bonus to be paid annually depending on 3rd Party Monitoring report. The annual bonus will be paid at the rate of 25 %, 25% and 50 % for the first, second and third year respectively. Of the total annual bonus payment, 40 % will go to NRM fund of VDC/WO while 20 % each equal portion will be paid to Management committee, Community warden and concern Forest/Forest Guard. Above information is clearly define in below Table 18:

Year wise %age & %age of different Sector			
Year wise %age of the total	Year-1 (25%)	Year-2 (25%)	Year-3 (50%)
%age of different sectors	VDC/WO NRM Fund (40 %)	VDC/WO NRM Fund (40 %)	VDC/WO NRM Fund (40 %)
	Management Committee (20%)	Management Committee (20%)	Management Committee (20%)
	Community Warden (20%)	Community Warden (20%)	Community Warden (20%)
	Forester/ Forest Guard (20%)	Forester/ Forest Guard (20%)	Forester/ Forest Guard (20%)

Table 18: Year wise bonus payment

For further clarification of the bonus payment, take an example of two plantations A & B and each one has area of 50 ha. Plantation-A falls in Grade-1 and Plantation-B in Grade-2 based on survival rate according to 3rd party monitoring report. According to the above bonus payment system, both plantations will be rewarded with bonus according to below formula.

Plantation-A (Grade-1) = Total Bonus Payment = 50 x 10000 = **500,000 PKRs Overall for 3 years**

Plantation-B (Grade-2) = Total Bonus Payment = 50 x 6000 = **300,000 PKRs Overall for 3 years**

Year wise distribution of Bonus payment is shown in Table 19 below.

Year	Year-1	Year-2	Year-3
Plantation-A Grade 1	125000 (25%)	125000 (25%)	250000 (50%)
Plantation-B Grade 2	75000 (25%)	75000 (25%)	150000 (50%)

Table 19: Bonus payment distribution per grade and year

Example distribution of annual bonus and beneficiaries' groups

Beneficiaries	Grade-A Plantation Year Wise			Grade-B Plantation Year Wise		
	Year-1 (25%)	Year-2 (25%)	Year-3 (50%)	Year-1 (25%)	Year-2 (25%)	Year-3 (50%)
VDC/WO NRM Fund (40 %)	50000	50000	100000	30000	30000	60000
Management Committee (20%)	25000	25000	50000	15000	15000	30000
Community Warden (20%)	25000	25000	50000	15000	15000	30000
Forester/Forest Guard (20%)	25000	25000	50000	15000	15000	30000
Total	125000	125000	250000	75000	75000	150000

Table 20: Distribution of annual bonus and beneficiaries' groups

The calculation is also attached in the cost norm Annex 1 No.:16

6. COST NORMS

For different types of plantations (Block Plantation with different spacing, wood lots and close plantation for slope stabilization), as also prescribed in the separate agreement, the **revised cost norms of 10BTTP according to revised PC-1 Dec 2024 were adopted**. For forest warden wadges, the existing rates were used with 15 % increase. Regarding livelihood and rangeland activities the open market rates per unit cost were taken. To incentivise the women related activity like women nurseries, the unit cost has been increased by 25% from 10BTAP. An integrated approach has been adopted for estimating the cost norms for training and capacity building, which includes existing cost norms of I&HRD&M directorate and KP Government revised rates of TA/DA along with open market rates and experience of DFOs in the pilot sites.

The detailed cost norms and PFRMP measures budget are attached in Annex 1. Table 21 provides an overview of the costs and norms per standard PU.

Overall Cost Norms per PU

Table 21: PFRMP Frame work cost norms and proposed budgets

Overall PU cost norms							
Item	Unit	Quantity	Unity Cost PKR	Total Cost (PKRs)	BTASP	Community Contribution	Contribution to Budget Line
Forest and Rangeland Resources							
Community Wardens (MM) for Existing Plantation and Enclosures	MM	96	30000	2880000	2880000	0	Forest Wardens
Update Plantation and Enclosure Journals	No	10	22400	224000	224000	0	Afforestation
Enrichment ex. Plantations (20% of Ha)	Ha	25	9546	238650	219490	19160	Afforestation
Block Plantation (Ha) 10x10	Ha	35	50200	1757000	1622880	134120	Afforestation
Block Plantation (Ha) 8x8	Ha	35	97380	3408300	3088645	319655	Afforestation
Community Wardens (MM) for New Plantation	MM	72	30000	2160000	2160000	0	Forest Wardens
Wood Lots (ha)	Ha	10	80640	806400	732480	73920	Afforestation
Farm Forestry	No	10000	20	200000	100000	100000	Afforestation
Spring Shed Measures per PU	PU	1	849520	849520	827418	22102	Afforestation
Seedlings	No	187338	15,77	2953450	2953450	0	

Fire Lines (km)	Km	10	20000	200000	200000	0	Fire fighting and compensation
Livelihood and community interventions							
Women Nursery (No)	No	3	250000	750000	750000	0	Value Chains & nurseries by women
No of fire fighting tool kit	Set	5	10000	50000	50000	0	Fire fighting and compensation
Tools for the communit Wardns	Set	5	10000	50000	50000	0	Capacity Development
Provide fodder (ha)	Ha	50	12500	625000	500000	125000	Value Chains & nurseries by women
Women Livelihood Project	No	1	400000	400000	400000	0	Value Chains & nurseries by women
VDC Project	No	1	600000	600000	600000	0	Value Chain - equipment & inputs
Bonus for Plantation (ha)	Ha	70	10000	700000	700000	0	Afforestation
Capacity Building							
Women Nursery Training	No	1	50000	50000	50000	0	Capacity Development
Pre- and post-harvest (NTPFs) at PU level	No	1	150000	150000	150000	0	Capacity Development
Training for Rotational Grazing at PU Level	No	1	300000	300000	300000	0	Capacity Development
CMST, LMST and finance training for VDC/VO at PU level	No	1	300000	300000	300000	0	Capacity Development
Basic Forest Management Training (PU share for Division Level)	No	6	50000	300000	300000	0	Capacity Development
Spring Shed management Training (PU share for Division Level)	No	6	50000	300000	300000	0	Capacity Development
Total in PKRs				20.252.320	19.458.363	793.957	
Total in Euro (@ 300 PKRs = 1 Euro)				67.508	64.861	2.647	

6.1 Cost estimation for Result 4 PFRMP measures implementation

Based on the detailed calculation of each measure in Annex 1 No.: 2 and in the detailed worksheets No 4 ff., the summery cost is shown in table 22. The costs are distributed across 3 major budget lines.

Budget Line	Amount in Euro	Available Amount in Euro	Balance	% of measure costs
1 Capacity Development	300.000	732.004	432.004	4,9
4.1 Forest Wardens	1.761.600	1.369.423	-392.177	28,7
4.2 Afforestation	2.933.154	4.016.993	1.083.839	47,8
4.3 Value Chains & women nurseries	1.061.953	1.233.571	171.619	17,3
6 Firefighting and compensation	83.667	414.857	331.190	1,4
Total	6,140,373	7,766,849	1626475	100

Table 22: PFRMP summery cost table

The budgets calculated for PFRMP measures per PU range between EUR 56,000 and 64,000 depending on the new planation planting model. However, in Gari Chandan and Azakel the PU budgets are substantially higher due to the amount of total of 48 forest warden. A budget amendment to balance that budget line 4.1 is required.

The Guidelines provide limits per PU for each unit of planting actions and costings. It is understood that not in all PUs all measures will be implemented as planned. At least each year a review of the technical implementation and financial spending has to be done.

It is expected that the training costs at PU level for technical trainings, at the moment quite moderately calculated might increase.

7. AGREEMENTS AND CONTRACT

The PFRMP Cooperation agreement signed between the Forest Department and the VDC/WO is the overall and most significant framework agreement for implementation of the selected measures described in chapter 4.

A three- party agreements will be signed between DFO Office, VDC/WO and either management committee or individuals depending on the nature of the measure including a performance-based agreement will be signed with the following PFRMP measures:

1. Women nursery
2. Forest Warden (chokidar, negheban) for enclosures and new plantations
3. New plantation, Enrichment planting and Wood lots

For all other implementation of measures, a simple sperate 3 party agreements will be signed with the concerned stakeholders involved describing the nature of activity and responsibilities.

Specimens of Agreements/contracts are available in Annex 5.1 -5.5.

8. VDC/WO NATURAL RESOURCE MANAGMENT FUND

One of the pillars for successful community-based forest management is the availability of funds after project end for required management and protection of the investments done by the FD projects BTAP, 10BTTAP and BTASP.

A **community fund** for sustainable forest management is essential to ensure long-term conservation, organizational growth, and economic benefits for local communities. Key reasons include:

1. **Sustaining Investments:** Ensures maintenance of initial projects like reforestation or forest restoration.
2. **Funding Silvicultural Activities:** Provides resources for planned activities such as thinning, pruning, and regeneration.
3. **Building Reserves:** Creates funds for new investments in natural resource management, such as soil conservation or biodiversity enhancement.
4. **Supporting Organizational Development:** Strengthens Village Development Committees (VDCs) and Women's Organizations (WOs) by funding capacity-building, training, and effective governance.
5. **Economic Stability:** Enables sustainable livelihoods through eco-tourism, agroforestry, and sustainable harvesting practices.
6. **Self-Reliance:** Reduces reliance on external funding, empowering communities to manage forests autonomously.
7. **Climate and Ecosystem Health:** Funds projects that mitigate climate change and protect forest ecosystems.

In summary, the fund sustains forests, supports community organizations, and enables long-term conservation through financial and institutional stability.

BTASP approach to NRM fund establishment

1. During the PFRMP planning process the issue of the community NRM Fund is been discussed and part of the Cooperation Agreement,
2. BTASP facilitates the process via VDC/VO consultation with the land owners and Management Committees to define the PUs own NRM Fund approach,
3. The NRM Fund is owned by the community members and managed by the VDC/VO on agreed bylaws/SOP (to be developed and agreed during 1st year of PFRMP implementation),
4. Different NRM Fund approaches can be tested in different regions due to cultural differences,
5. BTASP simple proposal for NRM Fund would be a saving account on VDC level managed by selected individual on behalf of the VDC/VO and supervised and monitored by the VDC/VO.

Sources for NFR Fund

1. BTASP bonus payment for successful planation management,
2. 5%-10% of each cash sale of forest products out of the plantations and enclosures established under the Forest Department,
3. External local donations,
4. Other project funding supporting sustainable forest management.

9. COMMUNITY CONTRIBUTION FOR PFRMP IMPLEMENTAION

BTASP will apply during PFRMP implementation a community contribution of 10% in kind—such as labor, materials, or local expertise—invaluable during project implementation. They foster **ownership** and a sense of **shared responsibility**, ensuring the community is actively invested in the project's success. By utilizing local resources and skills, costs are reduced, and the project becomes more sustainable and culturally relevant.

The positive effects include:

- **Empowerment:** Engages community members, building pride and commitment to maintaining the project.
- **Cost-Effectiveness:** Reduces financial burdens while maximizing local resources.
- **Capacity Building:** Enhances local skills and strengthens community organizations.
- **Sustainability:** Encourages long-term stewardship, as communities feel directly connected to the outcomes.

During the PFRMP planning process the issue of a 10% community contribution in kind **for all forest, plantation and enclosure activities** will be discussed and will be part of the Cooperation Agreement. Involving the community collaborative efforts, ensuring greater impact and lasting benefits.

10. ESMF DURING IMPLEMENTATION

The overall BTASP Project Implementation Manual (PIM) consist of 4 documents and are interlinked.

1. Financial Management Guidelines
2. PFRMP Manual
3. PFRMP Implementation Guideline; and
4. The Combined PFRMP and ESMF Steps for Project Designing, Planning, and Implementation (hereinafter the “Combined Steps Document”)

The Combined Steps Document, which is a companion document to the ESMF and the PFRMP, defines a step-by-step process on how to ensure the integration of ESMF considerations into the project lifecycle. The ESMF, like the PFRMP, is also part of the planning process. The only difference between the two is that the ESMF seeks to identify risks that could arise, and harm that could be caused to the project community, and the KP FD project staff and provide adequate mitigation measures to minimise the same. The steps have been reproduced below for ease of reference, and which provides an overview of the combined ESMF and the PPRMF steps to showcase how this shall be done in the project rollout – ESMF specific tasks have been highlighted, however, all steps shall contribute to its rollout:

Step 1: Preparation for Participatory Forest Resource Management Planning

Objectives

- Planning Team prepared for the Participatory Forest Resource Management Planning Process
- Eligible Planning Unit selected confirmed and its boundaries defined
- Awareness created with the community members in the planning unit on Participatory Forest Resource Management Planning.

Result

- Written resolution (Consent of Cooperation) adopted by representatives of all community sections (owners and users of forest and rangeland resources) in the planning unit on cooperation in the development and implementation of a Participatory Forest Resource Management Plan.

Activities

1. DFO notifies Planning Team (SDFO, CDO (m&f), Forester, Forest Guards, GIS expert).
2. Preparation of a workplan for PFRMP in the selected planning unit.
3. Preparation of GIS base map and collection of secondary data.
4. Community mobilization meetings (introductory meeting, and general community meetings including stakeholder analysis and social mapping)
5. Meetings with local communities are held to discuss the non-technical summary of the ESMF, provide them with information on the project, risks and mitigations, and provide with the space to give feedback and sufficient time to decide if they wish to be part of the project or not through a written resolution
6. Go/no-go decision on working in planning unit.
7. Additional awareness raising meetings regarding BTASP, including initial FPIC Stakeholder engagement activities to inform the communities about ESMF (Free, Prior, and Informed Consent).

-
8. Community meetings on resolution of cooperation (Consent of Cooperation)
 9. Reporting to DFO.

Step 2: Participatory rapid assessment of forest and rangeland resources

Objective

- Participatory rapid assessment of availability and condition of forest and rangeland resources.
- Identification of potential project sites and a list of activities.

Results

- Maps of forest and rangeland resources: landcover map; past interventions map; land use map; land ownership and control map;
- Table of overview of existing forest and rangeland resources
- Table with condition, risks and potential of each vegetation type
- Table with calculation of use of mature plantations
- Table of use of forest resources established by the Forest Department

Activities

1. Preparation of for data collection
2. Data collection and analysis
3. Sharing and cross-checking with community
4. Reporting to DFO

Step 3: Participatory socio-economic analysis and social organization

3a. Participatory socio-economic analysis

Objectives

1. Analysis of socio-economic situation of the Planning Unit, including analysis of ownership, use and control of forest and rangeland resources
2. Assessment of the social economic setups, including social institutions and relationship among stakeholders related to natural resources within the potential project sites and proposed list of activities

Results

- Tables on socio-economic situation, and ownership, use and control of forest and rangelands

Activities

1. Preparation for data collection
2. Data collection and analysis
3. Sharing and cross-checking with community
4. Reporting to DFO

3b. Social Organization

Objective

- Village Development Committee and Women Organization established or strengthened

Result

- Inclusive and registered Village Development Committee and Women Organization

The following monitoring tools for ESMF will also form part of the monitoring process:

Tool 4: Framework for Participatory Monitoring per Intervention,
Tool 5: GRFM Logbook,
Tool 6: Digital grievance resolution case file (once MIS is functional)
Tool 7: Work safety and environment checklist,
Tool 8: Serious incident reporting template,
Tool 9: Reporting template from PMU to KfW.

Monitoring of ESMF action plan: recommended indicators:

- Number of FD staff participating in BTASP community work in each division.
- Number of planning units with signed cooperation agreements.
- Number of participating VDC, WO, other VDC / WOs in BTASP-supported planning units (existing, newly established)
- Number of planning units where FPIC has started, has been signed.
- Number of community by-laws / naghas issued.
- Types and duration of restrictions in the by-laws / naghas
- Number of ESMF workshops, number of participants.
- Number of VDCs/WO contacted during the current monitoring period, cumulated.
- Number of complaints and positive feedback received, answered, during the current monitoring report, cumulated.

Women are important stakeholders in BTASP. Consistent with KfW's gender strategy²⁴, gender sensitivity and equity is an issue integrated in all project activities including the ESMF. It is not a stand-alone safeguard. Hence, in order to ensure the integration of Gender related considerations, the project draft and rollout a Gender Action Plan.

ESMF budget relevant for implementation²⁵

As shown in Table 24 budget allocations for ESMF have been made for implementation are estimates and will be defined by the applied cost norms. The activities proposed will be implemented simultaneously with the identified PFRMP measures.

Table 23: ESMF budget

Activity	Unit	Unit cost	Quantity	Amount (PKRs) Budget	Budget Head (BTASP)	How to budget this activity
Capacity building ESMF - Travel Costs - workshops – Divisional and Sub-Divisional Level incl. VDC/WO reps.	Round trips	60000	26	1.560.000	1.0 (Capacity Building)	2 workshops per district, 250 km round trip on average, cost per km 240
Capacity building on ESMF workshops DFO Field Team on Region level	Person 104 day	15500	104	1.612.000	1.0 (Capacity Building)	2 nights per workshop, accommodation and per diem costs per night
Equipment Forest Warden	Set per person	12000	600	7.200.000	4.0 (Local Labor Cost)	each set contains: glove, boots, torch, helmet, hoe, simple back pack
HSSE Training Forest Guards, Foresters, and Forest Warden +Training	100 FG +50 Foresters	3000	150	450.000	1.0 (Capacity Building)	Training on health, safety security and environment
HSSE Training HSSE for Chainsaw operators	5 persons per division	3000	175	525.000	1.0 (Capacity Building)	Training on health, safety security and environment, 5 person per division
Human Wildlife Conflict (HWC) Assessment & proposed mitigations	Lumpsum	600000	1	600.000	4.0 (Local Labor Cost)	Discussion with FEWD + WWF-Pakistan consultations (PU Specific)/ As and when required -

²⁴ KfW's Gender Strategy (2011) calls for identifying and seizing opportunities for contributing through the project to women's economic empowerment and other means of reducing the gap between men and women.

²⁵ ESMF Budget taken from the BTASP ESMF Guideline Report

Activity	Unit	Unit cost	Quantity	Amount (PKRs)	Budget Head (BTASP)	How to budget this activity
ESMF Monitoring of field work at divisions for ESMF actions in communities	Round trips	3000	300	900.000	4.0 (Local Labour Cost)	100 communities, 3 days per community, 20 km round trip from district town on average
Fire study	Days	45000	30	1.350.000	4.0 (Local Labour Cost)	Consultant contract with 30 days and 45500 PKRs/day
Fire Compensation Fund or insurance	Lumpsum	75.000.000,00	1	75.000.000,00	6.0 (Fire-fighting & compensation)	Requires further specification
Fire Response and mgt. Equipment	per Division	3461538	13	44.999.994	6.0 (Fire-fighting & compensation)	Firefighting guidelines to be developed and List of equipment for each Forest Division
Fire Mgt and response trainings	per Region	1000000	2	2.000.000	6.0 (Fire-fighting & compensation)	2 training per regions (2) inc. Peshawar
Fire response training on divisional training	per Division	600000	13	7.800.000	6.0 (Fire-fighting & compensation)	13 trainings one in each Forest Division
Stakeholder engagement	Round trips	6000	325	1.950.000	1.0 (Capacity Building)	13 FD, 5 years, 5 days per FD per year; cost per fieldwork day
Stakeholder workshops on Divisional level	Workshops	600000	52	31.200.000	1.0 (Capacity Building)	1 Workshop each year for 4 years per Forest Division,
Translator of ESMF related documents (Urdu)	(per page) Lumpsum	3000000	1	3.000.000	1.0 (Capacity Building)	Contract per translated page
Project Brief Leaflets	Copies	50	10000	500.000	1.0 (Capacity Building)	5 leaflets per planning unit for 2 years, printing costs
Print Outs of ESMF Guidelines	Copies	350	200	70.000	1.0 (Capacity Building)	N staff (province and district level)

Activity	Unit	Unit cost	Quantity	Amount (PKRs)	Budget Head (BTASP)	How to budget this activity
GRFM Cost	Lumpsum per year	400000	5	2.000.000	1.0 (Capacity Building)	need to be further specified Sessions, complaints, posters + effectiveness review/ Boxes / review visit
Communication with stakeholders via DFO office	per year	24000	65	1.560.000	1.0 (Capacity Building)	1 cell phone subscription per district, 2000 PKRs/per month, 12 month, 5 years
Travel costs for ESMF monitoring PMU and CDE&GED level	Days	18000	60	1.080.000	4.0 (Local Labour Cost)	6 days per semester, 200 km per day on average, plus accommodation and driver
Peace & Conflict study	Days	45000	30	1.350.000	4.0 (Local Labour Cost)	Consultant contract with 30 days and 45000 PKRs/day
PFRMP incl. community profile	per PU	240000	100	24.000.000	1.0 (Capacity Building)	1 PFRMP per PU, 4 staff, cost of fieldwork day (50%)
Gender Action Plan + Assessment	Days	45000	30	1.350.000	4.0 (Local Labour Cost)	Consultant contract with 30 days and 45000 PKRs/day
Mitigation Measures						
Rotational Grazing system training	PU	250000	100	25.000.000	4.0 (Local Labour Cost)	average across all Planning Units
Provision of Fodder Seed	PU	600000	100	60.000.000	4.0 (Local Labour Cost)	average across all Planning Units
Use of Organic pesticide instead of chemicals for nurseries and kitchen-garden	PU	1600	100	160.000	4.0 (Local Labour Cost)	average across all Planning Units

Activity	Unit	Unit cost	Quantity	Amount (PKRs)	Budget Head (BTASP)	How to budget this activity
Use of Biodegradable bags (piloting) for nursery raising	Number of bags	4	7000000	28.000.000	4.0 (Local Labour Cost)	About 50 % extra cost will be needed for Biodegradable bags instead of normal Plastic bags
Pre- and Post-harvest Training of NTFPs	PU	300000	100	30.000.000	4.0 (Local Labour Cost)	average across all Planning Units
Training of VDCs and WOs	PU	300000	100	30.000.000	1.0 (Capacity Building)	average across all Planning Units
TOTAL in PKRs				410.216.994		
TOTAL in EURO				1.367.390		

11. MONITORING

BTASP will apply project monitoring of all activities and ESMF requirements (chapter 10) at three levels:

1. First level Monitoring VDC/VO

At the **primary level**, the **VDC/VO** along with the local Forester/Forest Guard will conduct weekly or even daily monitoring in the case of plantation works. The VDC/VO will nominate 3 to 5 representatives and along with representatives of DFO Office (Forester/Forest Guard), they will conduct 100 % monitoring of all activities and will submit weekly report to DFO Office. A comprehensive but simple monitoring Proforma for the VDC/VO will be developed based on the DFO monitoring and reporting Proformas shown in Annex 6.1-6.5

2. Second Level Monitoring DFO Level

At **DFO level** (SDFO & CDO (M/F) will carry out monthly monitoring of the progress and will report to the PMU/PIC. The DFO office will also conduct a 100 % monitoring. The DFO office will report to PMU/PIC on monthly bases, which is also needed for the monthly financial report and fund request for the next month. Specimen of monitoring Performa are attached in Annex-6ff. However monthly reporting to PMU/PIC will be done according to PFRMP measures on PU level.

The DFO office for the **preliminary plantation inventory** and the **survival rate assessment** will carried out at the end of plantation and later on at annual bases. This inventory is also needed to determine the species composition, planning for replanting of failures as well as provide bases for 3rd party monitoring. For preliminary inventory the following methodology will be adopted see chapter 11.1. For the survival rate assessment, a standard method used already for 10BTAP will be applied see chapter 11.2.

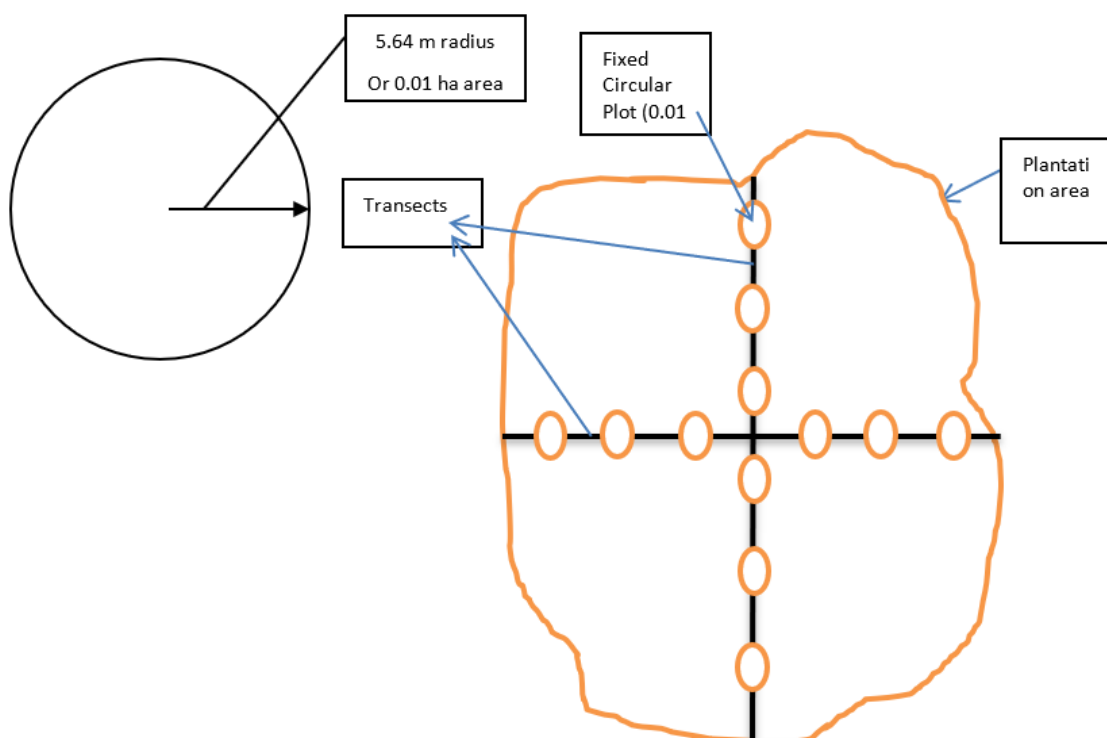
3. Third Level External Monitoring (3rd Party Monitoring)

Under the BTASP project an annual external monitoring will be carried out by a 3rd party. The organization for the 3rd party monitoring will be hired by PMU/PIC through comparative bidding process after the KfW approval of the PFRMP Implementation Guideline. The 3rd party will conduct 50 % monitoring of all activities except new block plantations, which will be 100%, linked with the Bonus payment system for new plantation. Along with other parameters, the 3rd party monitoring team will especially focus on internal monitoring system at DFO office level. At this time, it is proposed that the 3rd Party Monitoring also adopt the FD standard procedure for the survival rate assessment.

11.1 Preliminary forest inventory

For block plantations, woodlots, close plantations etc., a fixed area circular plot 5.64 m radius have an area of 0.01 ha or 100 m² will be executed to collect data on various parameters in the plot (species composition, total number of plants planted, survival rate, spacing, pit size and signs of damage) with a sampling intensity of 0.2 % i.e. one plot per 5 ha (20 plots will be taken from 100 ha plantation).

1. Layout two transects i.e. one vertical and one horizontal in such a manner that both cross each other approximately at the center of the plantation,
2. Circular plot with a radius of 5.64 m (1/100 ha) plots were used in forests and plantations.
3. One sample plot for every 5 ha were taken
4. Min 5 plots and Max 50 Plots
5. Information collected from Sample Plot (see below tally sheet)
 - Name of Plantation
 - GPS Coordinates
 - Specie Composition
 - Density



Tally Sheet

Village: _____

Name of Plantation: _____ Forest Division _____ Sub Division _____

Plot No: _____ Date: _____ Total area of Plantation: _____

GPS coordinated: _____ Legal Status _____

 Upper story / Trees:

S. No	Specie	Number/DBH	S. No	Specie	Number/DBH

11.2 Methodology of Survival Rate Assessment

Survival rate rates per year

As per SOPs framed for guidance of the field units for 10-BTTP, the below tabulated plant survival categories will be followed for calculating/assessing the survival percentage of various interventions.

Year	Plant Survival Category indicating Overall Condition					
Planting year	Outstanding	Excellent	Good	Average	Poor	Worst
1 st	90-100	80-89	71-79	60-70	40-59	Below 40
2 nd	85-100	75-84	65-74	50-64	35-49	Below 35
3 rd	75-100	65-74	55-64	45-54	30-44	Below 30

Table 24: Categories of survival rate per year

Overall Methodology: A combination of stratified random and systematic sampling techniques will be used. The area/ sites to be surveyed will be stratified according to the administrative, geographic and ecological characteristics and it will be ensured that all forest regions, forest divisions and ecological zones/ forest types are selected using stratified random sampling technique.

For **sampling plantations, wood lots and linear planting**, a circular is generally recommended. A survey pole and chain will be used to establish a circular plot with a radius of 7.9 m (1/50th ha). The circle will be visually divided into 4 equal parts for sapling count. For linear plantations 10 x 20 m rectangular quadrats will be established using survey pole and chain. In case the width of plantation is less than 10 m, then the width and length will be adjusted to establish 1/50th ha quadrats. The rectangular quadrats will be divided into 4 equal parts for sapling count. However, the final choice of the size and shape of the quadrat and its suitability for different techniques of afforestation will be made through field testing.

Number of Samples: The survey procedures are based on the assumption that seedling distributions are relatively homogenous within planted and seeded areas, and that areas where differences do exist will be grouped into **strata** before beginning a survey. While the number of samples are usually calculated using statistical techniques, however, in practice, normally **1 sample for every 2 ha** when using 1/50th ha quadrats.

Sampling Methods: Reconnaissance surveys: It is important to walk through the planted stand to verify size, shape, blank areas, and general observations on the health and condition of saplings. If necessary, stratify the planted stand into separate sampling units. Strata should be a minimum of 5 ha. Strata are identified where variations exist in species and soil moisture conditions. Topography, aspect, drainage class, and soil type are logical stratification variables. Moderate to steep slopes can also be used as a strata layer, if it is apparent that differences in regeneration exist. The strata will be roughly drawn on the map, and every stratum would be sampled separately.

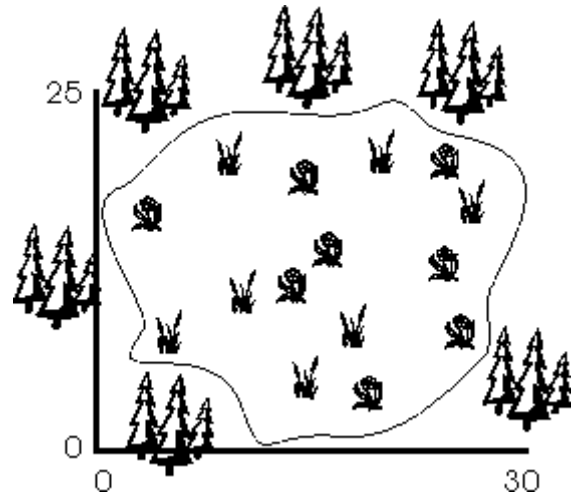
Locating the sample plot

It is important to eliminate personal bias in locating quadrats in the area to be sampled. The sample plot locations must be selected randomly across the area to be sampled. On a map of the area, divide the area into a grid with at least 2 times the area intersections within the boundary. Number these intersections and then using random number table, select the points for locating sample plots.

Counting Surviving Seedlings

In the sample plot data on various parameters in the plot is been collected. The data include: species composition, total number of plants planted, survival rate, spacing, pit size and signs of damage)

The survey sheet is available from the 10 BTTP.



11.3 Monthly financial reporting

The financial reporting is stipulated in the Financial Management Guidelines. All thirteen BTASP forest division will submit the monthly financial report to PMU/PIC. The financial report will be submitted separately at PU level along with accumulative report for the entire Forest Division. Detail of financial reporting including formats is available in Financial Management Guidelines.



Annex 1 Cost Norms BTASP

The cost norms of BTASP are documented in a separate Excel file attached to this report.

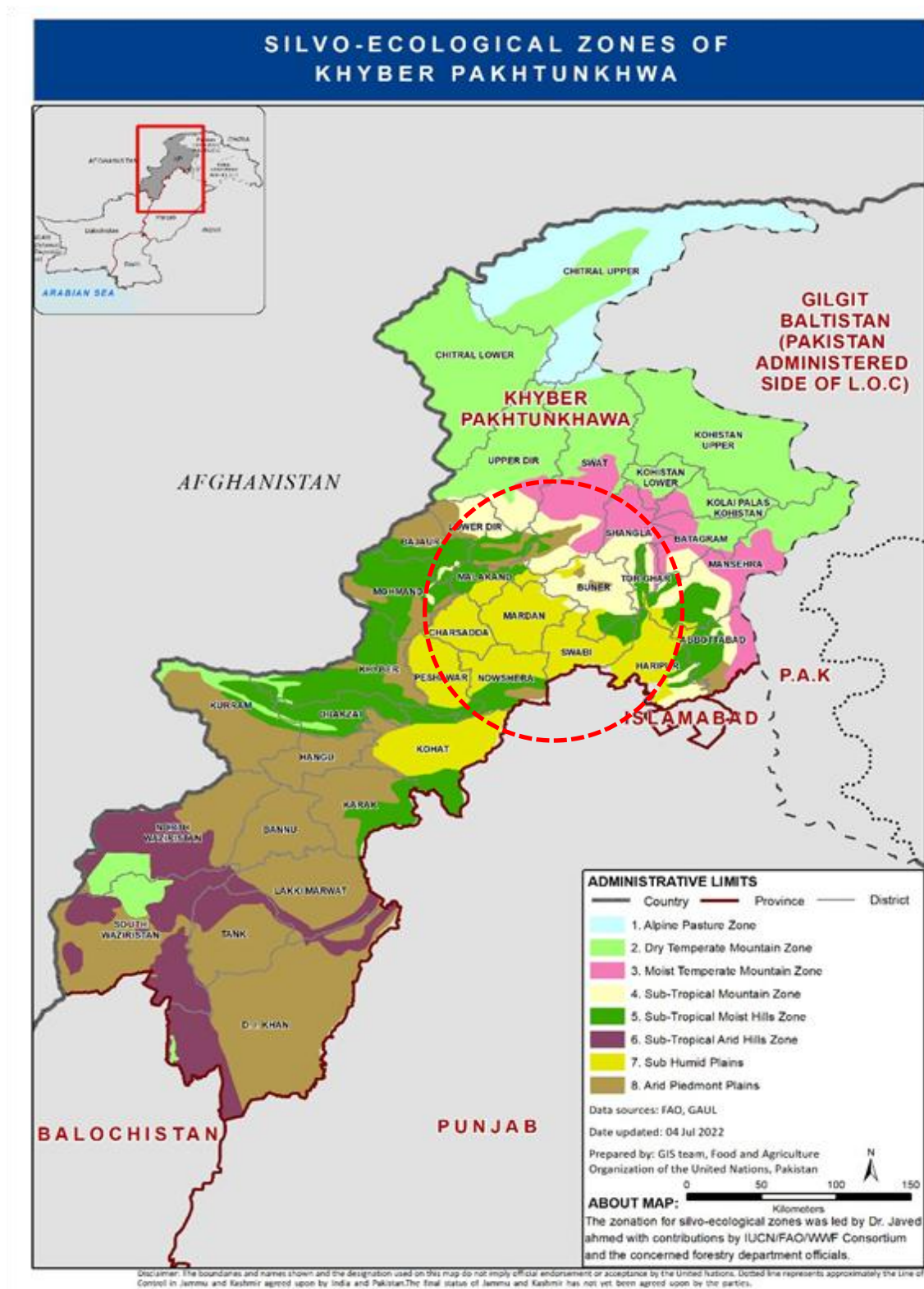
Annex 2 Plantation and Enclosures Journal

The plantation and enclosures Journal is a separate document to be used for documentation of planning and monitoring of PU forest sites as part of the Implementation Guideline.

Annex 3 BTAP and 10BTTAP plantation and enclosure details for 100 Planning Units

The Excel file is added to this report as a separate document.

Annex 4 Map of Silvo-Ecological Zones in KP



Map: Silvo-Map Ecological Zones in KP;

Source: Guidelines and Lists of Suitable Tree Species for Different Silvo-Ecological Zones in Pakistan; IUCN/FAO/WWF August, 2022

Annex 4.1: Species List to be used in BTASP

List of Species for Tropical Zone

Scientific Name	Local Name	Native/Exotic
Main Species		
<i>Dalbergia sissoo</i>	Shisham	Native
<i>Albizia lebbek</i>	Black Siris	Native
<i>Bombax ceiba</i>	Simal	Native
<i>Terminalia arjuna</i>	Arjun	Native
Subsidiary Species		
<i>Ficus carica</i>	Fig	Native
<i>Millettia pinnata</i>	Sukh Chain	Exotic
<i>Moringa oleifera</i>	Sohanjna	Native
<i>Syzygium cumini</i>	Jamun	Native
<i>Senegalia modesta</i>	Phulai	Native
<i>Tamarix aphylla</i>	Frash	Native
<i>Ziziphus mauritiana</i>	Ber	Native
<i>Gmelina arborea</i>	Gumhar	Exotic
<i>Parkinsonia aculeata</i>	Parkinsonia	Exotic
<i>Eucalyptus camaldulensis</i>	Sufeda, Lachi	Exotic

Table 25: List of species for Tropical Zone

List of Species for Sub-Tropical Broadleaved Evergreen

Scientific Name	Local Name	Native/Exotic
Main Species		
<i>Olea ferruginea</i>	Kahu	Native
<i>Acacia modesta</i>	Phulai	Native
Subsidiary Species		
<i>Bombax ceiba</i>	Simal	Native
<i>Bauhinia variegata</i>	Kachnar	Native
<i>Cassia fistula</i>	Amaltas	Native
<i>Cordia dichotoma</i>	Lasuri	Native
<i>Cordia myxa</i>	Lasura	Native
<i>Dalbergia sissoo</i>	Shisham	Native
<i>Morus alba</i>	Mulberry	Native
<i>Phyllanthus emblica</i>	Amla	Native
<i>Ailanthus altissima</i>	Ailanthus	Naturalized
<i>Robinia psuedoacacia</i>	Robinia	Naturalized
<i>Sapindus mukorossi</i>	Ritha	Exotic
<i>Eucalyptus camaldulensis</i>	Sufeda, Lachi	Exotic

Table 26: List of Species for Sub-Tropical Broadleaved Evergreen

List of Species for Sub-Tropical Chir Pine

Scientific Name	Local Name	Native/Exotic
Main Species		
<i>Pinus roxburghii</i>	Chir pine	Native
<i>Quercus incana</i>	White oak, bunj	Native
Subsidiary Species		
<i>Alnus nitida</i>	West Himalayan alder	Native
<i>Bauhinia variegata</i>	Kachnar	Native
<i>Bombax ceiba</i>	Simal	Native
<i>Cordia dichotoma</i>	Lasuri	Native
<i>Cordia myxa</i>	Lasura	Native
<i>Dalbergia sissoo</i>	Shisham	Native
<i>Ficus palmate</i>	Injeer	Native
<i>Leucaena leucocephala</i>	Ipil Ipil	Native
<i>Morus alba</i>	Mulberry	Native
<i>Phyllanthus emblica</i>	Amla	Native
<i>Pistacia integerrima</i>	Guli-pista	Native
<i>Salix tetrasperma</i>	Indian Willow	Native
<i>Sapindus mukorossi</i>	Ritha	Exotic
<i>Terminalia belerica</i>	Bahera	Native
<i>Punica granatum</i>	Annar	Native
<i>Ziziphus jujube</i>	Unab	Native
<i>Ailanthus altissima</i>	Ailanthus	Naturalized
<i>Robinia psuedoacacia</i>	Robinia	Naturalized
<i>Sapindus mukorossi</i>	Ritha	Exotic
<i>Eucalyptus camaldulensis</i>	Sufeda, lachi	Exotic

Table 27: List of Species for Sub-Tropical Chir Pine

List of Species for Moist Temperate Forest

Scientific Name	Local Name	Native/Exotic
Main Species		
<i>Abies pindrow</i>	Fir	Native
<i>Pinus wallichiana</i>	Blue pine	Native
<i>Picea smithiana</i>	Spruce	Native
<i>Taxus baccata</i>	Burmi	Native
<i>Quercus dilatata</i>	Barungi	Native
<i>Quercus glauca</i>	Banni	Native
Subsidiary Species		
<i>Acer oblongum</i>	Panhgor	Native
<i>Aesculus indica</i>	Bankhor	Native
<i>Juglans regia</i>	Walnut	Native

Table 28: List of Species for Moist Temperate Forest

List of species for Dry Temperate Forest

Scientific Name	Local Name	Native/Exotic
A. Main Species		
<i>Cedrus deodara</i>	Deodar	Native
<i>Pinus gerardiana</i>	Chilgoza	Native
<i>Pinus wallichiana</i>	Kail, Biar, Blue pine	Native
<i>Quercus semecarpifolia</i>	Brown oak	Native
<i>Quercus baloot</i>	Holy oak	Native
<i>Salix spp.</i>	Willow	Native
B. Subsidiary Species		
<i>Acer caesium</i>	Maple	Native
<i>Aesculus indica</i>	Indian Horse chestnut	Native
<i>Elaeagnus angustifolia</i>	Russian Olive	Native
<i>Fraxinus hookeri</i>	Ash	Native
<i>Juglans regia</i>	Walnut	Native
<i>Morus nigra</i>	Black Mulberry	Native
<i>Morus alba</i>	Mulberry	Native
<i>Pistacia khinjuk</i>	Khinjuk	Native
<i>Populus ciliata</i>	Himalayan popular	Native
<i>Prunus cornuta</i>	Himalayan bird cherry	Native
<i>Celtis caucasica</i>	Honey Berry	Native
<i>Robinia pseudoacacia</i>	Black Locust	Naturalized
<i>Salix babylonica</i>	Weeping willow	Naturalized
<i>Ailanthus altissima</i>	Ailanthus	Exotic
<i>Gleditsia triacanthos</i>	Honey locust	Exotic

Table 29: List of species for Dry Temperate Forest



Annex 5 Agreements

Annex 5.1 Agreement for Block Plantation

Agreement for Block Plantation with VDC/WO, management committee and DFO office

THIS AGREEMENT for the Block Plantation intervention of the BTASP (hereinafter the “Agreement”) is being entered on this ____ day of _____, 20..... (Hereinafter the “Signing Date”) by and between:

Forest Department through The Divisional Forest Officer (hereinafter referred to as “**Party A**”)

The Management committee established for the specific intervention /activities within the village jurisdiction in collaboration with VDC. (hereinafter referred to as “**Party B**”) And,

Village development communities (VDC) / Women Organization (WO) (hereinafter referred to as “**Party C**”)

(Party A, Party B and C shall hereinafter be collectively referred to as “Parties” and individually as Party)

This Agreement is being entered into by and between the Parties for the purpose of implementation of the Block Plantation activity in the Planning Unit _____/ total area _____ mentioned in the operational plan of the planning unit (appended herewith to this Agreement as “Annex ‘A’”) under the BTASP Project of the Khyber Pakhtunkhwa Forest Department. The role and responsibilities and terms and conditions of this agreement are as follows:

1. **TERM**

i. The term of this Agreement shall commence from the date _____, 20..... and shall continue for a period of five (_____) years (hereinafter the “Term”).

2. **SCOPE OF THE AGREEMENT**

This Agreement will be a legal document between the Management Committee, VDC/ WO of the Planning unit _____ and Divisional Forest office _____ for the implementation of the Block Plantation prescribed in the operational plan of Participatory Forest Resource Management Plan of Planning unit _____,

3. **ACTIVITIES**

The Block Plantation activities to be undertaken under this agreement are specified in the 5-year PFRMP of the _____ Planning Unit, which includes the detailed operational plan for each intervention:

4. Roles, Responsibilities and obligations of the contractual parties

i. **Obligations of “Party A”**

a) **Monitoring, Supervision**

- Monitoring of the application of ESMF especially the risk analysis and mitigation mechanisms, the use of child labour, equal pay between men and women, proper consultation with the land users, FPIC, the use of harmful fertilizers, etc.
- Provide payment for labor and planting stock along with transportation,
- Pay the watch and ward for three years as per the operational plan of PFRMP,
- Technical layout of planting design, plantation model and species composition according to the PFRMP,
- Provide technical assistance on implementation of block plantation to the VDC / Management committee,
- Provide orientation to the Management Committee and VDC/WO on the implementation guidelines for compliance,
- Monitor the plantation on semi-annual bases and report to the PMU,
- Survival rate assessment on annual basis,
- Keep the record of the implementation,
- Provide the payroll list to party C at the end of block plantation,
- Develop a plantation journal for the block plantation with sustainable control use for 20 years.

ii. **Obligation of “Party B”**

- Allocate the required land free of cost prescribed in the PFRMP for the block plantation,
- Shall agree with the sustainable control use of the block plantation as prescribed in the plantation journal,
- Shall agree on the 10 % contribution from the cash revenue generated of the block plantation to the NRM funds,
- Shall agree not to change the land use of the allocated land for up to 20 years.

iii. **Obligations of “Party C”**

- Ensure community contribution from the management committee (party B) as prescribed in the PFRMP,
- Shall jointly with the Management Committee (party B) identify the labors for plantation activity and provide a list of the identified labors with the required information (Name, Father Name, Village and copy of CNIC and) to party A,
- Shall notify a person from the VDC for monitoring of the plantation progress and submission of the completion report to the party A,
- Shall identify the Chowkidar and ward persons jointly with the management committee (party B) and shall share the list along with the required information with party A.

Performance based bonus

The performance-based bonus is a payment for survival rate of the block plantation. The bonus is going to be paid on annual basis for up to three years, on the third-party monitoring report as per the table given below;

GENERAL COMMITMENT

- i. The Parties agree and acknowledge that this Agreement cannot be used as documentary evidence by any Party to assert their rights locally and before the courts for their internal disputes not related to this Agreement.
- ii. The Parties agree and acknowledge that they will ensure compliance with ESMF framework of the BTASP.

6. CONDITIONS OF THE AMENDMENT AND EXTENSION OF THE AGREEMENT

- i. Either Party may amend the Agreement if required. However, such amendments must be mutually agreed by all the Parties in writing.
- ii. After 5-years implementation, the agreement may be extended with mutual consent of all the Parties.
- iii. By the end of the project (.....), the BTASP project will not have any financial obligations

7. TERMINATION OF THE AGREEMENT

- i. If Party A wishes to terminate the Agreement, they may invoke Rule 22 of the Joint Forest Management (Community Participation), Rules, 2004 ("**CP Rules**").
- ii. If Party B wishes to terminate the Agreement, they may invoke Rule 19(4) read with Rule 11(f) of the **CP Rules**, 2004.
- iii. Neither Party may unilaterally terminate the Agreement without first seeking to resolve any disputes amicably.

8. DISPUTE RESOLUTION

- i. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement.
- ii. In the event where any dispute, difference or question arises out of or in connection with this Agreement, and / or arising due to a breach of this Agreement, the aggrieved Party shall promptly serve upon the other Party a notice stating such disputes, differences or questions (the "Dispute Notice"). Upon the receipt of the Dispute Notice by the other Party, the Parties shall use their best efforts to amicably resolve such disputes, differences or questions through mutual consultation and negotiation within [30 (thirty)] days from the receipt of the same. If the Parties are unable to resolve such disputes, differences or questions within the above stipulated 30 (thirty) days period, the matter shall be referred to mediation.

-
- iii. In the event where any dispute, controversy or claim arising out of or relating to this Agreement or the breach, termination or invalidity thereof cannot be amicably settled through mutual consultation and negotiation, the Parties shall refer the same to BTASP PMU who shall nominate two (2) mediators. The Mediators shall attempt to resolve the dispute within [thirty (30)] days from the date of the appointment of the mediators.

The decision of the Mediators shall be final and binding.

iv. Contact person for dispute or GRM within the BTASP:

Until the rollout of the GRFM, all complaints and feedback related to the shall be directed towards the following BTASP project office contact for feedback and complaints:

Project Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009591191 ,
Email: pd.btaspforest@kp.gov.pk

Deputy Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009596367,
Email: ibrahimrazmian@gmail.com,

10. DISPUTE RESOLUTION WITHIN THE COMMUNITIES

- a. If any community member (man, woman or child; residential or semi-residential) feels disproportionately disadvantaged because of this Agreement, the community member may refer to the following procedure:
- a. **Step 1:** Presentation of issue to VDC/VO, Management Committee and SDFO/CDO (m/f). The person concerned or his/her representative can ask to discuss his/her case in a meeting of the VDC/VO, management committee and the SDFO/CDO. This request can be in oral or written form. If the case is not satisfactorily solved, a next step can be taken on initiative of the community member. (Time limit for redressal at this stage: **seven (7) Days**)
 - b. **Step 2: Presentation of the issue to the DFO:** The person concerned or his/her representative can ask to discuss his/her case in a meeting with the DFO and the (AD)CDE&GAD (m/f). This request can be in oral be or written form. If the case is not solved satisfactorily, a next step can be taken on initiative of the community member. (Time limit for redressal at this stage- (fourteen)14 Days)
 - c. **Step 3: Presentation of the issue to the CF:** The person concerned or his/her representative can ask to discuss his/her case in a meeting with the CF and PD or authorized representative from PMU and Director or Deputy Director CDEGAD. This request shall be in written form. The decision of this forum is final. (Time limit for redressal at this stage thirty (30) days)

11. PERFORMANCE BASED PAYMENT PROCEDURE

Beside Provision of Planting Stock and Transportation from Nursery to Plantation Site By BTASP, The VDC Will Be Paid **Rs. 31** (10x10 feet spacing i.e. 1075 plants per ha) and Rs. 45 (with 8 x 8 feet spacing i.e. 1680 plants per ha) Per Plant for Digging of Pits, Planting and Internal Carriage. The payment will be made in three instalments as given below:

For 10 x 10 feet spacing (1075 plants per ha)

Procedure for payment of Instalment to the VDC			
Cost per Plant 31 PKRs for Survived Plant			
Instalment	1st	2nd	3rd
Milestone	Mobility cost	After DFO Monitoring	After 3rd party Monitoring
Amount (PKRs) /plant	15	8	No. of Survived plants x 31 - (1st and 2nd Instalment)
%age of Labor Cost	48	26	26

For 8 x 8 feet spacing (1680 plants per ha)

Procedure for payment of Instalment to the VDC			
Cost per Plant 45 PKRs for Survived Plant			
Instalment	1st	2nd	3rd
Milestone	Mobility cost	After DFO Monitoring	After 3rd party Monitoring
Amount (PKRs) /plant	21	12	No. of Survived plants x 45 - (1st and 2nd Instalment)
%age of Labor Cost	48	26	26

SIGNATURE

Signature Party-A

Divisional Forest Officer_____

Dated_____

SDFO/RFO_____

Dated_____

CDO (M/F)_____

Dated_____

Signature Party-B

Representative Name: _____

Position: _____

CNIC: _____

Signature: _____

Date: _____

Signature Party-C

Chairman of the VDC

Name:

CNIC:

Date:

Chairperson of the WO

Name:

CNIC:

Date:

Vice Chairman of the VDC

Name:

CNIC:

Date:

Vice Chairperson of the WO

Name:

CNIC:

Date:

Treasurer of the VDC

Name:

CNIC:

Date:

Treasurer of the WO

Name:

CNIC:

Date:



Annex 5.2 Contract of Forest Warden

Agreement between Forest Warden with VDC/VO, management committee and DFO office

This contract is for the protection and watch and ward of the Block Plantation in planning unit (_____) (hereinafter the “contract”) is being entered on this ____ day of _____, 20..... (Hereinafter the “Signing Date”) by and between:

Forest Department through the Divisional Forest Officer (hereinafter referred to as “Party A”), and the individual forest warden _____ S/O _____ bearing CNIC: _____ (Name / Father name and CNIC) nominated by the VDC as per the criteria for this specific purpose within the village jurisdiction (hereinafter referred to as “Party B”) And, Village development communities (VDC) / Women Organization (WO) (hereinafter referred to as “Party C”)

(Party A, Party B and C shall hereinafter be collectively referred to as “Parties” and individually as Party)

This contract agreement is being entered into by and between the Parties for the purposes of protection, watch and ward of the block plantation in Planning Unit _____/ location _____ mentioned in the operational plan of the planning unit (appended herewith to this Agreement as “Annex ‘A’”) under the BTASP Project of the Khyber Pakhtunkhwa Forest Department. The roles, responsibilities, terms and conditions of this agreement are as follows:

1. TERM

i. The term of this Contract Agreement shall commence from the _____, 20..... and shall continue for a period of five () years (hereinafter the “Term”).

2. SCOPE OF THE AGREEMENT

This Contract Agreement will be a legal document between Party B, Party C of planning unit _____ and Divisional Forest officer _____ for the effective forest conservation, protection, and sustainable forest management within the Planning unit _____.

3. ACTIVITIES

The forest warden will perform the activities under this agreement as specified in the 5-year PFRMP of the _____ Planning Unit.

4. Roles, Responsibilities and obligations of the contractual parties

i. **Obligations of “Party A”**

- Monitoring of the application of ESMF especially the risk analysis and mitigation mechanisms, the use of child labour, equal pay between men and women, proper consultation with the land users, FPIC, the use of harmful fertilizers, etc.
- Pay the salary of the Negheban (**men or women**) through cross cheque for the period of years
- Provide trainings to the Negheban on watch and ward, monitoring and Fire management techniques
- Keep the record of the Negheban in the respective Planning unit file to be available for third party monitoring, if needed
- Provide the necessary tools, equipment's, and resources to Negheban to perform his/her duties effectively (e.g., uniforms, safety gear, fire management tools).
- Monitoring of the application of ESMF especially the risk analysis and mitigation mechanisms, the use of child labour, equal pay between men and women, proper consultation with the land users, FPIC, the use of harmful fertilizers, etc.
-

ii. **Obligations of “Party B”**

The Negheban agree to undertake the following tasks and follow the requirements of the SOPs listed below, ESMF requirements and PFRMP actions.

- **Patrolling:** Conduct daily patrols within the designated forest areas to prevent illegal activities such as logging, encroachment and grazing.
- **Monitoring:** Monitor the health and growth of trees and natural regeneration in enclosures managed by the VDC ensuring that the plant survival targets are met.
- **Incident Reporting:** Report any incidents of illegal activity, fire or environmental damage to the VDC and the local Forest Department.
- **Fire Management:** Assist in fire management and prevention measures as outlined by the VDC and CCFEWD protocols.
- **Community Engagement:** Liaise with local communities and report on any community concerns, including disputes related to forest use and management.
- **Maintenance Tasks:** Carry out tasks such as clearing paths, maintaining enclosure, and performing necessary upkeep of forest areas under the VDC's/Forest department supervision.

iii. **Obligations of “Party C”**

- Will identify the Negheban jointly with the management committee and will share the list of the proposed Negheban with the party A, with details including (Name /Father Name, Khail / Tribe, Location, CNIC Number and Account number etc.) through proper resolution of the VDCs duly Signed and Stamped by the Chairman and Vice Chairman of VDC/WO.
- Shall monitor the performance of the Negheban and will provide feedback to party A.
- Facilitate communication between the Negheban, the CCFEWD and local community members, ensuring transparency in forest protection activities.

5. GENERAL COMMITMENT

- i. The parties agree and acknowledge that this Agreement cannot be used as documentary evidence by any Party to assert their rights locally and before the courts for their internal disputes not related to this Agreement.
- ii. The Parties agree and acknowledge that they will ensure compliance with ESMF framework of the BTASP.

6. COMPENSATION AND PAYMENT STRUCTURE:

- I. The forest warden (men or women) will receive a monthly salary of PKR [30,000], to be paid at the end of each month.
- II. **Performance Based Incentives:** In addition to the salary, performance-based financial or non-financial incentives may be offered for achieving specific goals, such as:
PKR [20 % of overall Bonus for plantation] for maintaining plant survival rates above [80 % after 1st year] & [75 % after 2nd Year] for the duration of the agreement for new plantation under BTASP only. Non-financial incentives may cover recognition rewards; additional capacity building; additional access to resources; enhanced terms and conditions.
- III. **Payment Method:** Payments will be made through cheque's bank transfer to the forest warden.

7. PERFORMANCE METRICS AND EVALUATION:

The forest warden performance will be evaluated based on the following criteria:

- **Patrol Frequency:** Number of patrols completed per week as per the agreed schedule.
- **Incident Response:** Timeliness and accuracy in reporting incidents of illegal activities, fire or other threats to forest health.
- **Plant Survival Rates:** Achievement of plant survival rates in designated enclosures or plantations (e.g., 40% for arid regions, 90% for irrigated regions).
- **Community Feedback:** Feedback from local community members regarding the Negheban role in preventing illegal activities and resolving conflicts.
- Quarterly performance reviews will be conducted by the VDC and feedback will be provided to the Negheban. Poor performance may lead to additional training or reassignment.

6. CONDITIONS OF THE AMENDMENT AND EXTENSION OF THE AGREEMENT

- i. If the VDC/WO is not satisfied with the forest warden performance, they may issue a one-month notice for improvement. If there is no improvement within that period, the VDC/WO will nominate a replacement and send a resolution to the FD to terminate the forest warden contract and finalize the new appointment.
- ii. By the end of the project (.....) the BTASP project will not have any financial obligations.

7. TERMINATION OF THE AGREEMENT

- i. Party A will terminate this contract if the party A is not satisfied with the performance of Party B or violated the rules or misconduct. The Party A will give one month termination notice.
- ii. If party B wants to terminate the Agreement, he / she will give a 1-month prior notice to the Party A with a solid reason
- iii. Neither of the Parties may unilaterally terminate the Agreement without first seeking to resolve any disputes amicably.

8. DISPUTE RESOLUTION WITHIN BTASP

Contact person for dispute or GRM within the BTASP:

Until the rollout of the GRFM, all complaints and feedback related to the shall be directed towards the following BTASP project office contact for feedback and complaints:

Project Director BTASP,

Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,

Tel: 923009591191 ,

Email: pd.btasppforest@kp.gov.pk

Deputy Director BTASP,

Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,

Tel: 923009596367,

Email: ibrahimrazmian@gmail.com,

9. SOP for Neghaban- Mgt. Committee and VDC/WO

Preamble:

Patrolling personnel play a vital role in safeguarding the forests. This Code of Conduct outlines the ethical and professional standards expected of all personnel engaged in patrolling activities. Adherence to this code is crucial for maintaining the integrity of our forests, ensuring the safety of personnel, and fostering positive relationships with local communities.

The forest warden are recruited persons from within the community to carry out forest work and patrolling. Forest wardens are not part of the Forest Divisional staff structure and part of the official payroll. Forest warden are either financed by forest projects or community structures to safeguard portions of forest. The forest warden do not have any power of force nor carry any weapons.

I. Core Principles:

Integrity:	Act with honesty, transparency, and accountability in all actions. Avoid any form of corruption, bribery, or abuse of power.
Respect:	Treat all individuals, including colleagues, community members, and visitors, with respect and dignity, regardless of their background.
Professionalism:	Maintain a high standard of conduct, appearance, and performance. Be punctual, prepared, and diligent in carrying out assigned duties.
Safety:	Prioritize personal safety and the safety of others. Follow all safety protocols and use equipment responsibly.
Fairness:	Enforce regulations fairly and impartially. Avoid any form of discrimination or bias.
Confidentiality:	Maintain the confidentiality of sensitive information related to patrolling operations and investigations.

II. Specific Conduct Guidelines:

Use of force:	The forest warden does not possess the power of force and shall exercise restraint and de-escalation techniques whenever possible.
Interaction with the Public:	Engage with the public in a courteous and respectful manner. Provide accurate information and assistance when needed.
Reporting:	Accurately and promptly report all incidents, observations, and activities related to patrol duties to the VDC or Mgt. Committee.
Equipment Use:	Use all assigned equipment responsibly and according to established procedures. Maintain equipment in good working order.
Communication:	Maintain clear and effective communication with colleagues, supervisors, and other stakeholders.
Personal Conduct:	Refrain from any behaviour that could tarnish the reputation of the VDC, Mgt. Committee or the KP Forest Department. This includes substance abuse, misconduct, and any illegal activities.
Conflict of Interest:	Avoid any situation that could create a conflict of interest. Disclose any potential conflicts to supervisors.

III. Consequences of Violations:

Failure to comply with this Code of Conduct may result in disciplinary action, up to and including termination of employment.

Standard Operating Procedure (SOP) for Patrolling Personnel

I. Purpose:

This SOP outlines the standard procedures to be followed by patrolling personnel during forest patrols. Its purpose is to ensure consistency, effectiveness, and safety in patrol operations.

II. Pre-Patrol Procedures:

- **Equipment Check:** Ensure all necessary equipment is in good working order, including communication devices, first aid kits, navigation tools, and personal protective equipment.
- **Communication and Monitoring Check:** Test communication or Monitoring devices to ensure they are functioning properly.

III. Patrol Procedures:

- **Patrol Route:** Follow assigned patrol routes and document any deviations.
- **Observation:** Be observant of any signs of illegal activity, such as poaching, logging, or encroachment.
- **Data Collection:** Collect and record relevant data, including the location, time, and nature of observations. Use provided forms or electronic devices.
- **Evidence Gathering:** If evidence of illegal activity is found, carefully collect and preserve it according to established procedures.
- **Incident Reporting:** Report all incidents, observations, and activities to the designated supervisor promptly.
- **Interaction with Suspects:** If encountering suspected illegal actors, maintain a safe distance and observe their behavior. Do not engage unless it is safe to do so. Contact VDC/ Mgt. Committee members or law enforcement if necessary.
- **Community Engagement:** Engage with local communities to gather information and build relationships.
- **Navigation:** Use appropriate navigation tools and techniques to stay oriented and on course.
- **Communication:** If possible, maintain regular communication when in the field.

IV. Post-Patrol Procedures:

- **Debriefing:** Participate in post-patrol debriefings to discuss observations, challenges, and recommendations.
- **Report Submission:** Submit patrol reports and any collected evidence to the designated person.
- **Equipment Maintenance:** Clean and maintain all used equipment.
- **Emergency Procedures:**
 - **Medical Emergencies:** Follow established procedures for handling medical emergencies, including administering first aid and contacting emergency services.
 - **Security Incidents:** In case of security incidents, such as encounters with armed individuals, prioritize personal safety and contact law enforcement immediately.
 - **Communication Failure:** If communication devices fail, follow pre-determined contingency plans.
 -

VI. Training and Review:

- All patrolling personnel will receive thorough training on this SOP and the Code of Conduct.
- This SOP will be reviewed and updated periodically to ensure its effectiveness and relevance.

VII. Enforcement:

Supervisors are responsible for ensuring that patrolling personnel adhere to this SOP. Any deviations from the SOP will be addressed appropriately. This Code of Conduct and SOP provide a framework for ethical and effective patrolling activities. Adherence to these guidelines is essential for the protection of our forests and the safety of our personnel.

The SOPs describe the standard operation procedures between the forest warden, the VDC/WO, the Management Committee and the forest guard.

1. The forest warden, once contracted, shall receive a forest warden ID card, for clear identification as forest guard towards outsiders,
2. The forest warden will receive a set of equipment (see ESMF budget) incl. a vest to easily identify the forest warden from a distance, hand gloves and boots for protection during patrol and fire events.

Fire outbreak: in the event of fire, FW will contact and/or report immediately to the VDC chairperson and the Mgt. Committee chairperson; The FW will also inform the forest guard, who then will inform the DFO office. The immediate report shall contain location and size of fire and wind and fire directions; the FW will observe the fire development. The VDC chairperson will mobilise support from the community for firefighting and will deploy the fire patches provided by BTASP.

Encroachment by animals: in the event of animal encroachment, the FW will confine the animals in a secure location and will contact and/or report immediately to the VDC chair and forest guard.

Illegal logging: in the event of illegal logging, FW will try to take pictures from distance and will contact VDC, Mgt. Committee and the forest guard. The Mgt. Committee representative will contact police for further investigation.

Identified calamities: in the event of wind break, pest, flooding's, lands slides or any other calamity occurred on site the FW will report and document to the Management Committee. The Mgt. committee will seek advice and discuss possible solutions with the forest guard and forester.

Overuse of agreed use quotas per forest site Journal (Management Plan): in the event of identified over use of forest resources the forest warden will document and report to the Management committee. The Mgt. committee will then have to take actions towards the involved persons.

Observed replanting needs: in the event of identified planting need the FW will inform the Management Committee. The Mgt. committee will then after seeking advice of the forest guard or forester decide how, when and area size of seed sourcing or replanting shall be done. The Mgt. Committee has to provide the financial or in-kind inputs required and can request the FW for seed collection from the ground or lower branches.

SIGNATURE

Signature Party-A

Divisional Forest Officer_____

Dated_____

SDFO/RFO_____

Dated_____

CDO (M/F)_____

Dated_____

Signature Party-B

Individual beneficiary Name: _____

CNIC: _____

Signature: _____

Date: _____

Signature Party-C

Chairman of the VDC

Name: _____

CNIC: _____

Date: _____

Chairperson of the WO

Name: _____

CNIC: _____

Date: _____

Vice Chairman of the VDC

Name: _____

CNIC: _____

Date: _____

Vice Chairperson of the WO

Name: _____

CNIC: _____

Date: _____

Treasurer of the VDC

Name: _____

CNIC: _____

Date: _____

Treasurer of the WO

Name: _____

CNIC: _____

Date: _____



Annex 5.3 Agreement for Wood Lot

Agreement for Wood Lot Activities with VDC/WO, management committee and DFO office

This agreement for the wood lot intervention of the BTASP (hereinafter the “Agreement”) is being entered on this ____ day of _____, 20____ (Hereinafter the “Signing Date”) by and between: Forest Department through Divisional Forest Office(hereinafter referred to as “Party A”) The individual owner (Name / Father name and CNIC) nominated as the criteria for these specific intervention /activities within the village jurisdiction in collaboration with VDC. (hereinafter referred to as “Party B”) And, Village development communities (VDC) / Women Organization (WO) (hereinafter referred to as “Party C”)

(Party A, Party B and C shall hereinafter be collectively referred to as “Parties” and individually as Party)

This Agreement is being entered into by and between the Parties for the purposes of implementation of the wood lot activity in Planning Unit _____/ location _____ mentioned in the operational plan of the planning unit (appended herewith to this Agreement as “Annex ‘A’”) under the BTASP Project of the Khyber Pakhtunkhwa Forest Department. The role and responsibility and terms and conditions of this agreement are as follows:

1. TERM

i. The term of this Agreement shall commence from the _____, 20____ and shall continue for a period of five () years (hereinafter the “Term”).

2. SCOPE OF THE AGREEMENT

This Agreement will be a legal document between Party B, Party C of Planning unit _____ and Divisional Forest officer _____ for the implementation of the wood lot activities prescribed in the operational plan of Participatory Forest Resource Management Plan of Planning unit _____.

3. ACTIVITIES

The wood lot activities to be undertaken under this agreement are specified in the 5-year PFRMP of the _____ Planning Unit, which includes the detailed operational plan for each intervention:

4. Role, Responsibilities and obligation of the contractual parties

iv. Obligations of “Party A”

- Monitoring of the application of ESMF especially the risk analysis and mitigation mechanisms, the use of child labour, equal pay between men and women, proper consultation with the land users, FPIC, the use of harmful fertilizers, etc.
- Will provide suitable species for woodlot,
- Will provide technical assistance on implementation/execution of wood lot activity to the individual owner jointly with the VDC / WO,
- Shall provide orientation to the individual owner and VDC/WO on the implementation guidelines for compliance,
- Shall keep the record of the woodlot activity.

•

v. **Obligation of “Party B”**

- Party B shall agree with the implementation guidelines and sustainable control of wood lot as prescribed in the plantation journal,
- Shall agree with the 10 % contribution for the NRM fund,
- Shall be responsible for watch and ward of the wood lot.

vi. **Obligations of “Party C”**

- VDC and WO will identify the beneficiaries as per the prescribe criteria and provide the list of individuals / owners with details including (Name /Father Name, Khail / Tribe, Location, CNIC Number etc.)
- Shall ensure community contribution from the beneficiary (party B) as prescribed in the PFRMP
- Shall monitor the progress of the wood lot and provide feedback to Party A

5. GENERAL COMMITMENT

- iii. The Parties agree and acknowledge that this Agreement cannot be used as documentary evidence by any Party to assert their rights locally and before the courts for their internal disputes not related to this Agreement.
- iv. The Parties agree and acknowledge that they will ensure compliance with ESMF framework of the BTASP.

10. CONDITIONS OF THE AMENDMENT AND EXTENSION OF THE AGREEMENT

- iii. Either of the Party may amend the Agreement, if required. However, such amendments must be mutually agreed by all parties in writing.
- iv. After 5-years implementation the agreement may be extended with mutual consent of all Parties.
- v. By the end of the project (.....) the BTASP project will not have any financial obligations.

11. TERMINATION OF THE AGREEMENT

- iv. If Party A wishes to terminate the Agreement, they may invoke Rule 22 of the Joint Forest Management (Community Participation), Rules, 2004 (“**CP Rules**”).
- v. If Party B wishes to terminate the Agreement, they may invoke Rule 19(4) read with Rule 11(f) of the **CP Rules**, 2004.
- vi. Neither of the parties may unilaterally terminate the Agreement without first seeking to resolve any disputes amicably.

12. DISPUTE RESOLUTION

- i. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement.
- ii. In the event where any dispute, difference or question arises out of or in connection with this Agreement, and / or arising due to a breach of this Agreement, the aggrieved Party shall promptly serve upon the other Party a notice stating such disputes, differences or questions (the "Dispute Notice"). Upon the receipt of the Dispute Notice by the other Party, the Parties shall use their best efforts to amicably resolve such disputes, differences or questions through mutual consultation and negotiation within [30 (thirty)] days from the receipt of the same. If the Parties are unable to resolve such disputes, differences or questions within the above stipulated 30 (thirty) day period, the matter shall be referred to mediation.
- iii. In the event where any dispute, controversy or claim arising out of or relating to this Agreement or the breach, termination or invalidity thereof cannot be amicably settled through mutual consultation and negotiation, the Parties shall refer the same to BTASP PMU who shall nominate two (2) mediators. The Mediators shall attempt to resolve the dispute within [thirty (30)] days from the date of the appointment of the mediators.

The decision of the Mediators shall be final and binding.

iv. Contact person for dispute or GRM within the BTASP:

Until the rollout of the GRFM, all complaints and feedback related to the shall be directed towards the following BTASP project office contact for feedback and complaints:

Project Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009591191 ,
Email: pd.btaspforest@kp.gov.pk

Deputy Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009596367,
Email: ibrahimrazmian@gmail.com,

10. DISPUTE RESOLUTION WITHIN THE COMMUNITIES

- a. If any community member (man, woman or child; residential or semi-residential) feels disproportionately disadvantaged because of this Agreement the community member may refer to the following procedure:
 - a. **Step 1:** Presentation of issue to VDC/WO, Management Committee and SDFO/CDO (m/f). The person concerned or his/her representative can ask to discuss his/her case in a meeting of the VDC/WO, Management Committee and the SDFO/CDO. This request can be in oral or written form. If the case is

not satisfactorily solved, a next step can be taken on initiative of the community member. (Time limit for redressal at this stage: **seven (7) Days**)

- b. Step 2: Presentation of issue to DFO:** The person concerned or his/her representative can ask to discuss his/her case in a meeting with the DFO and the (AD)CDE&GAD (m/f). This request can be in oral be or written form. If the case is not solved satisfactorily, a next step can be taken on initiative of the community member. (Time limit for redressal at this stage-(fourteen)14 Days)
- c. Step 3: Presentation of the issue to CF:** The person concerned or his/her representative can ask to discuss his/her case in a meeting with the CF and PD or authorized representative from PMU and Director or Deputy Director CDEGAD. This request shall be in written form. The decision of this forum is final. (Time limit for redressal at this stage thirty (30) days)

11. PERFORMANCE BASED PAYMENT PROCEDURE

Beside Provision of Planting Stock and Transportation from Nursery to Plantation Site By BTASP, For Woodlot establishment, the owner/s will be paid Rs. 40 per plant (for earthwork, plantation and protection beside free provision of Plants). The payment will be made in three instalments as given below:

For wood lot with 8 x 8 feet spacing (1680 plants per ha)

Procedure for payment of Instalment to the VDC			
Cost per Plant 40 PKRs for Survived Plant only			
Instalment	1st	2nd	3rd
Milestone	After digging of Pits and Plantation	After DFO Monitoring	After 3rd party Monitoring
Amount (PKRs) /plant	20	10	No. of Survived plants x 40 - (1st and 2nd Instalment)
%age of Labor Cost	50	25	25

SIGNATURE

Signature Party-A

Divisional Forest Officer _____

Dated _____

SDFO/RFO _____

Dated _____

CDO (M/F) _____

Dated _____

Signature Party-B

Individual beneficiary Name: _____

CNIC: _____

Signature: _____

Date: _____

Signature Party-C

Chairman of the VDC

Name:

CNIC:

Date:

Chairperson of the WO

Name:

CNIC:

Date:

Vice Chairman of the VDC

Name:

CNIC:

Date:

Vice Chairperson of the WO

Name:

CNIC:

Date:

Treasurer of the VDC

Name:

CNIC:

Date:

Treasurer of the WO

Name:

CNIC:

Date:



Annex 5.4 Agreement Enrichment Planting

Agreement for Enrichment planting with VDC/WO, management committee and DFO office

This agreement for the Gap closure and enrichment planting intervention of the BTASP (hereinafter the "Agreement") is being entered on this ____ day of _____, 20..... (Hereinafter the "Signing Date") by and between:

Forest Department through Divisional Forest Officer (hereinafter referred to as "Party A")
The Management committee established for the specific intervention /activities within the village jurisdiction in collaboration with VDC. (hereinafter referred to as "Party B") And,
Village development communities (VDC) / Women Organization (WO) (hereinafter referred to as "Party C")

(Party A, Party B and C shall hereinafter be collectively referred to as "Parties" and individually as Party)

This Agreement is being entered into by and between the Parties for the purposes of implementation of the Gap closure and enrichment planting in Planning Unit _____/ Area _____ mentioned in the operational plan of the planning unit (appended herewith to this Agreement as "Annex 'A'") under the BTASP Project of the Khyber Pakhtunkhwa Forest Department. The role and responsibilities and terms and conditions of this agreement are as follows:

1. TERM

i. The term of this Agreement shall commence from the _____, 20__ and shall continue for a period of five () years (hereinafter the "Term").

2. SCOPE OF THE AGREEMENT

This Agreement will be a legal document between Management Committee, VDC/ WO of planning unit _____ and Divisional Forest office _____ for the implementation of the gap closure and enrichment planting prescribed in the operational plan of Participatory Forest Resource Management Plan of Planning unit _____.

3. ACTIVITIES

The Gap closure and enrichment planting activities to be undertaken under this agreement are specified in the 5-year PFRMP of the _____ Planning Unit, which includes the detailed operational plan for each intervention:

4. Roles, Responsibilities and obligations of the contractual parties:

vii. **Obligations of “Party A”**

a) **Monitoring & Supervision**

- Monitoring of the application of ESMF especially the risk analysis and mitigation mechanisms, the use of child labour, equal pay between men and women, proper consultation with the land users, FPIC, the use of harmful fertilizers, etc.
- Shall provide payment for labor and planting stock along with transportation,
- Shall pay the watch and ward for three years as per the operational plan,
- Technical layout of planting design, plantation model and species composition according to the PFRMP of Planning unit _____,
- Shall provide technical assistance on implementation/ execution on Gap closure and enrichment planting to the VDC / Management committee,
- Shall provide orientation to the Management Committee and VDC/WO on the implementation guidelines for compliance,
- Shall monitor the plantation on semi-annual basis and report to PMU,
- Survival rate assessment on annual basis,
- Shall keep the record of the implementation,
- Shall provide the payroll list to party C at the end of the afforestation activity,
- Shall develop a plantation journal for the activity with sustainable control use for 20 years.

viii. **Obligation of “Party B”**

- Party B shall allocate the land prescribed in the PFRMP for the Gap closure and enrichment planting,
- Shall agree with the sustainable control use of the Gap closure and enrichment planting as prescribed in the plantation journal,
- Shall agree on the 10 % contribution from the cash revenue generated of the activity to the NRM funds,
- Shall agree not to change the land use of the allocated land for up to 20 years.

ix. **Obligations of “Party C”**

- Party C shall ensure community contribution from the management committee (party B) as prescribed in the PFRMP,
- Shall jointly with the Management Committee (party B) identify the labors for plantation activity and provide a list of the identified labors with the required information (Name, Father Name, Village and copy of CNIC and) to party A,
- Shall notify a person form the VDC for monitoring of the plantation progress and submission of the completion report to the party A,
- Shall identify the watch and ward persons jointly with the management committee (party B) and share the list along with the required information with party A.

Performance based bonus

The performance-based bonus is a payment for survival rate of the activity. The bonus is going to be paid on annual basis for up to three years, on the third-party monitoring report as per the table given below;

6. GENERAL COMMITMENT

- v. The Parties agree and acknowledge that this agreement cannot be used as documentary evidence by any Party to assert their rights locally and before the courts for their internal disputes not related to this Agreement.
- vi. The Parties agree and acknowledge that they will ensure compliance with ESMF framework of the BTASP.

13. CONDITIONS OF THE AMENDMENT AND EXTENSION OF THE AGREEMENT

- vi. Either of the party may amend the Agreement, if required. However, such amendments must be mutually agreed by all Parties in writing.
- vii. After 5-years of implementation the agreement may be extended with mutual consent of all Parties.
- viii. By the end of the project (.....) the BTASP project will not have any financial obligations.

14. TERMINATION OF THE AGREEMENT

- vii. If Party A wishes to terminate the agreement, they may invoke Rule 22 of the Joint Forest Management (Community Participation), Rules, 2004 ("**CP Rules**").
- viii. If Party B wishes to terminate the Agreement, they may invoke Rule 19(4) read with Rule 11(f) of the **CP Rules**, 2004.
- ix. Neither of the parties may unilaterally terminate the Agreement without first seeking to resolve any disputes amicably.

15. DISPUTE RESOLUTION

- v. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement.
- vi. In the event where any dispute, difference or question arises out of or in connection with this Agreement, and / or arising due to a breach of this Agreement, the aggrieved Party shall promptly serve upon the other Party a notice stating such disputes, differences or questions (the "Dispute Notice"). Upon the receipt of the Dispute Notice by the other Party, the Parties shall use their best efforts to amicably resolve such disputes, differences or questions through mutual consultation and negotiation within [30 (thirty)] days from the receipt of the same. If the Parties are unable to resolve such disputes, differences or questions within the above stipulated 30 (thirty) day period, the matter shall be referred to mediation.

-
- vii. In the event where any dispute, controversy or claim arising out of or relating to this Agreement or the breach, termination or invalidity thereof cannot be amicably settled through mutual consultation and negotiation, the Parties shall refer the same to BTASP PMU who shall nominate two (2) mediators. The Mediators shall attempt to resolve the dispute within [thirty (30)] days from the date of the appointment of the mediators.

The venue for the mediation proceedings shall be at _____, Pakistan. The language of mediation and the appropriate documentation shall be _____.

The decision of the Mediators shall be final and binding.

- iv. **Contact person for dispute or GRM within the BTASP:**
Until the rollout of the GRFM, all complaints and feedback related to the shall be directed towards the following BTASP project office contact for feedback and complaints:

Project Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009591191 ,
Email: pd.btaspforest@kp.gov.pk

Deputy Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009596367,
Email: ibrahimrazmian@gmail.com,

12. DISPUTE RESOLUTION WITHIN THE COMMUNITIES

- b. If any community member (man, woman or child; residential or semi-residential) feels disproportionately disadvantaged because of this Agreement the community member may refer to the following procedure:
- d. **Step 1:** Presentation of issue to VDC/VO, management committee and SDFO/CDO (m/f). The person concerned or his/her representative can ask to discuss his/her case in a meeting of the VDC/VO, management committee and the SDFO/CDO. This request can be in oral or written form. If the case is not satisfactorily solved, a next step can be taken on initiative of the community member. (Time limit for redressal at this stage: **seven (7) Days**)
- e. **Step 2: Presentation of issue to DFO:** The person concerned or his/her representative can ask to discuss his/her case in a meeting with the DFO and the (AD)CDE&GAD (m/f). This request can be in oral or written form. If the case is not solved satisfactorily, a next step can be taken on initiative of the community member. (Time limit for redressal at this stage-(fourteen)14 Days)
- f. **Step 3: Presentation of the issue to CF:** The person concerned or his/her representative can ask to discuss his/her case in a meeting with the CF and PD or authorized representative from PMU and Director or Deputy Director CDE&GAD. This request shall be in written form. The decision of this forum is final. (Time limit for redressal at this stage thirty (30) days)

SIGNATURE

Signature Party-A

Divisional Forest Officer _____

Dated _____

SDFO/RFO _____

Dated _____

CDO (M/F) _____

Dated _____

Signature Party-B

Representative Name: _____

Position: _____

CNIC: _____

Signature: _____

Date: _____

Signature Party-C

Chairman of the VDC

Name:

CNIC:

Date:

Chairperson of the WO

Name:

CNIC:

Date:

Vice Chairman of the VDC

Name:

CNIC:

Date:

Vice Chairperson of the WO

Name:

CNIC:

Date:

Treasurer of the VDC

Name:

CNIC:

Treasurer of the WO

Name:

CNIC:



Annex 5.5 Women nurseries

Agreement for Women nursery activities with VDC/WO, management committee and DFO office

This contract for the raise of seedlings in the women nursery (.....Name) (hereinafter the “contract”) is being entered on this ____ day of _____, 20..... (Hereinafter the “Signing Date”) by and between:

Forest Department through Divisional Forest Office(hereinafter referred to as “Party A”), and the individual Nursery owner(Name / Father name and CNIC) nominated by the Forest Department as per the criteria specified in the PFRMP (hereinafter referred to as “Party B”) And, Village development communities (VDC) / Women Organization (WO) of the concern PU (hereinafter referred to as “Party C”)

(Party A, Party B and C shall hereinafter be collectively referred to as “Parties” and individually as Party)

This contract agreement is being entered into by and between the Parties for the purposes of the purchase of plant (from Part B by party A) for plantation in Planning Unit _____/ location _____ mentioned in the operational plan of the planning unit (appended here with to this Agreement as “Annex ‘A’”) under the BTASP Project of the Khyber Pakhtunkhwa Forest Department. The role and responsibility and terms and conditions of this agreement are as follows:

1. Terms of Contract:

The terms of this Contract Agreement shall begin from the _____, 20..... and shall continue for a period max 2 years (hereinafter the “Term”).

2. SCOPE OF THE AGREEMENT

This Contract Agreement will be a legal document between Parties for the raising of the plants in the nursery for the plantation purpose as per the species, varieties, quality, quantity and time specified in annexures - I & II to be purchased by the party A.

3. Role, Responsibilities and obligation of the contractual parties

A. Obligations of “Party A”

The forest department shall be bound to :

- Monitoring of the application of ESMF especially the risk analysis and mitigation mechanisms, the use of child labour, equal pay between men and women, proper consultation with the land users, FPIC, the use of harmful fertilizers, etc.
- Proposed will variety of plants which are agreed in Annex I, provided the quality of these plants meet the conditions specified in Annex II.
- Varieties of plants should have been Planted in plastic bags with fixed price tags which is PKR.10/- per fit plant.
- AS per requirement the forest department shall provide a 2 day trainings to owners of nurseries regarding Planting species, handling and record keeping.
- Defined payment as given below shall be made at the time of hand overing of plants.
- SDFO inspects nursery very 3 month and before each installment

- The 3rd Party Monitoring will monitor the nursery and especially the result before the final payment of the fit plant.

I. Terms and conditions of the advance payments:

Each nursery will contain 25,000 seedlings and the nursery owner will get 250,000 PKRs @ 10 Rs per plant in three different instalments. Below is the procedure for women nursery raising:

Women Nurseries			
Milestone	After signing of agreement	After 3 months 1 st Monitoring	During shifting of fit seeding for Plantation
Instalment	1 st Instalment (25 %)	2 nd Instalment (25 %)	Final payment according to available Fit seedlings
Amount in PKRs	62500	62500	(10 PKR x no of Fit plants) - 50 %
In each nursery 25000 seedlings will be raised with a rate of 10 PKRs per plant for fit plant. The nursery owner can get up to 250,000 PKR. Calculation: 10 PKR x 25.000 plants =250.000 PKR (max) Last instalment: based on the availability of fit plants: (10PKR *XYZ number of fit plants) minus 1. and 2. Instalment. Criteria for Fit Plants are ● 18 inches for broad leaved and 12 inches for Conifers ● The root shoot ratio should be 1:2 ● About 60 % foliage (leaves) are intact (evergreen) ● Pest free, not damaged			

B. Obligations of “Party B”

The Nursery owners shall be bound to:

- Hand over, particular species of plants in required quantity in accordance with the time, and quality mentioned in Annex I and Annex II.
- Ensure standards of the plants production be in accordance with Annex II.
- Nursery owners shall be bound to participate in the trainings concerning nursery raising.
- Nursery owners shall ensure to use the 7x3 size of the polythene bag .
- The Chairman and Vice Chairman VDC / Chairperson WOs the concerned VDC/WOs shall be the guarantor of nursery owners.
- In case of growing more amount of plant then the agreed, the nursery shall have the right to sell the excess amount in the market.

- The nursery owner shall be required to prepare the monthly records of his nursery in the form of nursery journal and shall demonstrate it to the Forest officials during the inspection.
- Apart from any natural calamity if the nursery owner fails to provide the agreed plant species the nursery owner shall be charged an appropriate penalty in addition to amount paid in advance which shall be determined by VDC/WOs and PMU. In case of natural calamity, the confirmation of the district administration is mandatory.
- Uprooting of plants from the soil, their classification and bundling of the species shall be the responsibility of the nursery owners,
- The transportation of the plant species from the nursery area to the Forest department/plantation site shall be the responsibility of the contractor.
- When the plants shall ready, the counting shall be done in the presence of the officials of the Forest Department, the VDC/WOs representatives and the owner of the nursery.

C. Obligations of “Party C”

Being the social organization of the Planning unit VDC/WOs shall be bound to:

- The Chairman and Vice Chairman VDC / Chairperson WOs the concerned VDC/WOs shall be the guarantor of nursery owners
- Shall monitor the performance of the Nursery owners and will provide feedback to party A .
- Shall ensure transparency and accountability in the process of purchasing the plants.
- In case of penalty the VDC/WOs shall be responsible for recovery of plenty amount if any, as guarantor. ☐

NOTE: In case of any dispute, PMU Peshawar shall take final decision which has to be accepted by the both parties.

D: DISPUTE RESOLUTION

- v. The Parties shall use their best efforts to settle amicably all disputes arising out of or in connection with this Agreement.
- vi. In the event where any dispute, difference or question arises out of or in connection with this Agreement, and / or arising due to a breach of this Agreement, the aggrieved Party shall promptly serve upon the other Party a notice stating such disputes, differences or questions (the “Dispute Notice”). Upon the receipt of the Dispute Notice by the other Party, the Parties shall use their best efforts to amicably resolve such disputes, differences or questions through mutual consultation and negotiation within [30 (thirty)] days from the receipt of the same. If the Parties are unable to resolve such disputes, differences or questions within the above stipulated 30 (thirty) day period, the matter shall be referred to mediation.
- vii. In the event where any dispute, controversy or claim arising out of or relating to this Agreement or the breach, termination or invalidity thereof cannot be amicably settled through mutual consultation and negotiation, the Parties shall refer the same to BTASP

PMU who shall nominate two (2) mediators. The Mediators shall attempt to resolve the dispute within [thirty (30)] days from the date of the appointment of the mediators.

The venue for the mediation proceedings shall be at _____, Pakistan. The language of mediation and the appropriate documentation shall be _____.

The decision of the Mediators shall be final and binding.

D: Contact person for disput or GRM within the BTASP:

Until the rollout of the GRFM, all complaints and feedback related to the shall be directed towards the following BTASP project office contact for feedback and complaints:

Project Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009591191 ,
Email: pd.btaspforest@kp.gov.pk

Deputy Director BTASP,
Palosi Road, Aman Abad, Opposite PFI, Near Peshawar Zoo, Peshawar,
Tel: 923009596367,
Email: ibrahimrazmian@gmail.com,

SIGNATURE

Signatories Party-A

Divisional Forest Officer _____

Dated _____

SDFO/RFO _____

Dated _____

CDO (M/F) _____

Dated _____

Signature Party-B

Individual beneficiary Nursery owner Name: _____

CNIC: _____

Signature: _____

Date: _____

Signature Party-C

Chairman of the VDC

Name:

CNIC:

Date:

Chairperson of the WO

Name:

CNIC:

Date:

Vice Chairman of the VDC

Name:

CNIC:

Date:

Vice Chairperson of the WO

Name:

CNIC:

Date:

Treasurer of the VDC

Name:

CNIC:

Date:

Treasurer of the WO

Name:

CNIC:

Date:

Annex-I The number of plants and the time of handing over plant species:

S/No.	Types of Plant Species	Number of Plants	Date of handing over of Plant Species
	Total Plant Species		

Annex-II Quality of Plant Species:

S/No.	Types of Plant Species	Height	Width

Annex 6 Monitoring Formts

Annex 6.1 Monitoring Format Plantation establishment

Name of Field Supervisor: _____ Date: _____

Plantation Location: _____ Type of Plantation: Block Plantation

/ Woodlot carried out: _____ Area in Ha _____

Activities detail and quantity:

S. No	Activity	Quantity
1.	Number of Pits Marked	
2.	Number of Pits dugout	
3.	Number of Plants shifter to Plantation site	
4.	Number of Plants planted	
5.	Watering of Plants (Number)	
6.	Number of supports installed. If needed	
7	Pits mulching (Number)	

Composition of Species Planted

S. No	1.	2.	3.	4.	5.	6.
Species						
Number						

Spacing of planting: 10x10 8x8 5x5 others

☐
☐
☐
☐

Damage Report

Type of Damage if any: _____ Quantity of Damage _____

Causes of Damage: _____

Signature / Thumb impression of Field Supervisor:

Report Received by:



Annex 6.2 Field Monitoring Form Performance of Forest Warden

Name of Chowkidar: _____ CNIC: _____

Date of Monitoring: _____ Name of Plantation: _____

Location of Plantation: _____

Does the forest warden has a specified workplan?

Sign of damages, if any. Please tick in the below with ranking from 1 - 5

Damage	<i>Cutting</i>	<i>Dead/Dry</i>	<i>Grazing</i>	<i>trampling</i>	<i>lopping</i>	<i>Fire</i>	<i>Others</i>
Ranking							

Key (1-very low, 2-Low, 3-Medium, 4-High, 5-Sevear)

General Remarks

1. Officer In Charge

2. Sub-Divisional Forest Officer:

Monitoring conducted by:

Forest Guard

Sub-Divisional Forest Officer

Report Checked & Verified by

Divisional Forest Officer

_____ **Forest Division**



Annex 6.3 Data sheet for Planting of multi-purpose fast growing tree species Enrichment planting , Wood Lots

Name of Plantation in charge & Designation # : _____ Date: _____

1) Contact #:

				-							
--	--	--	--	---	--	--	--	--	--	--	--

2) Date of establishment: _____ Division _____, Subdivision/Range _____

Block _____ Location: _____

3) Area Coordinates: N _____, E _____ Elevation: _____m

4) Area Claimed _____ ha & Area verified: _____ ha Aspect : _____

5) Ecological zone: _____ Site quality: Good ☐ Fair ☐ Poor ☐

6) Agreement exists: ☐ Yes ☐ No, if yes, please provide a copy of the agreement. **Only one time**

7) Name of VDC: _____ Name of Mgt. committee _____

8) Survival rate _____, Causes of failure _____

9) Number of Total plants _____ Tube Plants: _____ Bare rooted Plants _____

Species:

Species					
Number					

10) Beating up of failure: Yes ☐ No ☐, Date _____ If yes, how much _____

11) Slope: Gentle ☐ Moderate ☐ Steep ☐

12) Site accessibility: Metal road ☐ Mud road ☐, Bridle path ☐, other _____

13) Area demarcated: Yes ☐ No ☐

If yes, specify signs of demarcation _____

14) Land status: _____

15) Dispute if any _____

16) Source _____ of _____ planting _____ stock

17) Protection

mechanism:

18) Future

plan:

19) Issues:



Annex 6.4 Monitoring Women Nursery

A. General Information:

- Name of nursery grower _____ Date: _____
- 1) Marital Status: Married ☐ Single ☐ Widow ☐ Divorced ☐ Age: _____ years
- 2) CNIC #: _____
- 3) Date of establishment: _____ Location _____ nursery
- 4) Division _____, Range _____, Block _____
- 5) Coordinates: N _____, E _____ Elevation: _____ m
- 6) Agreement exists: ☐ Yes ☐ No, if yes, please provide a copy of the agreement. **One time only**

B. Technical Information

Particulars	Status (Y/N)	Comments
Inspection paths		
Signboards		
Fencing		
Irrigation system (Please tick, pipe, water channel, tube well)		

1) Management Practices:

- b) Total No. of poly: _____ No. Filled: _____ No. sown _____ No.
- c) Seed treatment ☐ Yes ☐ No, if yes, please specify _____
- d) Germinated: _____ No. Established seedlings: _____ No.

2) Cultural Practices (pictorial verification):

Practice	Irrigation/watering	Transplanting	Root trimming	Shifting	Fertilizers	Pesticides
Yes/No						

3) Specie wise status of plants (Plantable / un-plantable): _____ Units

Species	Poly bags sown (No.)	Germination rate (%)	No. of plantable seedlings	No. of un-plantable seedlings	Survival rate (%)	Remarks



Annex 6.5 Monthly Progress Report – DFO

DFO Monthly Progress Reporting										
Region: _____ Division: _____										
S. No	Item/Activities	Progress for the Month _____								
		PU _____			PU _____			PU _____		
		Total Target	Target achieved during a month	Total Achievement	Total Target	Target achieved during a month	Total Achievement	Total Target	Target achieved during a month	Total Achievement
1	Endloser and Plantation Journals									
2	Community Wardens (MM)									
3	Enrichment (20% of ha Ha)									
4	Block Plantation (Ha) 10x10									
5	Block Plantation (Ha) 8x8									
6	Wood Lots (ha)									
7	Close Plantation 5x5									
8	Seedling for FF									
9	Women Nursery (No)									
10	Provide fodder (ha)									
11	Spring Shed Measures per ha									
12	Fire Lines (km)									
13	No of fire fighting tool kit									
14	Tools for the communit Wardns									
15	Women Livelihood Project									
16	VDC Project									
17	Capacity Building									
	Training 1									
	Training 2									
Prepare and Submitted by (Name and Sign of DFO)					Verified by (name and sign of Conservator)					



6.6 Certificate of Plantation Establishment

Date: / /20

Certified that a total of _____ hectare/s plantation has been completed in Planning Unit (PU)

_____ with a following specification:

Forest Division: _____ Sub Division: _____

Union Council: _____ Village Council: _____

VDC/WO: _____ Mgt. Committee: _____

Start Date: _____ End Date: _____

Total Plants Planted: _____ B/Rooted: _____ P-Bag: _____

Spacing: _____ Pit Size: _____ Pit Design: _____

Specie Composition:

Specie						
Number						

President VDC

SDFO,

DFO,

Management Committee / Owner of the planation



Annex 6.7 Seedling handing over sheet

Certified that a total of seedlings has been handed over from:

Date:

Nursery owner:

Contract Number:

Planning Unit (PU) _____ with a following specification:

Forest Division: _____ Sub Division: _____

Recipient of the seedlings

DFO /SDFO: _____ Forest Division Name: _____ Sub-Division

VDC/WO : _____ Name of Planning Unit

Total Plants handed over: _____ B/Rooted: _____ P-Bag: _____

Specie Composition:

Specie						
Number						

All handed over plants fulfill the requirements of "FIT" plant.

Owner of the planation

DFO/ or VDC/WO or Management Committee /



Annex 7 Average cost and net returns Eucalyptus

The below table has been taken from the WWF Report: ²⁶

Table 13. Average cost of production for Eucalyptus / ha

Activity	Cost per ha (PKR)
Layout of pits	2,470
Digging of pits	13,832
Cost of Planting Stock (Rs. 5 per plant)	10,760
Carriage up to planting site	2000
Internal carriage of plants	7,410
Planting	9,880
Watering	64,400
Watch and Ward	26,680
Post plantation Care Year-2	51,520
Post plantation Care Year-3	34,346
Lease of land for 5 years	40,000
Total cost per ha	263,298
Marketing or Harvesting Cost (10% of TC)	26,329.8
Grand Total Cost for 5 years	289,628

Table 14. Average net returns for Eucalyptus / ha

No of trees at a 80% survival rate	1,721
Average weight per tree in kg	120
Total weight of all trees in ha	206,520
Price per kg	8
Gross Returns from 1 ha in 5 years	1,652,160
Net Returns from 1 ha in 5 years	1,652,160 - 289,628 = 1,362,532
Net Returns from 1 ha for 1 year	1,362,532/5 = 272,506
Net Returns from 1 Acre for 1 year	272,506/ 2.47 = 110,326

Table 30: Average cost and net return for Eucalyptus plantation

²⁶ Source: PAK- WWF Report 2020 Role of MDF & Particleboard Industry in protection of Natural Forests and Promotion of Agroforestry in Pakistan.



Annex 8 Seed Collection Calendar

Season for ripening of different Plant Specie Seeds and No. of seeds in per KG

S/No	Botanical Names	English Names	Collection of Seed (Months)	No. of seeds Per KG
1.	Acacia nilotica	Acacia/ Kikar	May & June	6600
2.	Acacia catechu	Cutch	December-February	32000-35000
3.	Acacia modesta	Phulai	January and February	21160-215000
4.	Albizia lebbek	Siris	January- March	4940-12430
5.	Albizia procera	White Siris	February-May	17640-29980
6.	Azadirachta indica	Neem	June-August	2880-4400
7.	Bombax ceiba	Red silk-cotton tree	April-May	26000-32000
8.	Bauhinia variegata	Kachnar / Orchid tree	May-June	3520-2470
9.	Ehretia officinalis	Koda Tree	November-February	46000
10.	Bauhinia purpurea	Purple Orchid Tree	January-March	3600
11.	Bischofia jananica	Java Cedar	December – January	91710
12.	Callistemon lanceolatus	Bottlebrushes	June-July	Numerous
13.	Casuarina equisetifolia	Horsetail	June-December	660000
14.	Cassia fistula	Golden Shower	January- April	5500-6070
15.	Cedrela toona	Red Cedar	May-June	282000-423000
16.	Cordia myxa	Assyrian plum	July-August	4230-5460
17.	Dalbergia sissoo	Shisham	December-January	13220-15860
18.	Dodonea viscosa	Broadleaf hopbush	May-June	70400-105600
19.	Eucalyptus camaldulensis	River Red Gum/ Sufaida	AugustOctober-February	n.n.
20.	Eucalyptus citriodora	lemon eucalyptus/ citron-scented gum	August-October	57000-99000
21.	Eucalyptus tereticornis	Forest Red Gum	August-October	367400
22.	Syzygium cumini	Black plum	July- August	1166
23.	Ficus bengalensis	Fig	July-August	Numerous
24.	Ficus religiosa	Sacred Fig	March-May	Numerous
25.	Gmelina arborea	White Teak	May-June	1460-2640
26.	Jacoranda ovalfolia	Fern Tree	February-March	38800

27.	Leucaena leucocephala	Ipil Ipil	May-December	28570
28.	Melia azedarach	China Berry/ White Cedar	January-February	1440
29.	Morus Alba	Mulberry	April-May	42300-45800
30.	Olea cucpidata	Wild Olive	September-December	5820-11780
31.	Parkinsonia aculeata	Horse Bean	May-July	12320
32.	Pongamia glabra	Sukh Chain/ Indian Beech	April-May	1540-1650
33.	Pinus roxburghii	Chire Pine	March-May	2000-11000
34.	Pinus wallichiana	Blue Pine/ Himalayan white Pine	September-October	6000-15000
35.	Prosopis juliflora	Mesquite	May-June	22000-52000
36.	Prosopis cineraria	Persian mesquite / Ghaf	June-August	27160-28600
37.	Robinia pseudoacacia	Black locust	October-January	52800
38.	Alstonia scholaris	Devil Tree	May-June	620000
39.	Crataeva religiosa	Sacred Garlic Pear / Temple Plant	June-July	195000-225000
40.	Delonix regia	Flamboyant/ Gulmohar Tree	July- August	
41.	Moringa oleifera	Moringa/ Drumstick Tree	April-May	
42.	Saraca indica	Ashoka Tree	June-July	
43.	Tamarix aphylla	Athel Pine	September-December	481000
44.	Tecoma stans	Yellow Trumpetbush	May-July	85000
45.	Terminalia arjuna	Arjuna/ Arjun Tree	February-May	520
46.	Terminalia bellirica	Baheda	November-February	1760-2420
47.	Terminalia chebula	Black Myrobalan	November-March	150-200
48.	Ziziphus mauritiana	Indian jujube / Indian Plum	March-April	1020
49.	Ziziphus nummularia	Wild Jujube / Jharberi	October-November	2360



Annex 9: Implementation Calendar PFRMP Measures

Task	Project Year / Months, Low land & Foothills (Peshawar, Malakand, Haripur, Lower Swat, AT, and part of Buner)												Project Year / Months, High hills (Bailies, Upper Swat, Ajour, Kaghan, Hazara Tribal, Siran and Tor Ghar)												Risk
	PY 2												PY 2												
	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	
Timetable implementation																									
Climatical conditions																									
Monsoon rains																									Somehow damage due to F ash floods
Spring Season																									
Snowfall																									
Main community/farm operations																									
Field preparation time wheat, corn etc.																									Two crop, unavailability
Harvest time communities																									Single Crop, unavailability
Gross Cutting																									Community unavailability
BTASP measures																									
1. Update of the plantation and enclosures Journals																									
1.1. Review of existing journals																									
1.2. Collection of necessary info/data																									
1.3. Update the plantation and enclosures journals																									
2. Afforestation New Block Plantations																									
2.1. Aggrement with VDC/Owners																									
2.2. Site preperation and earthwork																									
2.3. Plantation																									
2.4. Replacement of Failure																									
3. Nurse ries																									
3.1. Agreement with women and preperation																									
3.2. Nursery Raising																									
3.3. Transplantation of seedling from Nursery	After one year P-bag plants and one more seedlings during planting season												After 18 to 24 months P-bag plants during planting season												
4. Enrichment planting																									
4.1. Site preperation and earthwork																									
4.2. Plantation																									
5. Spring protection planting																									Damage due to grazing in intel stage
5.1. Site preperation and earthwork																									
5.2. Plantation																									
6. Wood Lots																									Desirable planting stock is available
6.1. Site preperation and earthwork																									
6.2. Plantation																									
7. Thinning																									Community agreed on sustainable and single tree harvesting
7.1. Marking of trees																									
7.2. Harvesting																									
8. Range Management																									
8.1. Training of Rotational Grazing system																									
8.2. Distribution of Fodder Seeds																									
9. Training of Olive Grafting																									
10. Vegetable Gardening																									
11. VDC/WO Trainings																									
12. Community Technical Trainings																									
13. Fire Fighting Activities																									

Annex 10 Revision of Gov Rates TA/DA



GOVERNMENT OF KHYBER PAKHTUNKHWA FINANCE DEPARTMENT (REGULATION WING)

NO. FD/SOSR-II/8-52/2017
Dated Peshawar the: 14/07/2017

From:

The Secretary to Govt. of Khyber Pakhtunkhwa,
Finance Department, Peshawar.

To:

1. The Senior Member, Board of Revenue, Khyber Pakhtunkhwa.
2. All Administrative Secretaries to Govt. of Khyber Pakhtunkhwa.
3. The Principal Secretary to Governor, Khyber Pakhtunkhwa
4. The Principal Secretary to Chief Minister, Khyber Pakhtunkhwa.
5. The Secretary, Provincial Assembly, Khyber Pakhtunkhwa
6. All Heads of Attached Departments in Khyber Pakhtunkhwa.
7. All Commissioners in Khyber Pakhtunkhwa.
8. All Deputy Commissioners in Khyber Pakhtunkhwa.
9. All Political Agents / District & Sessions Judges in Khyber Pakhtunkhwa
10. The Registrar, Peshawar High Court, Peshawar.
11. The Chairman, Public Service Commission, Khyber Pakhtunkhwa.
12. The Chairman, Services Tribunal, Khyber Pakhtunkhwa.
13. The Chairman, Provincial Ombudsman Secretariat, Khyber Pakhtunkhwa

Subject:

**REVISION OF RATES OF DAILY ALLOWANCE ON OFFICIAL DUTY WITHIN
THE COUNTRY**

In continuation of this Department Notification of even number dated 21/05/2005, 17/12/2005, 05/10/2006, 13/07/2010 & subsequent amendment in rules 2.36 in clause (i) of TA/DA Rules vide this department notification dated 31/10/2012 the Provincial Government of Khyber Pakhtunkhwa has been pleased to enhance / revise the rates of Daily Allowance admissible to all Provincial Servants, Govt of Khyber Pakhtunkhwa working in BPS-01 to BPS-22 with effect from 01/07/2017 as per details given below:-

BPS	Existing Ordinary Rates	Special Rates	Revised Ordinary Rates	Special Rates
1-4	310	500	496	800
5-11	390	550	624	880
12-16	700	900	1120	1440
17-18	1250	1600	2000	2560
19-20	1550	2050	2480	3280
21	1750	2500	2800	4000
22	1750	3000	2800	4800

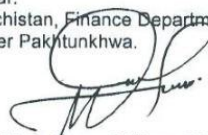
SECRETARY TO GOVERNMENT
OF KHYBER PAKHTUNKHWA
FINANCE DEPARTMENT

Endst: NO. FD/SO(SR-II)/8-52/2017

Dated Peshawar the 14/07/2017

A Copy is forwarded for information to the:-

1. Accountant General, Khyber Pakhtunkhwa, Peshawar.
2. Secretaries to Government of Punjab, Sindh & Balochistan, Finance Department.
3. All Autonomous / Semi Autonomous Bodies in Khyber Pakhtunkhwa.


(Musharraf Khan Marwat)
Additional Secretary (Reg.)



Annex 11:BTASP Hand-Over Agreement

BTASP Hand-Over Agreement of Plantations and Enclosers

The **Hand-Over contract** shall after the end of the project/ or earlier (tbd) ensure a binding agreement for further technical collaboration and responsibilities between the DFOs office, VDC and its PFRMP and the Mgt. Committee with its plantation and enclosure Journal. The hand over contracts shall contain in general the following (not complete and needs to be further developed) content:

Mgt. Committee will have the obligation:

The Mgt. Committee agrees/commit to the continuation and implementation of the participatory developed site-specific development plans “Journals”.

The Mgt. Committee agree to report to the VDC/WO on (tbd) basis on the developments and actions carried out

The Mgt. Committee agree that representatives of the VDC/WO and the DFOs staff are allowed to monitor the condition of the site and carried out activities.

VDC will have the task:

To check and monitor the silvicultural activities and controlled use as documented in the “journals” and the overall PFRMP.

Review the Overall PFRMP after a period of 10 years with the involvement of the Management committees

DFO staff will have the task

To check and monitor the silvicultural activities and controlled use as documented in the “journals” and the overall PFRMP on VDC/WO level and Mgt. committee level

Check and monitor harvesting quotas on annual bases agreed in the PFRMP and subsequently in the “journals” for conifers and broadleaved species.

To support the VDC/WO on the update of the PFRMP after 10 years with technical advisory on silvicultural planning,



Annex 12 Seed Testing Costs

DIRECTOR RESEARCH & DIRECTORATE, PESHAWAR		GOVERNMENT OF KHYBER PAKHTUNKHWA FOREST DEPARTMENT PESHAWAR
SHAMI ROAD, PESHAWAR PHONE 091-9210584 FAX 091-9211051 EMAIL rd.dir2002@gmail.com Twitter: Research@Research62286502 Facebook: ResearchandDevelopment Directorate		
No. <u>887</u> /R&D Dated Peshawar the <u>19/02</u> /2025		

To

The Project Director
10-BTTP/ BTASP.

Subject: **RATES FOR SEED TESTING.**

Memo:

Kindly refer to subject noted above and it is stated that the following are the rates for different tests.

S. No	Test	Unit Cost	Total Per Test
01	Seed Viability	1250	5000
02	Seed Germination	2000	8000
03	Seed Health	2500	10000

The above rates are for four different replicates, please.

Director
Research & Development
Directorate, Peshawar